

15. Buses

15.1 Overview

EDMAC in this chapter means the GWCA function of ESWM. [Table 15.1](#) lists the bus masters and bus slaves, [Figure 15.1](#) shows the system bus configuration, and [Table 15.3](#) lists the addresses assigned for each bus.

Table 15.1 Bus specifications (1 of 2)

Bus master/ slave name	Sync clock	Description
Classification: Bus masters		
CPU0MAXIBI (Arm® Cortex®-M85)	CPUCLK0	Connected to the CPU0 Master-AXI (M-AXI) Interface
CPU0PAHBI (Arm® Cortex®-M85)	CPUCLK0	Connected to the CPU0 Peripheral AHB (P-AHB) Interface
CPU1CAHBI (Arm® Cortex®-M33)	CPUCLK1	Connected to the CPU1 Instruction Cache for instructions and operands
CPU1SAHBI (Arm® Cortex®-M33)	CPUCLK1	Connected to the CPU1 Data Cache for system
NPU0BI	NPULCK	Connected to the NPU0 Interface
NPU1BI	NPULCK	Connected to the NPU1 Interface (Read Only)
DMAC/DTC0BI	ICLK	Connected to the DMAC/DTC0 Interface
DMAC/DTC1BI	ICLK	Connected to the DMAC/DTC1 Interface
EDMACBI	ICLK	Connected to the EDMAC Interface
CEUBI	PCLKA	Connected to the CEU Interface (Write Only)
GLCDC0BI	PCLKA	Connected to the GLCDC0 Interface (Read Only)
GLCDC1BI	PCLKA	Connected to the GLCDC1 Interface (Read Only)
DRW0BI	ICLK	Connected to the DRW0 Interface
DRW1BI	ICLK	Connected to the DRW1 Interface
MIPIDSIBI	PCLKA	Connected to the MIPI-DSI Interface
MIPICSIBI	ICLK	Connected to the MIPI-CSI via VIN Interface (Write Only)
Classification: Bus slaves		
MRC0BI	MRICK	Connected to Code MRAM
MRE0BI	MRPCLK	Connected to Extra MRAM (Read Only)
CPU0SAHBI	ICLK	Connected to CPU0 S-AHB
CPU1TCMBI	CPUCLK1	Connected to CPU1 S-TCM and CPU1 C-TCM
S0BI	ICLK	Connected to SRAM0
S1BI	ICLK	Connected to SRAM1
S2BI	ICLK	Connected to SRAM2
S3BI	ICLK	Connected to SRAM3
ECBI	BCLK	Connected to the external devices (External Memory Interface)
OSPI0BI	ICLK	Connected to the OSPI (via DOTF) (External Memory Interface)
OSPI1BI	ICLK	Connected to the OSPI for SIP-Flash (External Memory Interface)
PBBi	PCLKB	Connected to peripheral modules synchronizes with PCLKB
PABi	PCLKA	Connected to peripheral modules synchronizes with PCLKA
PIBi	ICLK	Connected to peripheral modules synchronizes with ICLK
PS0BI	ICLK	Connected to ICU0 controller Connected to DMAC/DTC0 controller

Table 15.1 Bus specifications (2 of 2)

Bus master/ slave name	Sync clock	Description
PS1BI	ICLK	Connected to ICU1 controller Connected to DMAC/DTC1 controller
PSBI	CPUCLK1	Connected to peripheral system module. (CPU1-cache controller)
	ICLK	Connected to peripheral system modules. (MPU, CSC/SDRAM, SRAM, Debug component, Bus controller, common ICU controller, CPU controller, debug controller, NPU, IPC controller, and security attribution controller)
	MRPCLK	Connected to peripheral system module. (MRAM controller)
	PCLKB	Connected to peripheral system module. (system controller)

Note: CPU0MAXIBI: CPU0 M-AXI Bus Interface
 CPU0PAHBI: CPU0 P-AHB Bus Interface
 CPU1CAHBI: CPU1 C-AHB Bus Interface
 CPU1SAHBI: CPU1 S-AHB Bus Interface
 NPU0BI: NPU0 Bus Interface
 NPU1BI: NPU1 Bus Interface
 DMAC/DTC0BI: DMAC/DTC0 Bus Interface
 DMAC/DTC1BI: DMAC/DTC1 Bus Interface
 EDMACBI: EDMAC Bus Interface
 GLCDC0BI: GLCDC (graphic 1) Bus Interface
 GLCDC1BI: GLCDC (graphic 2) Bus Interface
 DRW0BI: DRW (texture) Bus Interface
 DRW1BI: DRW (data) Bus Interface
 CEUBI: CEU Bus Interface
 MIPIDSIBI: MIPI-DSI Bus Interface
 MIPICSI: MIPI-CSI via VIN Bus Interface
 MRC0BI: Code MRAM Bus Interface
 MRE0BI: Extra MRAM Bus Interface
 CPU0SAHBI: CPU0 S-AHB Interface (Bus Interface from Bus Masters except CPU0 to CPU0 ITCM and CPU0 DTCM)
 CPU1TCMBI: Bus Interface from Bus Masters except CPU1 to CPU1 S-TCM and CPU1 C-TCM
 S0BI: SRAM0 Bus Interface
 S1BI: SRAM1 Bus Interface
 S2BI: SRAM2 Bus Interface
 S3BI: SRAM3 Bus Interface
 ECBI: External memory CSC and SDRAM Bus Interface
 OSPI0BI: External memory OSPI Bus Interface
 OSPI1BI: External memory OSPI for SIP-Flash Bus Interface
 PBBI: Peripheral synchronizes with PCLKB Bus Interface
 PABI: Peripheral synchronizes with PCLKA Bus Interface
 PIBI: Peripheral synchronizes with ICLK Bus Interface
 PS0BI: ICU0, DMAC/DTC0 Controller Bus Interface
 PS1BI: ICU1, DMAC/DTC1 Controller Bus Interface
 PSBI: Peripheral System Bus Interface

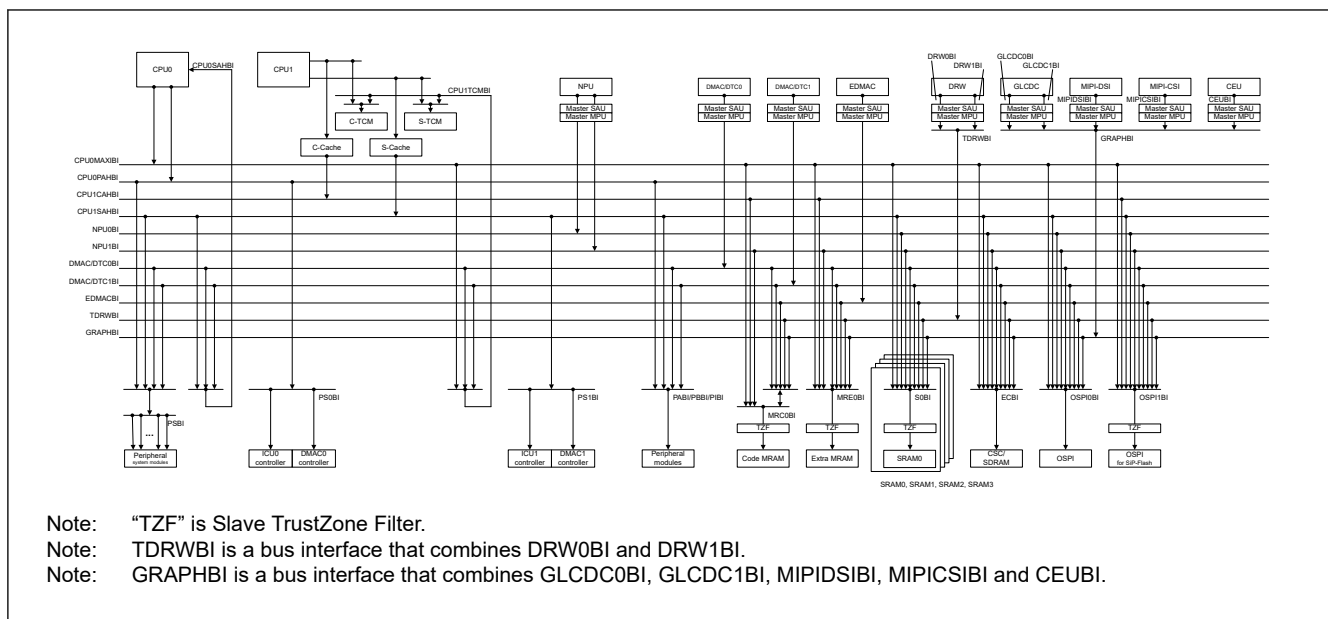


Figure 15.1 System bus connection

Table 15.2 System bus access path

	Name	Master													
		CPU0MAXBI	CPU0PAHBI	CPU1CAHBI	CPU1SAHBI	NPU0BI	NPU1BI (R-only)	DMAC/DTC0BI	DMAC/DTC1BI	EDMACBI	GLCDC0/1BI (R-only)	TDRW0/1BI	MIP4DSI	MIP4CSI (W-only)	CEUBI (W-only)
Slave	MRC0BI (Read)	T	F	T	F	F	T	T	T	T	T	T	T	F	F
	MRC0BI (Write)	T	F	T	F	F	F	T	T	F	F	F	F	F	F
	MRE0BI (Read)	T	F	T	F	F	T	T	T	T	T	T	T	F	F
	CPU0SAHBI	F	F	F	T	F	F	T	T	F	F	F	F	F	F
	CPU1TCMBI	T	F	F	F	F	F	T	T	F	F	F	F	F	F
	S0BI	T	F	F	T	T	T	T	T	T	T	T	T	T	T
	S1BI	T	F	F	T	T	T	T	T	T	T	T	T	T	T
	S2BI	T	F	F	T	T	T	T	T	T	T	T	T	T	T
	S3BI	T	F	F	T	T	T	T	T	T	T	T	T	T	T
	ECBI	T	F	F	T	T	T	T	T	T	T	T	T	T	T
	OSPI0BI	T	F	F	T	T	T	T	T	T	T	T	T	T	T
	OSPI1BI	T	F	T	T	T	T	T	T	T	T	T	T	T	T
	PBBI	F	T	F	T	F	F	T	T	F	F	F	F	F	F
	PABI	F	T	F	T	F	F	T	T	F	F	F	F	F	F
	PIBI	F	T	F	T	F	F	T	T	F	F	F	F	F	F
	PS0BI	F	T	F	F	F	F	F	F	F	F	F	F	F	F
	PS1BI	F	F	F	T	F	F	F	F	F	F	F	F	F	F
	PSBI	F	T	F	T	F	F	T	T	F	F	F	F	F	F

T: The master can access slaves

F: The master cannot access slaves

Table 15.3 Addresses assigned for each slave (1 of 2)

Region	Address	Bus Interface	Area
Code	0x0200_0000 to 0x0207_FFFF (Non-secure callable or Secure) 0x1200_0000 to 0x1207_FFFF (Non-secure)	MRC0BI	Code MRAM
	0x0208_0000 to 0x020F_FFFF (Non-secure callable or Secure) 0x1208_0000 to 0x120F_FFFF (Non-secure)	MRC0BI*2	Code MRAM
	0x02A0_0000 to 0x02CF_FFFF (Non-secure callable or Secure) 0x02E0_0000 to 0x02FF_FFFF (Non-secure callable or Secure) 0x12A0_0000 to 0x12CF_FFFF (Non-secure) 0x12E0_0000 to 0x12FF_FFFF (Non-secure)	MRE0BI	Extra MRAM
	0x0800_0000 to 0x0FFF_FFFF (Non-secure callable or Secure) 0x1800_0000 to 0x1FFF_FFFF (Non-secure)	OSPI1BI	OSPI for SIP-Flash
Data	0x2200_0000 to 0x2207_FFFF (Non-secure callable or Secure) 0x221A_0000 to 0x221A_FFFF (Non-secure callable or Secure) 0x3200_0000 to 0x3207_FFFF (Non-secure) 0x321A_0000 to 0x321A_FFFF (Non-secure)	S0BI	SRAM0
	0x2208_0000 to 0x220F_FFFF (Non-secure callable or Secure) 0x221B_0000 to 0x221B_FFFF (Non-secure callable or Secure) 0x3208_0000 to 0x320F_FFFF (Non-secure) 0x321B_0000 to 0x321B_FFFF (Non-secure)	S1BI	SRAM1
	0x2210_0000 to 0x2217_FFFF (Non-secure callable or Secure) 0x221C_0000 to 0x221C_FFFF (Non-secure callable or Secure) 0x3210_0000 to 0x3217_FFFF (Non-secure) 0x321C_0000 to 0x321C_FFFF (Non-secure)	S2BI	SRAM2
	0x2218_0000 to 0x2219_FFFF (Non-secure callable or Secure) 0x221D_0000 to 0x221D_3FFF (Non-secure callable or Secure) 0x3218_0000 to 0x3219_FFFF (Non-secure) 0x321D_0000 to 0x321D_3FFF (Non-secure)	S3BI	SRAM3
	0x2800_0000 to 0x2803_FFFF (Non-secure callable or Secure) 0x3800_0000 to 0x3803_FFFF (Non-secure)	CPU0SAHBI	CPU0 ITCM/DTCM
	0x2A00_0000 to 0x2A01_FFFF (Non-secure callable or Secure) 0x3A00_0000 to 0x3A01_FFFF (Non-secure)	CPU1TCMBI	CPU1 CTCM/STCM

Table 15.3 Addresses assigned for each slave (2 of 2)

Region	Address	Bus Interface	Area
Peripheral	0x4000_0000 to 0x4000_1FFF (Secure) 0x5000_0000 to 0x5000_1FFF (Non-secure)	PSBI	MPU controller
	0x4000_2000 to 0x4000_2FFF (Secure) 0x5000_2000 to 0x5000_2FFF (Non-secure)		SRAM controller
	0x4000_3000 to 0x4000_3FFF (Secure) 0x5000_3000 to 0x5000_3FFF (Non-secure)		CSC/SDRAM controller
	0x4000_4000 to 0x4000_4FFF (Secure) 0x5000_4000 to 0x5000_4FFF (Non-secure)		BUS controller
	0x4000_6000 to 0x4000_6FFF (Secure) 0x5000_6000 to 0x5000_6FFF (Non-secure)		Common ICU controller
	0x4000_8000 to 0x4000_8FFF (Secure) 0x5000_8000 to 0x5000_8FFF (Non-secure)		Security attribution controller
	0x4000_A000 to 0x4000_AFFF (Secure) 0x5000_A000 to 0x5000_AFFF (Non-secure) 0x4000_C000 to 0x4000_CFFF (Secure) 0x5000_C000 to 0x5000_CFFF (Non-secure)	PS0BI PS1BI	DMAC/DTC controller ICU controller
	0x4000_F000 to 0x4000_FFFF (Secure) 0x5000_F000 to 0x5000_FFFF (Non-secure)	PSBI	CPU controller
	0x4001_0000 to 0x4001_AFFF (Secure) 0x5001_0000 to 0x5001_AFFF (Non-secure)		Debug component
	0x4001_B000 to 0x4001_BFFF (Secure) 0x5001_B000 to 0x5001_BFFF (Non-secure)		Debug controller
	0x4001_C000 to 0x4001_CFFF (Secure) 0x5001_C000 to 0x5001_CFFF (Non-secure)		CPU1 cache controller
	0x4001_E000 to 0x4001_EFFF (Secure) 0x5001_E000 to 0x5001_EFFF (Non-secure)		System controller
	0x4002_0000 to 0x4002_0FFF (Secure) 0x5002_0000 to 0x5002_0FFF (Non-secure)		IPC controller
	0x4012_0000 to 0x4013_FFFF (Secure) 0x5012_0000 to 0x5013_FFFF (Non-secure)		MRAM controller
	0x4014_0000 to 0x4014_0FFF (Secure) 0x5014_0000 to 0x5014_0FFF (Non-secure)		NPU controller
	0x4020_0000 to 0x402F_FFFF (Secure) 0x5020_0000 to 0x502F_FFFF (Non-secure)	PBBI	Peripheral synchronizes with PCLKB
	0x4030_0000 to 0x403F_FFFF (Secure) 0x5030_0000 to 0x503F_FFFF (Non-secure)	PABI	Peripheral synchronizes with PCLKA
	0x4040_0000 to 0x404F_FFFF (Secure) 0x5040_0000 to 0x504F_FFFF (Non-secure)	PIBI	Peripheral synchronizes with ICLK
External RAM	0x6000_0000 to 0x6FFF_FFFF	ECBI	CSC/SDRAM (external bus)
	0x7000_0000 to 0x7FFF_FFFF	OSPI1BI*1	OSPI (via DOTF)
	0x8000_0000 to 0x9FFF_FFFF	OSPI0BI	OSPI (via DOTF)

Note 1. Products without SIP-FLASH can be accessed, but products with SIP-FLASH cannot.

Note 2. MRAM size 1 MB product is accessible, but MRAM size 512 KB product is not.

15.2 Description of Buses

15.2.1 Arbitration

For arbitration between masters in each slave, fixed-priority and round-robin methods can be selected. For details, see [section 15.3.23. BUSMABTGRAPHBI : Bus Master Arbitration Control Register](#), [section 15.3.24. BUSSABT0<slave> : Bus Slave Arbitration Control Register 0](#), and [section 15.3.25. BUSSABT1MRC0BI : Bus Slave Arbitration Control Register 1](#).

15.2.2 External Bus

The external bus controller arbitrates requests for bus access on the external address space from the CPU M-AXI bus, DMAC/DTC bus, EDMAC bus, GLCDC0 bus, GLCDC1 bus, DRW0 bus, DRW1 bus, MIPI bus and CEU bus. See [section 15.2.1. Arbitration](#) for the priority and arbitration method of each master to the External bus.

The bus system provides an external space for the OSPI (OSPI0BI, OSPI1BI). See [section 45, Octal Serial Peripheral Interface \(OSPI\)](#).

[Table 15.4](#) lists the external bus specifications and [Table 15.5](#) lists the I/O pins.

The external bus controller transfers between the AXI master and the external bus over the FIFO.

In External address space access, the internal bus is released by capturing the bus access request with the FIFO before the access to the external bus is complete. This allows the internal bus to start the next bus access. However, if the FIFO is full and the next access is an external address space access, it will wait for the previous external bus operation to complete.

[Figure 15.2](#) shows an example of operation when the FIFO function is in use. When this function is in use, if the next operation is peripheral module access after external device access, peripheral module access is executed in parallel without waiting for the completion of external device access.

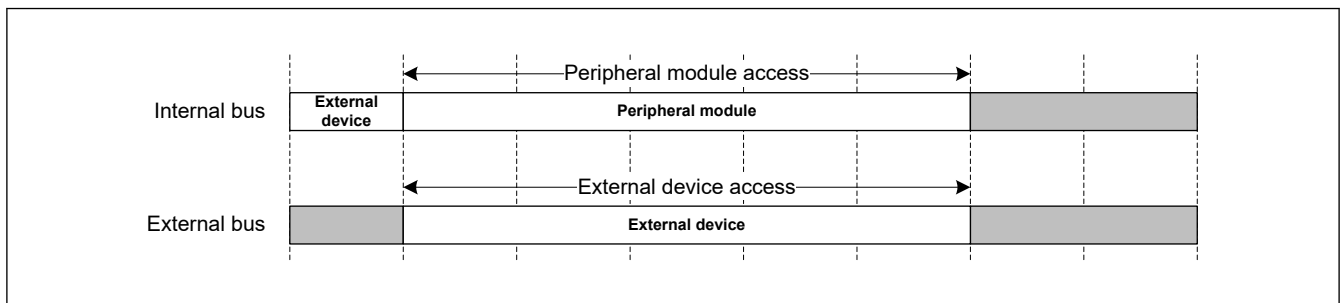


Figure 15.2 Example operation when the FIFO function is in use

Table 15.4 External bus specifications

Parameter	Specifications
External address space	- The external address space is divided into 8 CS areas (CS0 to CS7) and the SDRAM area (SDCS) for management. - Chip select signals can be output for each area. - The bus width can be set for each area. - Separate bus: Selectable to 8-bit, 16-bit or 32-bit bus space - Address/data multiplexed bus: Selectable to 8-bit or 16-bit bus space - Endian mode can be specified for each area. Big endian mode has the following restrictions. See section 15.2.5.1. Endianness Constraint for details.
CS area controller	- Recovery cycles can be inserted: - Read recovery: Up to 15 cycles - Write recovery: Up to 15 cycles - Cycle wait function: Wait for up to 31 cycles (for page access, up to 7 cycles) - Use wait control to set up: - Assertion and negation timing of chip select signals (CS0 to CS7) - Assertion timing of the read signal (RD) and write signals (WR0/WR and WR1) - Timing of data output starts and ends - Write access modes: - Single-write strobe mode and byte strobe mode Separate bus or address/data multiplexed bus can be set for each area.
SDRAM area controller	- Multiplexed output of row address and column address (8, 9, 10, or 11 bits) - Self-refresh and auto-refresh selectable - CAS latency can be specified from 1 to 3 cycles.
FIFO	FIFO number : 16 for Read Data and Write Data
Frequency	- The CS area controller (CSC) operates in synchronization with the external bus clock (BCLK)*1 - The frequency of the EBCLK pin output is the same as BCLK by default. Half of the BCLK cycles can be supplied by setting the EBCLK Pin Output Select bit, BCKCR.BCLKDIV, in the External Bus Clock Control Register. For more information, see section 9, Clock Generation Circuit. - The SDRAM area controller (SDRAMC) operates in synchronization with the SDRAM clock (SDCLK).

Note 1. BCLK and SDCLK must operate at the same frequency when the SDRAM is in use.

Table 15.5 External bus I/O pins (1 of 2)

Pin name	I/O	Related functions	Description
EBCLK, SDCLK ^{*1}	Output	CSC, SDRAMC	Clock output pin
A23 to A00 ^{*2}	Output	CSC, SDRAMC	Address output pins
D31 to D00 DQ31 to DQ00	I/O	CSC, SDRAMC	D31 to D00 are CSC data input/output pins. DQ31 to DQ00 are SDRAMC data input/output pins. - D31 to D00, DQ31 to DQ00 pins are enabled when the 32-bit bus space is specified. - D15 to D00, DQ15 to DQ00 pins are enabled when the 16-bit bus space is specified. - D07 to D00, DQ07 to DQ00 pins are enabled when the 8-bit bus space is specified.
BC0	Output	CSC	- Strobe signal that indicates (when low) that D07 to D00 are valid during access to an external address space in single-write strobe mode, active-low. - When an 8-bit bus space is specified, this output pin is always held low regardless of the write access mode.
BC1	Output	CSC	- Strobe signal that indicates (when low) that D15 to D08 are valid during access to an external address space in single-write strobe mode, active-low. - This pin is not used when the 8-bit bus space is specified.
BC2	Output	CSC	- Strobe signal that indicates (when low) that D23 to D16 are valid during access to an external address space in single-write strobe mode, active-low. - This pin is not used when the 8- or 16-bit bus space is specified.
BC3	Output	CSC	- Strobe signal that indicates (when low) that D31 to D24 are valid during access to an external address space in single-write strobe mode, active-low. - This pin is not used when the 8- or 16-bit bus space is specified.
CS0 ^{*3}	Output	CSC	Chip select signal for area 0 (CS0), active-low
CS1 ^{*3}	Output	CSC	Chip select signal for area 1 (CS1), active-low
CS2 ^{*3}	Output	CSC	Chip select signal for area 2 (CS2), active-low
CS3 ^{*3}	Output	CSC	Chip select signal for area 3 (CS3), active-low
CS4	Output	CSC	Chip select signal for area 4 (CS4), active-low
CS5	Output	CSC	Chip select signal for area 5 (CS5), active-low
CS6	Output	CSC	Chip select signal for area 6 (CS6), active-low
CS7	Output	CSC	Chip select signal for area 7 (CS7), active-low
RD	Output	CSC	Strobe signal that indicates that a read from an external address space (CS0 to CS7) is in progress, active-low.
WR0/WR ^{*4}	Output	CSC	- WR0 signal is a strobe signal that indicates that a write to an external address space is in progress in byte strobe mode, and D07 to D00 are valid, active-low. - WR signal is a strobe signal that indicates that a write to an external address space is in progress in single-write strobe mode, active-low. - When an 8-bit bus space is specified, this output pin is held low during a write access regardless of the write access mode.
WR1	Output	CSC	- Strobe signal that indicates that D15 to D08 are valid during a write to an external address space in byte strobe mode, active-low. - This signal is invalid in single-write strobe mode. - This pin is not used when the 8-bit bus space is specified.
WR2	Output	CSC	- Strobe signal that indicates that D23 to D16 are valid during a write to an external address space in byte strobe mode, active-low. - This signal is invalid in single-write strobe mode. - This pin is not used when the 8- or 16-bit bus space is specified.
WR3	Output	CSC	- Strobe signal that indicates that D31 to D24 are valid during a write to an external address space in byte strobe mode, active-low. - This signal is invalid in single-write strobe mode. - This pin is not used when the 8- or 16-bit bus space is specified.
ALE	Output	CSC	Address latch signal when address/data multiplexed bus is selected.
WAIT	Input	CSC	Wait request signal used when accessing the external address space (CS0 to CS7), active-low

Table 15.5 External bus I/O pins (2 of 2)

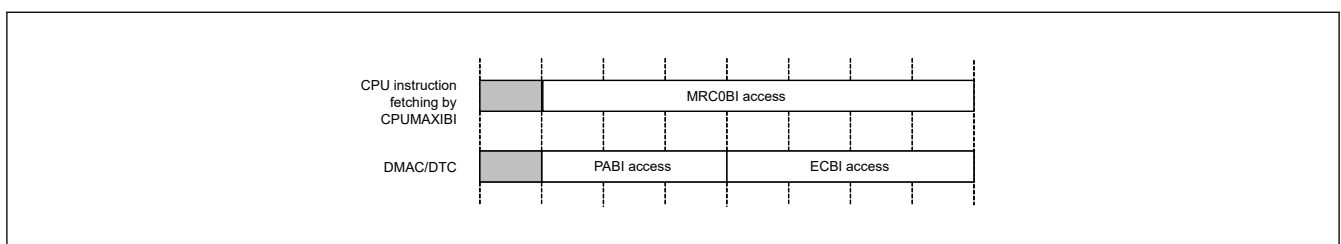
Pin name	I/O	Related functions	Description
CKE	Output	SDRAMC	Clock enable signal
SDCS	Output	SDRAMC	Chip select signal, active-low
RAS	Output	SDRAMC	Row address strobe signal, active-low
CAS	Output	SDRAMC	Column address strobe signal, active-low
WE	Output	SDRAMC	Write enable signal, active-low
DQM0	Output	SDRAMC	I/O data mask enable signal for DQ07 to DQ00
DQM1	Output	SDRAMC	I/O data mask enable signal for DQ15 to DQ08
DQM2	Output	SDRAMC	I/O data mask enable signal for DQ23 to DQ16
DQM3	Output	SDRAMC	I/O data mask enable signal for DQ31 to DQ24

- Note 1. The EBCLK and the SDCLK pin functions are shared by the CS area controller (CSC) and the SDRAM area controller (SDRAMC). When using the CSC and the SDRAMC simultaneously, the SDCLK pin function is valid.
- Note 2. The A23 to A00 pin functions are shared by the CSC and the SDRAMC.
 When using the CSC only:
 The A00 and BC0 pin functions share the same pin, and either becomes valid according to the area, with the function being A00 in byte strobe mode and BC0 in single-write strobe mode. Setting the 8-bit external bus width is prohibited in single-write strobe mode.
 When using the SDRAMC only:
 The A15 to A00 pin functions are valid.
 The A00 and DQM1 pin functions share the same pin, and either pin function becomes valid according to the external bus width.
 When selecting 8-bit bus width, the pin function is A00. When selecting 16-bit bus width, the pin function is DQM1.
 When using the CSC and the SDRAMC simultaneously:
 The A23 to A16 pin functions are valid for CSC. The A15 to A00 pin functions are shared by the CSC and the SDRAMC. In the SDRAMC functions, the A00 and the DQM1 pin functions work as described.
 In the CSC functions, the A00 and the BC0 pin function works as described.
- Note 3. The CS0 to CS3 (CSC) and SDRAMC pin functions share the same pin. When using the CSC and the SDRAMC simultaneously, the CS0 to CS3 pin functions are invalid.
- Note 4. The WR0 signal and WR signal are identical. The WR0 signal is referred to as WR in single-write strobe mode.

15.2.3 Parallel Operation

Parallel operation is possible when different bus-master modules are requesting access to different slave interface. For example, if the CPU is fetching an instruction from Code Flash and an operand from MRAM, the DMAC can handle transfers between a peripheral module and the external memory interface at the same time.

Figure 15.3 shows an example of parallel operations. In this example, the CPU uses CPUMAXIBI to access MRC0BI. Furthermore, the DMAC/DTC simultaneously accesses the peripheral bus or the external bus during access to MRC0BI by the CPU.

**Figure 15.3 Example of parallel operations**

15.2.4 Bus Settings

Set up the external bus with the following registers :

- Mode settings :
 - CSn Mode Register (CSnMOD)
 - CSn Wait Control Register 1 (CSnWCR1)
 - CSn Wait Control Register 2 (CSnWCR2)
 - CSn Control Register (CSnCR)
 - CSn Recovery Cycle Setting Register (CSnREC)

CS Recovery Cycle Insertion Enable Register (CSRECEN)

- I/O port assignments :
PmnPFS.PMR = 1 and PmnPFS.PSEL[4:0] = 0x0B
- Frequency of the external bus clock (BCLK) and SDRAM clock (SDCLK) :
SCKDIVCR register.

See [section 21, I/O Ports](#), for information on PmnPFS and [section Clock Generation Function](#) for information on SCKDIVCR.

15.2.5 Restrictions

15.2.5.1 Endianness Constraint

Memory space must be little-endian to execute code of the Arm Cortex-M85 and Arm Cortex-M33 core.

Big endian mode has the following restrictions.

- Only CPU0, CPU1, DMAC/DTC0 and DMAC/DTC1 can access to big endian area.
- The Arm Cortex-M85 core and Arm Cortex-M33 core is fixed at little-endian order, so instruction code can only be allocated to external spaces with little-endian specified. If an area is specified as big-endian, no instruction code can be allocated to it.
- No unaligned access
- Accesses with different sizes to the same address are prohibited.
- 64-bit data access is not possible in the case of areas with big-endian setting.
- The memory type must be 'device memory'. (Sparse-write is prohibited)
- For changing the memory type, see reference [1].
- CPU0 (Cortex-M85 core) cannot access to big-endian area when using M-profile Vector Extension (MVE).

15.2.5.2 Bufferable Write Access

In the following bufferable write access, the write response comes from the midpoint of the bus, not from the endpoint of the write access.

1. TDRW accesses any area.
2. CPU0 M-AXI accesses CPU1TCMBI.
3. CPU0 P-AHB accesses any area.
4. CPU1 C-AHB accesses any area.
5. CPU1 S-AHB accesses any area.
6. DMA accesses PABI, PBBI, PIBI, PSBI, CPU0SAHBI and CPU1TCMBI.

If an access violation occurs due to bufferable write access, an error response is not returned to the bus master, but an Interrupt request or Reset request is generated depending on the setting of the BUSOAD register. For more information, see [section 15.7.2. Operations When a Bus Error Occurs](#).

15.2.5.3 Notes on Writing Completion Timing

For the following accesses, a write response is returned to the bus master before the write is completed, regardless of the bufferable attribute.

When writing to MRC0BI, the write response is returned from the write buffer for MRAM, not from the MRAM itself, regardless of the bufferable/non-bufferable setting.

In the write access to ECBI, the write response comes from the CSC/SDRAMC controller when the write access starts to CS/SDRAM regardless of whether it is bufferable or non-bufferable setting.

15.3 Register Descriptions

15.3.1 BUSSARA : Bus Security Attribution Register A

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x100

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BUSS A0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BUSSA0	Bus Security Attribution A0 0: Secure 1: Non-secure	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

BUSSA0 bit (Bus Security Attribution A0)

Security attributes for arbitration control register. The target registers are as follows:

- BUSMABTGRAPHBI
- BUSSABT0<slave>
 <slave> = MRE0BI, S0BI, S1BI, S2BI, S3BI, ECBI, OSPI0BI, OSPI1BI, CPU0SAHBI, CPU1TCMBI, PBBI, PABI, PIBI, PSBI
- BUSSABT1MRC0BI

15.3.2 BUSSARB : Bus Security Attribution Register B

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x104

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BUSS B0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BUSSB0	Bus Security Attribution B0 0: Secure 1: Non-secure	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

BUSSB0 bit (Bus Security Attribution B0)

Security attributes of bus error related registers. The target registers are as follows:

- BUSERRCLR<Master Name>
- MBWERRCLR
- BUSOAD
- BUSOADPT
- BUSIRQEN<Master Name>

<Master Name>: CPU0/CPU1/DMAC0/DMAC1/NPU/EDMAC/GLCDC/TDRW/MIP10/MIP11/CEU

15.3.3 BUSSARC : Bus Security Attribution Register C

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x110

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BUSSC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BUSSC0	Bus Security Attribution C0 0: Secure 1: Non-secure	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

BUSSC0 bit (Bus Security Attribution C0)

Security attributes of registers for bus (SDRAMC/CSC) Control. The target registers are as follows:

- BUS.CSnCR (n = 0 to 7)
- BUS.CSnREC (n = 0 to 7)
- BUS.CSRECEN
- BUS.CSnMOD (n = 0 to 7)
- BUS.CSnWCR1 (n = 0 to 7)
- BUS.CSnWCR2 (n = 0 to 7)
- BUS.SDCCR
- BUS.SDCMOD
- BUS.SDAMOD
- BUS.SDSELF
- BUS.SDRFCR
- BUS.SDRFEN
- BUS.SDICR
- BUS.SDIR
- BUS.SDADR
- BUS.SDTR
- BUS.SDMOD

- BUS.SDSR
- CPSCU.BUSPARC

15.3.4 BUSPARC : Bus Privileged Attribution Register C

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x114

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BUSPA0
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	BUSPA0	External bus controller privilege attribution 0: Privileged 1: Unprivileged	R/W
31:1	—	These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-2, P-TYPE-1

BUSPA0 bit (External bus controller privilege attribution)

Privileged attributes of registers for Bus (SDRAMC/CSC) Control. The target registers are as follows:

- BUS.CSnCR (n = 0 to 7)
- BUS.CSnREC (n = 0 to 7)
- BUS.CSRECEN
- BUS.CSnMOD (n = 0 to 7)
- BUS.CSnWCR1 (n = 0 to 7)
- BUS.CSnWCR2 (n = 0 to 7)
- BUS.SDCCR
- BUS.SDCMOD
- BUS.SDAMOD
- BUS.SDSELF
- BUS.SDRFCR
- BUS.SDRFEN
- BUS.SDICR
- BUS.SDIR
- BUS.SDADR
- BUS.SDTR
- BUS.SDMOD
- BUS.SDSR

15.3.5 CSnCR : CSn Control Register (n = 0 to 7)

Base address: BUS = 0x4000_3000
 BUS_NS = 0x5000_3000

Offset address: 0x802 + 0x10 × n (n = 0 to 7)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	MPXEN	—	—	—	EMODE	—	—	BSIZE[1:0]	—	—	—	—	EXENB
Value after reset:	0	0	0	0	0	0	0	0	0	0	0 ¹	0	0	0	0	0 ¹

Bit	Symbol	Function	R/W
0	EXENB	Operation Enable 0: Disable operation 1: Enable operation	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
5:4	BSIZE[1:0]	External Bus Width Select 0 0: 16-bit bus space 0 1: 32-bit bus space 1 0: 8-bit bus space Others: Setting prohibited	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
8	EMODE	Endian Mode 0: Little endian 1: Big endian	R/W
11:9	—	These bits are read as 0. The write value should be 0.	R/W
12	MPXEN	Address/Data Multiplexed I/O Interface Select 0: Separate bus interface is selected for area n. 1: Address/data multiplexed I/O interface is selected for area n.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. When n = 0, the value after reset is 1.

The BSIZE, EMODE, and MPXEN bits are not rewritable when the EXENB bit is set to 1. It is possible to rewrite at the same time as the EXENB bit.

Do not attempt to write the CSnCR register while the external bus is being accessed.

EXENB bit (Operation Enable)

The EXENB bit enables operation of the associated CS area. On MCU reset, operation is enabled (EXENB = 1) only for area 0. Operation in other areas is disabled (EXENB = 0). Attempts to access disabled areas have no effect.

When the CSC and SDRAMC are in use at the same time, EBCLK and SDCLK must operate at the same frequency.

BSIZE[1:0] bits (External Bus Width Select)

The BSIZE[1:0] bits specify the data bus width for the associated area.

When the address/data multiplexed I/O interface is set with the MPXEN bit, it is prohibited to set it in 32-bit bus space, and the operation is not guaranteed when it is set.

EMODE bit (Endian Mode)

The EMODE bit specifies the endianness for the associated area. See [section 15.2.5.1. Endianness Constraint](#) for details.

MPXEN bit (Address/Data Multiplexed I/O Interface Select)

The MPXEN bit specifies separate bus interface or address/data multiplexed I/O interface of each area.

15.3.6 CSnREC : CSn Recovery Cycle Register (n = 0 to 7)

Base address: BUS = 0x4000_3000
 BUS_NS = 0x5000_3000

Offset address: 0x80A + 0x10 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	WRCV[3:0]				—	—	—	—	RRCV[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	RRCV[3:0]	Read Recovery 0 0 0 0: No recovery cycle is inserted. 0 0 0 1: 1 recovery cycle is inserted. 0 0 1 0: 2 recovery cycles are inserted. 0 0 1 1: 3 recovery cycles are inserted. 0 1 0 0: 4 recovery cycles are inserted. 0 1 0 1: 5 recovery cycles are inserted. 0 1 1 0: 6 recovery cycles are inserted. 0 1 1 1: 7 recovery cycles are inserted. 1 0 0 0: 8 recovery cycles are inserted. 1 0 0 1: 9 recovery cycles are inserted. 1 0 1 0: 10 recovery cycles are inserted. 1 0 1 1: 11 recovery cycles are inserted. 1 1 0 0: 12 recovery cycles are inserted. 1 1 0 1: 13 recovery cycles are inserted. 1 1 1 0: 14 recovery cycles are inserted. 1 1 1 1: 15 recovery cycles are inserted.	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W
11:8	WRCV[3:0]	Write Recovery 0 0 0 0: No recovery cycle is inserted. 0 0 0 1: 1 recovery cycle is inserted. 0 0 1 0: 2 recovery cycles are inserted. 0 0 1 1: 3 recovery cycles are inserted. 0 1 0 0: 4 recovery cycles are inserted. 0 1 0 1: 5 recovery cycles are inserted. 0 1 1 0: 6 recovery cycles are inserted. 0 1 1 1: 7 recovery cycles are inserted. 1 0 0 0: 8 recovery cycles are inserted. 1 0 0 1: 9 recovery cycles are inserted. 1 0 1 0: 10 recovery cycles are inserted. 1 0 1 1: 11 recovery cycles are inserted. 1 1 0 0: 12 recovery cycles are inserted. 1 1 0 1: 13 recovery cycles are inserted. 1 1 1 0: 14 recovery cycles are inserted. 1 1 1 1: 15 recovery cycles are inserted.	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

When the CSnCR.EXENB bit is set to 1, if a write access is made, the write is ignored.

Do not attempt to write to the CSnREC register while the external bus is being accessed.

When the preceding bus access is from a separate bus, CSnREC is valid when the recovery cycle insertion is enabled in the Separate Bus Recovery Cycle Insertion Enable bit (RCVEN_i (i = 0 to 7)) in CSRECEN. When the preceding bus access is an address/data multiplexed bus access, CSnREC is valid when the recovery cycle insertion is enabled with the Multiplexed Bus Recovery Cycle Insertion Enable bit (RCVENM_j (j = 0 to 7)) in CSRECEN. For more information, see [section 15.5.4. Insertion of Recovery Cycles](#).

RRCV[3:0] bits (Read Recovery)

The RRCV[3:0] bits specify the number of recovery cycles inserted after a read access on the external bus for CSn (n = 0 to 7). When recovery cycle insertion is enabled and a value other than 0000 is set, 1 to 15 recovery cycles are inserted when:

- After a read access to the external bus, a read access is made to the external bus in the same area.
- After a read access to the external bus, a read access is made to the external bus in a different area.
- After a read access to the external bus, a write access is made to the external bus in the same area.
- After a read access to the external bus, a write access is made to the external bus in a different area.

WRCV[3:0] bits (Write Recovery)

The WRCV[3:0] bits specify the number of recovery cycles inserted after a write access on the external bus for CS_n (n = 0 to 7). When recovery cycle insertion is enabled and a value other than 0000 is set, 1 to 15 recovery cycles are inserted when:

- After a write access to the external bus, a read access is made to the external bus in the same area.
- After a write access to the external bus, a read access is made to the external bus in a different area.
- After a write access to the external bus, a write access is made to the external bus in the same area.
- After a write access to the external bus, a write access is made to the external bus in a different area.

15.3.7 CSRECEN : CS Recovery Cycle Insertion Enable Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0x880

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RCVE NM7	RCVE NM6	RCVE NM5	RCVE NM4	RCVE NM3	RCVE NM2	RCVE NM1	RCVE NM0	RCVE N7	RCVE N6	RCVE N5	RCVE N4	RCVE N3	RCVE N2	RCVE N1	RCVE N0
Value after reset:	0	0	1	1	1	1	1	0	0	0	1	1	1	1	1	0

Bit	Symbol	Function	R/W
0	RCVEN0	Separate Bus Recovery Cycle Insertion Enable 0 0: Disabled 1: Enabled	R/W
1	RCVEN1	Separate Bus Recovery Cycle Insertion Enable 1 0: Disabled 1: Enabled	R/W
2	RCVEN2	Separate Bus Recovery Cycle Insertion Enable 2 0: Disabled 1: Enabled	R/W
3	RCVEN3	Separate Bus Recovery Cycle Insertion Enable 3 0: Disabled 1: Enabled	R/W
4	RCVEN4	Separate Bus Recovery Cycle Insertion Enable 4 0: Disabled 1: Enabled	R/W
5	RCVEN5	Separate Bus Recovery Cycle Insertion Enable 5 0: Disabled 1: Enabled	R/W
6	RCVEN6	Separate Bus Recovery Cycle Insertion Enable 6 0: Disabled 1: Enabled	R/W
7	RCVEN7	Separate Bus Recovery Cycle Insertion Enable 7 0: Disabled 1: Enabled	R/W
8	RCVENM0	Multiplexed Bus Recovery Cycle Insertion Enable 0 0: Disabled 1: Enabled	R/W

Bit	Symbol	Function	R/W
9	RCVENM1	Multiplexed Bus Recovery Cycle Insertion Enable 1 0: Disabled 1: Enabled	R/W
10	RCVENM2	Multiplexed Bus Recovery Cycle Insertion Enable 2 0: Disabled 1: Enabled	R/W
11	RCVENM3	Multiplexed Bus Recovery Cycle Insertion Enable 3 0: Disabled 1: Enabled	R/W
12	RCVENM4	Multiplexed Bus Recovery Cycle Insertion Enable 4 0: Disabled 1: Enabled	R/W
13	RCVENM5	Multiplexed Bus Recovery Cycle Insertion Enable 5 0: Disabled 1: Enabled	R/W
14	RCVENM6	Multiplexed Bus Recovery Cycle Insertion Enable 6 0: Disabled 1: Enabled	R/W
15	RCVENM7	Multiplexed Bus Recovery Cycle Insertion Enable 7 0: Disabled 1: Enabled	R/W

Note: S-TYPE-3, P-TYPE-3

Writes are ignored if the CSnCR.EXENB bit is set to 1. Write CSRECEN only when all CSnCR.EXENB bits are "0".

Do not attempt to write the CSRECEN register while the external bus is being accessed. For more information on insertion recovery cycles, see [section 15.5.4. Insertion of Recovery Cycles](#).

RCVENi Bit (Separate Bus Recovery Cycle Insertion Enable i) (i = 0 to 7)

This bit enables the insertion of read or write recovery cycles when, after a read or write access on the external bus, a read or write access is made on the external bus to the same or different area.

RCVENMj Bit (Multiplexed Bus Recovery Cycle Insertion Enable j) (j = 0 to 7)

This bit enables the insertion of read or write recovery cycles when, after a read or write access on the external bus, a read or write access is made on the external bus to the same or different area.

Table 15.6 Access type associations with the RCVENn bits

Access type	External Address Space	Insertion of recovery cycles	Associated bits (Separate/Multiplexed)
Read access after read access	Same area	Recovery cycles specified in the RRCV[3:0] bits are inserted for the priority access area	RCVEN0/RCVENM0
	Different area	Recovery cycles specified in the RRCV[3:0] bits are inserted for the priority access area	RCVEN1/RCVENM1
Write access after read access	Same area	Recovery cycles specified in the RRCV[3:0] bits are inserted for the priority access area	RCVEN2/RCVENM2
	Different area	Recovery cycles specified in the RRCV[3:0] bits are inserted for the priority access area	RCVEN3/RCVENM3
Read access after write access	Same area	Recovery cycles specified in the WRCV[3:0] bits are inserted for the priority access area	RCVEN4/RCVENM4
	Different area	Recovery cycles specified in the WRCV[3:0] bits are inserted for the priority access area	RCVEN5/RCVENM5
Write access after write access	Same area	Recovery cycles specified in the WRCV[3:0] bits are inserted for the priority access area	RCVEN6/RCVENM6
	Different area	Recovery cycles specified in the WRCV[3:0] bits are inserted for the priority access area	RCVEN7/RCVENM7

15.3.8 CSnMOD : CSn Mode Register (n = 0 to 7)

Base address: BUS = 0x4000_3000
 BUS_NS = 0x5000_3000

Offset address: 0x002 + 0x10 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PRMOD	—	—	—	—	—	PWENB	PRENB	—	—	—	—	EWENB	—	—	WRMOD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	WRMOD	Write Access Mode Select 0: Byte strobe mode 1: Single-write strobe mode	R/W
2:1	—	These bits are read as 0. The write value should be 0.	R/W
3	EWENB	External Wait Enable 0: Disabled 1: Enabled	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W
8	PRENB	Page Read Access Enable 0: Disabled 1: Enabled	R/W
9	PWENB	Page Write Access Enable 0: Disabled 1: Enabled	R/W
14:10	—	These bits are read as 0. The write value should be 0.	R/W
15	PRMOD	Page Read Access Mode Select 0: Normal access compatible mode 1: External data read continuous assertion mode	R/W

Note: S-TYPE-3, P-TYPE-3

When the CSnCR.EXENB bit is set to 1, if a write access is made, the write is ignored.

Do not write to the CSnMOD register while access to the CSn area is in progress.

WRMOD bit (Write Access Mode Select)

The WRMOD bit selects the write access operating mode. Writing 0 selects byte strobe mode, in which data writes are controlled by the WRn signals (n = 0 to 3) associated with the respective byte positions. Writing 1 selects single-write strobe mode, in which data writes are controlled by the BCn (n = 0 to 3) and WR signals associated with the respective byte positions.

Note: Setting the external bus width to 8 bits is prohibited in single-write strobe mode.

Table 15.7 Control signals for write access mode

Mode	Pin name							
Write access mode	WR3	WR2	WR1	WR0/WR	BC3	BC2	BC1	BC0
Byte strobe mode	✓	✓	✓	✓ (WR0)	—	—	—	—
Single write strobe mode	—	—	—	✓ (WR)	✓	✓	✓	✓

✓ : Enabled, — : Disabled

EWENB bit (External Wait Enable)

The EWENB bit enables external waits. Writing 0 disables the WAIT signal. Writing 1 selects external wait and allows the WAIT signal to control the number of waits per cycle. In this state, wait cycles are inserted when the WAIT signal is low.

PRENB bit (Page Read Access Enable)

The PRENB bit enables page read accesses.

Note: When the address/data multiplexed I/O interface is selected with the CSnCR.MPXEN bit, PRENB should not be set to enable page read accesses. Page read accesses are not supported in the address/data multiplexed I/O interface.

PWENB bit (Page Write Access Enable)

The PWENB bit enables page write accesses.

Note: When the address/data multiplexed I/O interface is selected with the CSnCR.MPXEN bit, PWENB should not be set to enable page write accesses. Page write accesses are not supported in the address/data multiplexed I/O interface.

PRMOD bit (Page Read Access Mode Select)

The PRMOD bit selects the operating mode for page read accesses. Writing 0 selects normal access compatible mode, in which the RD signal is negated and an RD assert wait is inserted each time a unit of data is read. When there is no RD assert wait, the RD signal is negated only in the final transfer of the external bus access.

Writing 1 selects external data read continuous assertion mode, in which an RD assert wait is inserted and the RD signal is continuously asserted during the wait.

15.3.9 CSnWCR1 : CSn Wait Control Register 1 (n = 0 to 7)

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0x004 + 0x10 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	CSRWAIT[4:0]					—	—	—	CSWWAIT[4:0]				
Value after reset:	0	0	0	0	0	1	1	1	0	0	0	0	0	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	CSPRWAIT[2:0]			—	—	—	—	—	CSPWWAIT[2:0]		
Value after reset:	0	0	0	0	0	1	1	1	0	0	0	0	0	1	1	1

Bit	Symbol	Function	R/W
2:0	CSPWWAIT[2:0]	Page Write Cycle Wait Select*1 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles is inserted. 0 1 1: Wait with a length of 3 clock cycles is inserted. 1 0 0: Wait with a length of 4 clock cycles is inserted. 1 0 1: Wait with a length of 5 clock cycles is inserted. 1 1 0: Wait with a length of 6 clock cycles is inserted. 1 1 1: Wait with a length of 7 clock cycles is inserted.	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
10:8	CSPRWAIT[2:0]	Page Read Cycle Wait Select*2 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles is inserted. 0 1 1: Wait with a length of 3 clock cycles is inserted. 1 0 0: Wait with a length of 4 clock cycles is inserted. 1 0 1: Wait with a length of 5 clock cycles is inserted. 1 1 0: Wait with a length of 6 clock cycles is inserted. 1 1 1: Wait with a length of 7 clock cycles is inserted.	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
20:16	CSWWAIT[4:0]	Normal Write Cycle Wait Select 0 0 0 0 0: No wait is inserted. 0 0 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 0 0 1 0: Wait with a length of 2 clock cycles is inserted. 0 0 0 1 1: Wait with a length of 3 clock cycles is inserted. ⋮ 1 1 1 0 1: Wait with a length of 29 clock cycles is inserted. 1 1 1 1 0: Wait with a length of 30 clock cycles is inserted. 1 1 1 1 1: Wait with a length of 31 clock cycles is inserted.	R/W
23:21	—	These bits are read as 0. The write value should be 0.	R/W
28:24	CSRWAIT[4:0]	Normal Read Cycle Wait Select 0 0 0 0 0: No wait is inserted. 0 0 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 0 0 1 0: Wait with a length of 2 clock cycles is inserted. 0 0 0 1 1: Wait with a length of 3 clock cycles is inserted. ⋮ 1 1 1 0 1: Wait with a length of 29 clock cycles is inserted. 1 1 1 1 0: Wait with a length of 30 clock cycles is inserted. 1 1 1 1 1: Wait with a length of 31 clock cycles is inserted.	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. The CSPWWAIT[2:0] value is only valid when the CSnMOD.PWENB bit is set to 1.

Note 2. The CSPRWAIT[2:0] value is only valid when the CSnMOD.PRENB bit is set to 1.

When the CSnCR.EXENB bit is set to 1, if a write access is made, the write is ignored.

Do not attempt to write to the CSnWCR1 register while the external bus is being accessed. Set each of these bits within a range of the restrictions described in [\(1\) Constraints on using separate bus interface](#) or [\(2\) Constraints on using address/data multiplexed bus interface](#), according to the bus interface used.

CSPWWAIT[2:0] bits (Page Write Cycle Wait Select)

The CSPWWAIT[2:0] bits specify the number of wait cycles to be inserted into the second and subsequent accesses during a page write cycle. The setting is enabled when the CSnMOD.PWENB bit is set to 1.

Note: The settings must satisfy $1 \leq \text{CSnWCR2.WDON}[2:0] \text{ value} \leq \text{CSnWCR2.WRON}[2:0] \text{ value} \leq \text{CSnWCR1.CSPWWAIT}[2:0] \text{ value}$, and $\text{CSnWCR2.CSON}[2:0] \text{ value} \leq \text{CSnWCR2.WRON}[2:0] \text{ value} \leq \text{CSnWCR1.CSPWWAIT}[2:0] \text{ value}$.

CSPRWAIT[2:0] bits (Page Read Cycle Wait Select)

The CSPRWAIT[2:0] bits specify the number of wait cycles to be inserted into the second and subsequent accesses during a page read cycle. The setting is enabled when the CSnMOD.PRENB bit is set to 1.

Note: The settings must satisfy $\text{CSnWCR2.CSON}[2:0] \text{ value} \leq \text{CSnWCR2.RDON}[2:0] \text{ value} \leq \text{CSnWCR1.CSPRWAIT}[2:0] \text{ value}$.

CSWWAIT[4:0] bits (Normal Write Cycle Wait Select)

The CSWWAIT[4:0] bits specify the number of wait cycles to be inserted into the first access during a normal write cycle or page write cycle.

Note: The settings must satisfy $1 \leq \text{CSnWCR2.WDON}[2:0] \text{ value} \leq \text{CSnWCR2.WRON}[2:0] \text{ value} \leq \text{CSnWCR1.CSWWAIT}[4:0] \text{ value}$, and $\text{CSnWCR2.CSON}[2:0] \text{ value} \leq \text{CSnWCR2.WRON}[2:0] \text{ value} \leq \text{CSnWCR1.CSWWAIT}[4:0] \text{ value}$.

CSRWAIT[4:0] bits (Normal Read Cycle Wait Select)

The CSRWAIT[4:0] bits specify the number of wait cycles to be inserted into the first access during a normal read cycle or page read cycle.

Note: The settings must satisfy $\text{CSnWCR2.CSON}[2:0] \text{ value} \leq \text{CSnWCR2.RDON}[2:0] \text{ value} \leq \text{CSnWCR1.CSRWAIT}[4:0] \text{ value}$.

15.3.10 CSnWCR2 : CSn Wait Control Register 2 (n = 0 to 7)

Base address: BUS = 0x4000_3000
 BUS_NS = 0x5000_3000

Offset address: 0x008 + 0x10 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	CS0N[2:0]			—	WD0N[2:0]			—	WR0N[2:0]			—	RD0N[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	AWAIT[1:0]		—	WD0FF[2:0]			—	CSW0FF[2:0]			—	CSR0FF[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1

Bit	Symbol	Function	R/W
2:0	CSR0FF[2:0]	Read Access CS Extension Cycle Select 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles is inserted. 0 1 1: Wait with a length of 3 clock cycles is inserted. 1 0 0: Wait with a length of 4 clock cycles is inserted. 1 0 1: Wait with a length of 5 clock cycles is inserted. 1 1 0: Wait with a length of 6 clock cycles is inserted. 1 1 1: Wait with a length of 7 clock cycles is inserted.	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
6:4	CSW0FF[2:0]	Write Access CS Extension Cycle Select 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles is inserted. 0 1 1: Wait with a length of 3 clock cycles is inserted. 1 0 0: Wait with a length of 4 clock cycles is inserted. 1 0 1: Wait with a length of 5 clock cycles is inserted. 1 1 0: Wait with a length of 6 clock cycles is inserted. 1 1 1: Wait with a length of 7 clock cycles is inserted.	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
10:8	WD0FF[2:0]	Write Data Output Extension Cycle Select 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles is inserted. 0 1 1: Wait with a length of 3 clock cycles is inserted. 1 0 0: Wait with a length of 4 clock cycles is inserted. 1 0 1: Wait with a length of 5 clock cycles is inserted. 1 1 0: Wait with a length of 6 clock cycles is inserted. 1 1 1: Wait with a length of 7 clock cycles is inserted.	R/W
11	—	This bit is read as 0. The write value should be 0.	R/W
13:12	AWAIT[1:0]	Address Cycle Wait Select 0 0: No wait is inserted. 0 1: Wait with a length of 1 clock cycle is inserted. 1 0: Wait with a length of 2 clock cycles is inserted. 1 1: Wait with a length of 3 clock cycles is inserted.	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
18:16	RDON[2:0]	RD Assert Wait Select 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles is inserted. 0 1 1: Wait with a length of 3 clock cycles is inserted. 1 0 0: Wait with a length of 4 clock cycles is inserted. 1 0 1: Wait with a length of 5 clock cycles is inserted. 1 1 0: Wait with a length of 6 clock cycles is inserted. 1 1 1: Wait with a length of 7 clock cycles is inserted.	R/W
19	—	This bit is read as 0. The write value should be 0.	R/W
22:20	WRON[2:0]	WR Assert Wait Select 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles is inserted. 0 1 1: Wait with a length of 3 clock cycles is inserted. 1 0 0: Wait with a length of 4 clock cycles is inserted. 1 0 1: Wait with a length of 5 clock cycles is inserted. 1 1 0: Wait with a length of 6 clock cycles is inserted. 1 1 1: Wait with a length of 7 clock cycles is inserted.	R/W
23	—	This bit is read as 0. The write value should be 0.	R/W
26:24	WDON[2:0]	Write Data Output Wait Select 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles is inserted. 0 1 1: Wait with a length of 3 clock cycles is inserted. 1 0 0: Wait with a length of 4 clock cycles is inserted. 1 0 1: Wait with a length of 5 clock cycles is inserted. 1 1 0: Wait with a length of 6 clock cycles is inserted. 1 1 1: Wait with a length of 7 clock cycles is inserted.	R/W
27	—	This bit is read as 0. The write value should be 0.	R/W
30:28	CSON[2:0]	CS Assert Wait Select 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles is inserted. 0 1 1: Wait with a length of 3 clock cycles is inserted. 1 0 0: Wait with a length of 4 clock cycles is inserted. 1 0 1: Wait with a length of 5 clock cycles is inserted. 1 1 0: Wait with a length of 6 clock cycles is inserted. 1 1 1: Wait with a length of 7 clock cycles is inserted.	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

When the CSnCR.EXENB bit is set to 1, if a write access is made, the write is ignored.

Do not attempt to write to the CSnWCR2 register while the external bus is being accessed. Set each of these bits within a range of the restrictions described in (1) [Constraints on using separate bus interface](#) or (2) [Constraints on using address/data multiplexed bus interface](#), according to the bus interface used.

CSROFF[2:0] bits (Read Access CS Extension Cycle Select)

The CSROFF[2:0] bits specify the number of wait cycles to be inserted during the period from the end of a wait cycle (RD signal negated) until the CSn signal (n = 0 to 7) is negated in read access mode.

CSWOFF[2:0] bits (Write Access CS Extension Cycle Select)

The CSWOFF[2:0] bits specify the number of wait cycles to be inserted during the period from the end of a wait cycle (WRn signal (n = 0 to 3) negated) until the CSn signal (n = 0 to 7) is negated in write access mode.

Note: The settings must satisfy CSnWCR2.WDOFF[2:0] value ≤ CSnWCR2.CSWOFF[2:0] value.

WDOFF[2:0] bits (Write Data Output Extension Cycle Select)

The WDOFF[2:0] bits specify the number of wait cycles to be inserted during the period from the end of a wait cycle (WRn signal (n = 0 to 3) negated) until the write-data output is complete in write access mode.

Note: The settings must satisfy CSnWCR2.WDOFF[2:0] value $\leq$ CSnWCR2.CSWOFF[2:0] value.

AWAIT[1:0] bits (Address Cycle Wait Select)

AWAIT[1:0] bits specify the number of wait cycles to be inserted into an address output cycle with the address/data multiplexed I/O interface.

Note: CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.AWAIT[1:0] value.

For read access, satisfy CSnWCR2.AWAIT[1:0] value + 2 $\leq$ CSnWCR2.RDON[2:0] value $\leq$ CSnWCR1.CSRWAIT[4:0] value.

For write access, satisfy CSnWCR2.AWAIT[1:0] value + 2 $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value and CSnWCR2.AWAIT[1:0] value + 2 $\leq$ CSnWCR2.WDON[2:0] value $\leq$ CSnWCR1.CSPWAIT[2:0] value.

RDON[2:0] bits (RD Assert Wait Select)

RDON[2:0] bits specify the number of wait cycles to be inserted before the RD signal is asserted.

Note: For normal read access, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.RDON[2:0] value $\leq$ CSnWCR1.CSRWAIT[4:0] value.

For page read access, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.RDON[2:0] value $\leq$ CSnWCR1.CSPWAIT[2:0] value.

When the address/data multiplexed I/O interface is selected, satisfy CSnWCR2.AWAIT[1:0] value + 2 $\leq$ CSnWCR2.RDON[2:0] value $\leq$ CSnWCR1.CSRWAIT[4:0] value.

WRON[2:0] bits (WR Assert Wait Select)

The WRON[2:0] bits specify the number of wait cycles to be inserted before the WRn signal (n = 0 to 3) is asserted.

Note: For normal write access, satisfy 1 $\leq$ CSnWCR2.WDON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value, and CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSPWAIT[2:0] value.

For page write access, satisfy 1 $\leq$ CSnWCR2.WDON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSPWAIT[2:0] value, and CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSPWAIT[2:0] value.

When the address/data multiplexed I/O interface is selected, satisfy CSnWCR2.AWAIT[1:0] value + 2 $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value.

WDON[2:0] bits (Write Data Output Wait Select)

The WDON[2:0] bits specify the number of wait cycles to be inserted before the write data is output.

Note: For normal write access, satisfy 1 $\leq$ CSnWCR2.WDON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value.

For page write access, satisfy 1 $\leq$ CSnWCR2.WDON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSPWAIT[2:0] value.

When the address/data multiplexed I/O interface is selected, satisfy CSnWCR2.AWAIT[1:0] value + 2 $\leq$ CSnWCR2.WDON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value.

CSON[2:0] bits (CS Assert Wait Select)

The CSON[2:0] bits specify the number of wait cycles to be inserted before the CSn signal (n = 0 to 7) is asserted.

Note: For normal read access, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.RDON[2:0] value $\leq$ CSnWCR1.CSRWAIT[4:0] value.

For page read access, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.RDON[2:0] value $\leq$ CSnWCR1.CSPWAIT[2:0] value.

For normal write access, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value.

For page write access, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSPWWAIT[2:0] value.

When the address/data multiplexed I/O interface is selected, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.AWAIT[1:0] value.

15.3.11 SDCCR : SDC Control Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0xC00

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	BSIZE[1:0]	—	—	—	—	EXENB
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EXENB	Operation Enable 0: Disable 1: Enable	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
5:4	BSIZE[1:0]	SDRAM Bus Width Select 0 0: 16-bit bus space 0 1: 32-bit bus space 1 0: 8-bit bus space Others: Setting prohibited	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The BSIZE bits is not rewritable when the EXENB bit is set to 1. It is possible to rewrite at the same time as the EXENB bit.

EXENB bit (Operation Enable)

The EXENB bit enables the operation of the SDRAM address space. On reset, operation is disabled (EXENB = 0). Attempts to access disabled areas have no effect.

When CSC and SDRAMC are in use at the same time, EBCLK and SDCLK must operate at the same frequency.

BSIZE[1:0] bits (SDRAM Bus Width Select)

The BSIZE[1:0] bits specify the data bus width for the associated area.

15.3.12 SDCCMOD : SDC Mode Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0xC01

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	EMODE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EMODE	Endian Mode 0: Little endian 1: Big endian	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

When the SDCCR.EXENB bit is set to 1, if a write access is made, the write is ignored.

Writing to this register is possible only once after release from reset. The second and subsequent writes to this register are prohibited. SDRAM access operation is not guaranteed if more than one write access is attempted.

EMODE bit (Endian Mode)

The EMODE bit specifies the endianness for the SDRAM address space. See [section 15.2.5.1. Endianness Constraint](#) for details.

15.3.13 SDAMOD : SDRAM Access Mode Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0xC02

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	BE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BE	Continuous Access Enable 0: Continuous access is disabled 1: Continuous access is enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

When the SDCCR.EXENB bit is set to 1, if a write access is made, the write is ignored.

Set SDAMOD while the conditions listed in [Table 15.33](#) are satisfied. The operation is not guaranteed if this register is set while these conditions are not satisfied.

BE bit (Continuous Access Enable)

This bit enables or disables continuous access to the SDRAM access space.

15.3.14 SDSELF : SDRAM Self-Refresh Control Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0xC10

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SFEN
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SFEN	SDRAM Self-Refresh Enable 0: Disable 1: Enable	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Set the SDSELF register only when the conditions in [Table 15.33](#) are satisfied. Otherwise, operation of SDRAM I/F function is not guaranteed.

SFEN bit (SDRAM Self-Refresh Enable)

The SFEN bit controls the self-refresh operation. Setting this bit to 1 initiates an auto-refresh cycle, after which self-refresh begins. Clearing this bit to 0 ends the self-refresh, and auto-refresh resumes. When the bit is set to 1, the write value takes

effect when the self-refresh operation starts. When it is cleared to 0, if the self-refresh operation has already finished, the write value takes effect before the auto-refresh starts. If the self-refresh operation is in progress, the write value takes effect after the self-refresh is finished and before the auto-refresh is started.

15.3.15 SDRFCR : SDRAM Refresh Control Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0xC14

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	REFW[3:0]				RFC[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
11:0	RFC[11:0]	Auto-Refresh Request Interval Setting 0 0 0 0 0 0 0 0 0 0 0 0: Setting prohibited 0 0 0 0 0 0 0 0 0 0 0 1: 2 cycles 0 0 0 0 0 0 0 0 0 0 1 0: 3 cycles ⋮ 1 1 1 1 1 1 1 1 1 1 1 1: 4096 cycles	R/W
15:12	REFW[3:0]	Auto-Refresh Cycle/Self-Refresh Clearing Cycle Count Setting 0 0 0 0: 1 cycle 0 0 0 1: 2 cycles 0 0 1 0: 3 cycles 0 0 1 1: 4 cycles 0 1 0 0: 5 cycles 0 1 0 1: 6 cycles 0 1 1 0: 7 cycles 0 1 1 1: 8 cycles 1 0 0 0: 9 cycles 1 0 0 1: 10 cycles 1 0 1 0: 11 cycles 1 0 1 1: 12 cycles 1 1 0 0: 13 cycles 1 1 0 1: 14 cycles 1 1 1 0: 15 cycles 1 1 1 1: 16 cycles	R/W

Note: S-TYPE-3, P-TYPE-3

RFC[11:0] bits (Auto-Refresh Request Interval Setting)

The RFC[11:0] bits specify the auto-refresh request interval. They can be written to at any time, regardless of the state of the Auto-Refresh Operation Enable bit (RFEN) in SDRFEN. If auto-refresh is enabled, the write value takes effect after the end of the auto-refresh cycles. The refresh counter uses SDCLK.

REFW[3:0] bits (Auto-Refresh Cycle/Self-Refresh Clearing Cycle Count Setting)

The REFW[3:0] bits specify the number of auto-refresh cycles and the number of self-refresh clearing cycles. They can be written to at any time, regardless of the state of the Auto-Refresh Operation Enable bit (RFEN) in SDRFEN. If an auto-refresh cycle is in progress, the value written to the bits while auto-refresh is enabled takes effect after the cycle completes.

Note: Auto-refresh requests are not accepted while the SDRAM is being accessed. This means they must sometimes wait until the access completes for the auto-refresh interval to be extended. Set the RFC[11:0] bits to an auto-refresh request interval value that meets the specifications of the SDRAM being used. Additionally, make sure to set the auto-refresh request interval to a duration longer than the auto-refresh cycle. The auto-refresh interval cannot be automatically adjusted when the frequency is changed during operation. In this case, perform a self-refresh operation and set the auto-refresh interval to an appropriate value for the frequency again.

15.3.15.1 Auto-refresh Request Interval and RFC Set Value

The SDRAMC (SDRAM area controller) includes a 12-bit refresh counter that generates auto-refresh requests at fixed intervals. Use the following equation to calculate the set value for the RFC[11:0] bits from the auto-refresh request interval:

$$\text{RFC} = (\text{Auto-refresh request interval} / \text{SDCLK cycle}) - 1$$

Note: Auto-refresh requests are not accepted while the SDRAM is being accessed. They must wait until the access completes. However, the counter value is updated regardless of whether or not the request was accepted. If two or more auto-refresh requests are generated while the SDRAM is being accessed, the second and subsequent requests are ignored.

15.3.16 SDRFEN : SDRAM Auto-Refresh Control Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0xC16

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	RFEN
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RFEN	Auto-Refresh Operation Enable 0: Disable 1: Enable	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

RFEN bit (Auto-Refresh Operation Enable)

The RFEN bit enables auto-refresh operation. When auto-refresh is required, set the RFEN bit to 1 before SDRAM access.

If the RFEN bit is set to 0 when auto-refresh is enabled, the RFEN bit becomes 0 after the auto-refresh cycle ends, and the auto-refresh operation stops. The interval at which refresh requests are generated is determined by the value in the Auto-Refresh Request Interval Setting bits (RFC[11:0]) in the SDRAM Refresh Control Register (SDRFCR).

15.3.17 SDICR : SDRAM Initialization Sequence Control Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0xC20

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	INIRQ
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	INIRQ	Initialization Sequence Start 0: Invalid 1: Start initialization sequence	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Writing to this register is possible only once after release from reset. Operation is not guaranteed if more than one write access is attempted.

INIRQ bit (Initialization Sequence Start)

Setting the INIRQ bit to 1 starts the SDRAM initialization sequence and automatically sets the Initialization Status bit (INIST) in the SDRAM Status Register (SDSR) to 1. The INIST bit clears automatically after the initialization sequence ends. The value written to the INIRQ bit is not retained.

Initialize the SDRAM according to the procedure in [section 15.6.11.1. SDRAMC Access Procedure](#).

Note: Set the INIRQ bit to start the SDRAM initialization sequence only when the conditions in [Table 15.33](#) are satisfied. Otherwise, operation is not guaranteed.

15.3.18 SDIR : SDRAM Initialization Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0xC24

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	PRC[2:0]		ARFC[3:0]				ARFI[3:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Function	R/W
3:0	ARFI[3:0]	Initialization Auto-Refresh Interval 0 0 0 0: 3 cycles 0 0 0 1: 4 cycles 0 0 1 0: 5 cycles 0 0 1 1: 6 cycles 0 1 0 0: 7 cycles 0 1 0 1: 8 cycles 0 1 1 0: 9 cycles 0 1 1 1: 10 cycles 1 0 0 0: 11 cycles 1 0 0 1: 12 cycles 1 0 1 0: 13 cycles 1 0 1 1: 14 cycles 1 1 0 0: 15 cycles 1 1 0 1: 16 cycles 1 1 1 0: 17 cycles 1 1 1 1: 18 cycles	R/W
7:4	ARFC[3:0]	Initialization Auto-Refresh Count 0 0 0 0: Setting prohibited 0 0 0 1: 1 time 0 0 1 0: 2 times 0 0 1 1: 3 times 0 1 0 0: 4 times 0 1 0 1: 5 times 0 1 1 0: 6 times 0 1 1 1: 7 times 1 0 0 0: 8 times 1 0 0 1: 9 times 1 0 1 0: 10 times 1 0 1 1: 11 times 1 1 0 0: 12 times 1 1 0 1: 13 times 1 1 1 0: 14 times 1 1 1 1: 15 times	R/W

Bit	Symbol	Function	R/W
10:8	PRC[2:0]	Initialization Precharge Cycle Count 0 0 0: 3 cycles 0 0 1: 4 cycles 0 1 0: 5 cycles 0 1 1: 6 cycles 1 0 0: 7 cycles 1 0 1: 8 cycles 1 1 0: 9 cycles 1 1 1: 10 cycles	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Writing to this register is possible only once after release from reset. Operation is not guaranteed if more than one write access is attempted.

ARFI[3:0] bits (Initialization Auto-Refresh Interval)

The ARFI[3:0] bits specify the interval at which the auto-refresh commands are issued in the SDRAM initialization sequence.

ARFC[3:0] bits (Initialization Auto-Refresh Count)

The ARFC[3:0] bits specify the number of times auto-refresh is to be performed in the SDRAM initialization sequence.

PRC[2:0] bits (Initialization Precharge Cycle Count)

The PRC[2:0] bits specify the number of precharged cycles in the SDRAM initialization sequence.

Note: Implement settings that satisfy the specifications of the connected SDRAM before starting the initialization sequence.

15.3.19 SDADR : SDRAM Address Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0xC40

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	MXC[1:0]	
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	MXC[1:0]	Address Multiplex Select 0 0: 8-bit shift 0 1: 9-bit shift 1 0: 10-bit shift 1 1: 11-bit shift	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

When the SDCCR.EXENB bit is set to 1, if a write access is made, the write is ignored.

Set SDADR only when the conditions in [Table 15.33](#) are satisfied. Otherwise, operation is not guaranteed.

MXC[1:0] bits (Address Multiplex Select)

The MXC[1:0] bits select the size of the shift towards the lower half of the row address in row address/column address multiplexing. For details, see [Table 15.38](#).

15.3.20 SDTR : SDRAM Timing Register

Base address: BUS = 0x4000_3000
 BUS_NS = 0x5000_3000

Offset address: 0xC44

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	RAI[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	RCD[1:0]		RP[2:0]			WR	—	—	—	—	—	CL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bit	Symbol	Function	R/W
2:0	CL[2:0]	SDRAMC Column Latency 0 0 0: Setting prohibited 0 0 1: 1 cycle 0 1 0: 2 cycles 0 1 1: 3 cycles Others: Setting prohibited	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
8	WR	Write Recovery Interval 0: 1 cycle 1: 2 cycles	R/W
11:9	RP[2:0]	Row Precharge Interval 0 0 0: 1 cycle 0 0 1: 2 cycles 0 1 0: 3 cycles 0 1 1: 4 cycles 1 0 0: 5 cycles 1 0 1: 6 cycles 1 1 0: 7 cycles 1 1 1: 8 cycles	R/W
13:12	RCD[1:0]	Row Column Latency 0 0: 1 cycle 0 1: 2 cycles 1 0: 3 cycles 1 1: 4 cycles	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
18:16	RAI[2:0]	Row Active Interval 0 0 0: 1 cycle 0 0 1: 2 cycles 0 1 0: 3 cycles 0 1 1: 4 cycles 1 0 0: 5 cycles 1 0 1: 6 cycles 1 1 0: 7 cycles 1 1 1: Setting prohibited	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

When the SDCCR.EXENB bit is set to 1, if a write access is made, the write is ignored.

The SDTR register specifies the timing for read and write accesses to the SDRAM. For more information, see [section 15.6.11.3. Timing Register Settings and Access Timing](#).

Set the SDTR register only when the conditions in [Table 15.33](#) are satisfied. Otherwise, operation is not guaranteed.

Writing to this register is possible only once after release from reset. Operation is not guaranteed if more than one write access is attempted.

CL[2:0] bits (SDRAMC Column Latency)

The CL[2:0] bits specify the column latency of the SDRAM controller. This setting only affects the latency setting on the SDRAM controller side. To specify the column latency for externally connected SDRAM, use the SDRAM mode register (SDMOD).

WR bit (Write Recovery Interval)

The WR bit specifies the interval that must elapse between the SDRAM write command (WRIT) and deactivation (PRE).

RP[2:0] bits (Row Precharge Interval)

The RP[2:0] bits specify the minimum number of cycles that must elapse between the SDRAM deactivation command (PRE) and the next valid command.

RCD[1:0] bits (Row Column Latency)

The RCD[1:0] bits set the SDRAM row-column latency.

RAI[2:0] bits (Row Active Interval)

The RAI[2:0] bits specify the minimum interval that must elapse between the SDRAM row activation command (ACTV) and deactivation (PRE). The value specified in these bits must be less than or equal to the sum of the row column latency (RCD[1:0]) and column latency (CL[2:0]) settings.

15.3.21 SDMOD : SDRAM Mode Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0xC48

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MR[14:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
14:0	MR[14:0]	Mode Register Setting Writing to these bits triggers a mode register set command	R/W
15	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The SDMOD register specifies the value to be written to the SDRAM mode register. Writing to SDMOD causes a mode register set command to be issued automatically to the SDRAM. Set SDMOD only when the conditions in [Table 15.33](#) are satisfied. Otherwise, operation is not guaranteed.

Writing to this register is possible only once after release from reset. Operation is not guaranteed if more than one write access is attempted.

MR[14:0] bits (Mode Register Setting)

Writing to the MR[14:0] bits causes a mode register set command to be issued to the SDRAM, and the setting in the MR[14:0] bits is output to the lower bits of the address. For more information, see [section 15.6.9. Setting the Mode Register](#).

Write the mode register according to the procedure in [section 15.6.11.1. SDRAMC Access Procedure](#).

- Note:
1. Set a burst length of 1 for the SDRAM. Operation is not guaranteed with any other burst length setting.
 2. The SDRAM column latency must match the setting in the SDRAMC Column Latency setting (CL[2:0]) in the SDRAM Timing Register (SDTR). Operation is not guaranteed if the latency settings do not agree.
 3. Make sure the SRFST, INIST, and MRSST status bits in the SDRAM Status Register (SDSR) are all 0.

15.3.22 SDSR : SDRAM Status Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0xC50

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	SRFS T	INIST	—	—	MRSS T
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MRSST	Mode Register Setting Status 0: Mode register setting not in progress 1: Mode register setting in progress	R
2:1	—	These bits are read as 0.	R
3	INIST	Initialization Status 0: Initialization sequence not in progress 1: Initialization sequence in progress	R
4	SRFST	Self-Refresh Transition/Recovery Status 0: Transition/recovery not in progress 1: Transition/recovery in progress	R
7:5	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

MRSST bit (Mode Register Setting Status)

When set to 1, the MRSST bit indicates that SDRAM mode register setting is in progress.

INIST bit (Initialization Status)

When set to 1, the INIST bit indicates that the SDRAM initialization sequence is in progress.

SRFST bit (Self-Refresh Transition/Recovery Status)

When set to 1, the SRFST bit indicates that a transition to or recovery from a self-refresh operation is in progress for the SDRAM. The in progress interval begins when the bits in [Table 15.8](#) are written to and lasts until the associated commands are issued.

Note: Execution of a self-refresh, initialization sequence, or mode register setting can only be performed when all the status bits are 0. Do not rewrite the registers and bits in [Table 15.8](#) when any of the SRFST, INIST, or MRSST status bits is set to 1.

Table 15.8 Registers and bits requiring status bit checking

Function	Register	Bits
Self-refresh	SDSELF	SFEN
Initialization sequence	SDICR	INIRQ
Mode register setting	SDMOD	MR[14:0]

15.3.23 BUSMABTGRAPHBI : Bus Master Arbitration Control Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0x1100

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ARBS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ARBS	Arbitration Select 0: Fixed priority 1: Round-robin	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

This register specifies the arbitration method for GRAPHBI.

ARBS bit (Arbitration Select)

This bit specifies the priority between bus masters connected to GRAPHBI. For fixed priority, see [Table 15.9](#). For round-robin, see [Table 15.10](#).

Table 15.9 Bus priorities with fixed-priority arbitration (ARBS = 0)

Bus Master Arbitration Control Register	Bus interface	Priority order ^{*1}
BUSMABTGRAPHBI	GRAPH	CEU > MIPI-CSI > GLCDC0 > GLCDC1 > MIPI-DSI

Note 1. ">" represents left side has high priority.

Table 15.10 Bus priorities with round-robin arbitration (ARBS = 1)

Bus Master Arbitration Control Register	Bus interface	Priority order ^{*1}
BUSMABTGRAPHBI	GRAPH	CEU ↔ MIPI-CSI ↔ GLCDC0 ↔ GLCDC1 ↔ MIPI-DSI

Note 1. "↔" represents round-robin.

15.3.24 BUSSABT0<slave> : Bus Slave Arbitration Control Register 0

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0x1208 (BUSSABT0MRE0BI), 0x1220 (BUSSABT0S0BI), 0x1228 (BUSSABT0S1BI), 0x1230 (BUSSABT0S2BI),
0x1238 (BUSSABT0S3BI), 0x1250 (BUSSABT0ECBI), 0x1258 (BUSSABT0OSPI0BI), 0x1260 (BUSSABT0OSPI1BI),
0x1268 (BUSSABT0CPU0SAHBI), 0x1270 (BUSSABT0CPU1TCMBI), 0x1278 (BUSSABT0PBBI), 0x1280 (BUSSABT0PABI),
0x1288 (BUSSABT0PIBI), 0x1290 (BUSSABT0PSBI)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ARBS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ARBS	Arbitration Select for <slave> 0: Fixed priority 1: Round-robin	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

This register specifies the arbitration method for <slave> buses.

(<slave> = MRE0BI, S0BI, S1BI, S2BI, S3BI, ECBI, OSPI0BI, OSPI1BI, CPU0SAHBI, CPU1TCMBI, PBBI, PABI, PIBI, PSBI)

ARBS bit (Arbitration Select for <slave>)

The ARBS bit specifies the priority between bus masters connected to <slave>. For fixed priority, see [Table 15.11](#). For round-robin, see [Table 15.12](#).

Table 15.11 Bus priorities with fixed-priority arbitration (ARBS = 0)

Bus Slave Arbitration Control Register	Slave interface	Priority order*1
BUSSABT0MRE0BI	MRE0BI	GRAPH > TDRW > NPU1 > EDMAC > DMAC/DTC0 > DMAC/DTC1 > CPU0MAXI > CPU1CAHB
BUSSABT0SnBI (n = 0 to 3) BUSSABT0ECBI BUSSABT0OSPI0BI	S0BI, S1BI, S2BI, S3BI ECBI OSPI0BI	GRAPH > TDRW > NPU0 > NPU1 > EDMAC > DMAC/DTC0 > DMAC/DTC1 > CPU0MAXI > CPU1SAHB
BUSSABT0OSPI1BI	OSPI1BI	GRAPH > TDRW > NPU0 > NPU1 > EDMAC > DMAC/DTC0 > DMAC/DTC1 > CPU0MAXI > CPU1SAHB > CPU1CAHB
BUSSABT0CPU0SAHBI	CPU0SAHBI	DMAC/DTC0 (R > W) > DMAC/DTC1 (R > W) > CPU1SAHB
BUSSABT0CPU1TCMBI	CPU1TCMBI	(DMAC/DTC0 (R > W) > DMAC/DTC1 (R > W)) > CPU0MAXI
BUSSABT0PBBI BUSSABT0PABI BUSSABT0PIBI BUSSABT0PSBI	PBBI PABI PIBI PSBI	DMAC/DTC0 (R > W) > DMAC/DTC1 (R > W) > CPU0PAHB > CPU1SAHB

Note 1. ">" represents left side has high priority.

Table 15.12 Bus priorities with round-robin arbitration (ARBS = 1) (1 of 2)

Bus Slave Arbitration Control Register	Slave interface	Priority order*1
BUSSABT0MRE0BI	MRE0BI	GRAPH ↔ TDRW ↔ NPU1 ↔ EDMAC ↔ DMAC/DTC0 ↔ DMAC/DTC1 ↔ CPU0MAXI ↔ CPU1CAHB

Table 15.12 Bus priorities with round-robin arbitration (ARBS = 1) (2 of 2)

Bus Slave Arbitration Control Register	Slave interface	Priority order*1
BUSSABT0SnBI (n = 0 to 3) BUSSABT0ECBI BUSSABT0OSPI0BI	S0BI, S1BI, S2BI, S3BI ECBI OSPI0BI	GRAPH ↔ TDRW ↔ NPU0 ↔ NPU1 ↔ EDMAC ↔ DMAC/DTC0 ↔ DMAC/DTC1 ↔ CPU0MAXI ↔ CPU1SAHB
BUSSABT0OSPI1BI	OSPI1BI	GRAPH ↔ TDRW ↔ NPU0 ↔ NPU1 ↔ EDMAC ↔ DMAC/DTC0 ↔ DMAC/DTC1 ↔ CPU0MAXI ↔ CPU1SAHB ↔ CPU1CAHB
BUSSABT0CPU0SAHBI	CPU0SAHBI	DMAC/DTC0 ↔ DMAC/DTC1 ↔ CPU1SAHB
BUSSABT0CPU1TCMBI	CPU1TCMBI	(DMAC/DTC0 ↔ DMAC/DTC1) ↔ CPU0MAXI
BUSSABT0PBBI BUSSABT0PABI BUSSABT0PIBI BUSSABT0PSBI	PBBI PABI PIBI PSBI	DMAC/DTC0 ↔ DMAC/DTC1 ↔ CPU0PAHB ↔ CPU1SAHB

Note 1. “↔” represents round-robin.

15.3.25 BUSSABT1MRC0BI : Bus Slave Arbitration Control Register 1

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0x1200

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ARBS[1:0]
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	ARBS[1:0]	Arbitration Select 0 0: Fixed priority 0 1: Setting prohibited 1 0: Combination of round-robin and fixed priority 1 1: Round-robin	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

This register specifies the arbitration method for MRC0BI.

ARBS[1:0] bits (Arbitration Select)

The ARBS bits specify the priority between bus masters connected to Code MRAM. For the priority of each setting, see [Table 15.13](#), [Table 15.14](#) and [Table 15.15](#).

Table 15.13 Bus priorities with fixed-priority arbitration (ARBS[1:0] = 00b)

Bus Slave Arbitration Control Register	Slave interface	Priority order*1
BUSSABT1MRC0BI	MRC0BI	GRAPH > TDRW > EDMAC > DMAC/DTC0 > DMAC/DTC1 > NPU1 > CPU0MAXI > CPU1CAHB

Note 1. “>” represents left side has high priority.

Table 15.14 Bus priorities with combination of round-robin and fixed priority arbitration (ARBS[1:0] = 10b)

Bus Slave Arbitration Control Register	Slave interface	Priority order*1
BUSSABT1MRC0BI	MRC0BI	(GRAPH ↔ TDRW ↔ EDMAC ↔ DMAC/DTC0 ↔ DMAC/DTC1) > NPU1 > CPU0MAXI > CPU1CAHB

Note 1. “↔” represents round-robin. “>” represents left side has high priority.

Table 15.15 Bus priorities with round-robin arbitration (ARBS[1:0] = 11b)

Bus Slave Arbitration Control Register	Slave interface	Priority order*1
BUSSABT1MRC0BI	MRC0BI	(GRAPH ↔ TDRW ↔ EDMAC ↔ DMAC/DTC0 ↔ DMAC/DTC1) ↔ NPU1 ↔ CPU0MAXI ↔ CPU1CAHB

Note 1. “↔” represents round-robin.

15.3.26 BUSERRSTAT<Master Name> : BUS Error Status Register

(Master Name: CPU0, CPU1, DMAC0, DMAC1, NPU, EDMAC, GLCDC, TDRW, MIPI0, MIPI1, CEU)

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0x1A00 (BUSERRSTATCPU0), 0x1A10 (BUSERRSTATCPU1), 0x1A20 (BUSERRSTATDMAC0), 0x1A30 (BUSERRSTATDMAC1), 0x1A40 (BUSERRSTATNPU), 0x1A50 (BUSERRSTATEDMAC), 0x1A60 (BUSERRSTATGLCDC), 0x1A70 (BUSERRSTATTDRW), 0x1A80 (BUSERRSTATMIPI0), 0x1A90 (BUSERRSTATMIPI1), 0x1AA0 (BUSERRSTATCEU)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	MSERRSTAT	ILERRSTAT	MMERRSTAT	—	—	SLERRSTAT
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SLERRSTAT	Slave Bus Error Status 0: No error occurred 1: Error occurred	R
2:1	—	These bits are read as 0.	R
3	MMERRSTAT	Master MPU Error Status 0: No error occurred 1: Error occurred	R
4	ILERRSTAT	Illegal Address Access Error Status 0: No error occurred 1: Error occurred	R
5	MSERRSTAT	Master Security Attribution Unit Error Status 0: No error occurred 1: Error occurred	R
7:6	—	These bits are read as 0.	R

Note: S-TYPE-5, P-TYPE-2

Note: This register is initialized after returning from Software Standby mode while the debugger is connected.

This register is cleared by reset other than Bus error reset, Common memory reset, and CPU local memory reset.

When the bus error occurred to plural at the same time, the STAT bit becomes effective only 1 by the following order of priority. The left side is high priority.

Master Security attribution unit Error > Slave bus Error, Slave TrustZone filter Error > Master-MPU Error > Illegal Address Access Error

After one of MSERRSTAT, ILERRSTAT, MMERRSTAT, and SLERRSTAT are set once, MSERRSTAT, ILERRSTAT, MMERRSTAT, and SLERRSTAT are not renewed.

SLERRSTAT bit (Slave Bus Error Status)

When slave error or TrustZone filter error occurs by bus, BUSERRSTAT.SLERRSTAT<Master Name>.SLERRSTAT is set to 1.

There is clearing conditions.

- Set BUSERRCLR<Master Name>.SLERRCLR to 1.

Slave error is an error that occurs on a slave such as Time-out.

The SLERRSTAT bit is not set when debugger violate.

For detail of slave error that occurs by bus, see [section 15.7. Bus Error Monitoring Section](#).

MMERRSTAT bit (Master MPU Error Status)

When master mpu error occurs by bus, BUSERRSTAT<Master Name>.MMERRSTAT is set to 1.

There is clearing conditions.

- Set BUSERRCLR<Master Name>.MMERRCLR to 1.

For detail of master mpu error that occurs by bus, see [section 16, Memory Protection Unit \(MPU\)](#).

ILERRSTAT bit (Illegal Address Access Error Status)

When illegal address access error occurs by bus and write access to MRE0BI occurs, BUSERRSTAT<Master Name>.ILERRSTAT is set to 1.

There is clearing conditions.

- Set BUSERRCLR<Master Name>.ILLERRCLR to 1.

The ILERRSTAT bit is not set when debugger violate.

For detail of illegal address access error that occurs by bus, see [section 15.7. Bus Error Monitoring Section](#).

MSERRSTAT bit (Master Security Attribution Unit Error Status)

When MSAU error occurs by bus, BUSERRSTAT<Master Name>.MSERRSTAT is set to 1.

There is clearing conditions.

- Set BUSERRCLR<Master Name>.MSERRCLR to 1.

For detail of MSAU error that occurs by bus, see [section 15.7. Bus Error Monitoring Section](#).

Table 15.16 Associations between bus types and registers

n	Bus master type	Bus Error Status Register (0x1A00 + 0x10 × n)	Bus Error Clear Register (0x1A04 + 0x10 × n)	Bus Error IRQ Enable Register (0x1A08 + 0x10 × n)
0	CPU0	BUSERRSTATCPU0	BUSERRCLRCPU0	BUSIRQENCPU0
1	CPU1	BUSERRSTATCPU1	BUSERRCLRCPU1	BUSIRQENCPU1
2	DMAC/DTC0	BUSERRSTATDMAC0	BUSERRCLRDMAC0	BUSIRQENDMAC0
3	DMAC/DTC1	BUSERRSTATDMAC1	BUSERRCLRDMAC1	BUSIRQENDMAC1
4	NPU	BUSERRSTATNPU	BUSERRCLRNPU	BUSIRQENNPU
5	EDMAC	BUSERRSTATEDMAC	BUSERRCLREDMAC	BUSIRQENEDMAC
6	GLCDC	BUSERRSTATGLCDC	BUSERRCLRGLCDC	BUSIRQENGLCDC
7	TDRW	BUSERRSTATDRW	BUSERRCLRTDRW	BUSIRQENTDRW
8	MIPI-DSI	BUSERRSTATMIPI0	BUSERRCLRMPI0	BUSIRQENMIPI0
9	MIPI-CSI	BUSERRSTATMIPI1	BUSERRCLRMPI1	BUSIRQENMIPI1
10	CEU	BUSERRSTATCEU	BUSERRCLRCEU	BUSIRQENCEU

15.3.27 BUSERRCLR<Master Name> : BUS Error Clear Register

(Master Name: CPU0, CPU1, DMAC0, DMAC1, NPU, EDMAC, GLCDC, TDRW, MIPI0, MIPI1, CEU)

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0x1A04 (BUSERRCLRCPU0), 0x1A14 (BUSERRCLRCPU1), 0x1A24 (BUSERRCLRDMAC0), 0x1A34 (BUSERRCLRDMAC1),
0x1A44 (BUSERRCLRNPU), 0x1A54 (BUSERRCLRDMAC), 0x1A64 (BUSERRCLRGLCDC), 0x1A74 (BUSERRCLRTDRW),
0x1A84 (BUSERRCLRMPI0), 0x1A94 (BUSERRCLRMPI1), 0x1AA4 (BUSERRCLRCEU)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	MSER RCLR	ILERR CLR	MMER RCLR	—	—	SLER RCLR

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	SLERRCLR	Slave Bus Error Clear 0: No effect 1: Clear the BUSERRSTAT<Master Name>.SLERRSTAT flag	W
2:1	—	The write value should be 0.	W
3	MMERRCLR	Master MPU Error Clear 0: No effect 1: Clear the BUSERRSTAT<Master Name>.MMERRSTAT flag	W
4	ILERRCLR	Illegal Address Access Error Clear 0: No effect 1: Clear the BUSERRSTAT<Master Name>.ILERRSTAT flag	W
5	MSERRCLR	Master Security Attribution Unit Error Clear 0: No effect 1: Clear the BUSERRSTAT<Master Name>.MSERRSTAT flag	W
7:6	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-2

When writing 1 to BUSERRCLR<Master Name>, stop the bus access that causes an error in the corresponding bus master.

15.3.28 BUSOAD : BUS Operation After Detection Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0x1000

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	BWER ROAD	SLER ROAD	ILERR OAD

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	ILERRROAD	Illegal address access error operation after detection All bus master 0: Interrupt 1: Reset	R/W
1	SLERRROAD	Slave bus error operation after detection All bus master 0: Interrupt 1: Reset	R/W
2	BWERROAD	Bufferable write error operation after detection All bus master 0: Interrupt 1: Reset	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Note: When an error occurs in speculative access issued by CPU0 and CPU1, interrupt or a system reset request may occur depending on the setting values of the OAD bit and BUSIRQEN<Master Name>.EN. In that case, it can be avoided by using Arm-MPU. See [section 15.8. References](#)[3] for the detail.

ILERROAD bit (Illegal address access error operation after detection)

The OAD bit specify operation when the illegal address access error is detected. When OAD bit is 0, error response is returned and if BUSIRQEN<Master Name>.EN = 1, and interrupt request is generated. When OAD bit is 1, error response is returned and reset request is generated.

SLERROAD bit (Slave bus error operation after detection)

The OAD bit specify operation when the slave bus error is detected. When OAD bit is 0, error response is returned and if BUSIRQEN<Master Name>.EN = 1, and interrupt request is generated. When OAD bit is 1, error response is returned and reset request is generated.

BWERROAD bit (Bufferable write error operation after detection)

The OAD bit specify operation when the bufferable write error is detected.

When OAD bit is 0, interrupt request is generated.

When OAD bit is 1, reset request is generated.

Interrupts due to bus errors (including bufferable errors) can also be used as NMI. See the [section 14, Interrupt Controller Unit \(ICU\)](#) NMIER register specifications for details.

15.3.29 BUSOADPT : BUS Operation After Detection Protect Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0x1004

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: BUSOAD register writing is possible. 1: BUSOAD register writing is protected. Read is possible.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key code These bits enable or disable writes to the PROTECT bit.	W

Note: S-TYPE-3, P-TYPE-2

PROTECT bit (Protection of register)

The PROTECT bit enables or disables writes to the associated registers to be protected.

BUSOADPT.PROTECT controls the following registers:

- BUSOAD

When the PROTECT bit is set simultaneously, write 0xA5 to the KEY[7:0] bits using half word access.

KEY[7:0] bits (Key code)

The KEY[7:0] bits enable or disable writes to the PROTECT bit. When writing to the PROTECT bit simultaneously, write 0xA5 to the KEY[7:0] bits. When other values are written to the KEY[7:0] bits, the PROTECT bit is not updated. The KEY[7:0] bits are always read as 0x00.

15.3.30 MBWERRSTAT : Master Bufferable Write Error Status Register

Base address: BUS = 0x4000_3000
 BUS_NS = 0x5000_3000

Offset address: 0x1B00

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	MBWE RR21	MBWE RR20	MBWE RR19	MBWE RR18	MBWE RR17	MBWE RR16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	MBWE RR9	MBWE RR8	—	—	—	—	—	—	MBWE RR1	MBWE RR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MBWERR0	Master Bufferable Write Error 0: No bufferable write error in Master #0 1: Bufferable write error occurs in Master #0	R
1	MBWERR1	Master Bufferable Write Error 0: No bufferable write error in Master #1 1: Bufferable write error occurs in Master #1	R
7:2	—	These bits are read as 0.	R
8	MBWERR8	Master Bufferable Write Error 0: No bufferable write error in Master #8 1: Bufferable write error occurs in Master #8	R
9	MBWERR9	Master Bufferable Write Error 0: No bufferable write error in Master #9 1: Bufferable write error occurs in Master #9	R
15:10	—	These bits are read as 0.	R
16	MBWERR16	Master Bufferable Write Error 0: No bufferable write error in Master #16 1: Bufferable write error occurs in Master #16	R
17	MBWERR17	Master Bufferable Write Error 0: No bufferable write error in Master #17 1: Bufferable write error occurs in Master #17	R
18	MBWERR18	Master Bufferable Write Error 0: No bufferable write error in Master #18 1: Bufferable write error occurs in Master #18	R
19	MBWERR19	Master Bufferable Write Error 0: No bufferable write error in Master #19 1: Bufferable write error occurs in Master #19	R
20	MBWERR20	Master Bufferable Write Error 0: No bufferable write error in Master #20 1: Bufferable write error occurs in Master #20	R
21	MBWERR21	Master Bufferable Write Error 0: No bufferable write error in Master #21 1: Bufferable write error occurs in Master #21	R
31:22	—	These bits are read as 0.	R

Note: S-TYPE-5, P-TYPE-2

Note: This register is initialized after returning from Software Standby mode while the debugger is connected.

Note: This register is cleared by reset other than Bus error reset, Common memory reset, and CPU local memory reset.

MBWERRn bit (Master Bufferable Write Error n)

When bufferable write access error occurs, MBWERRSTAT.MBWERRn is set to 1.

There are clearing conditions.

- Set MBWERRCLR. MBWECLR_n to 1

If an error is detected on the slave side, the error flag of the master with the access path is set at the same time in addition to the flag of the master that issued the access.

Table 15.17 lists the registers associated with each bus master type.

Table 15.17 Associations between bus master types and bits

Bus master type	Master bufferable write error status bit	Master bufferable write error clear bit
CPU0	MBWERR0	MBWECLR0
CPU1	MBWERR1	MBWECLR1
DMAC/DTC0	MBWERR8	MBWECLR8
DMAC/DTC1	MBWERR9	MBWECLR9
NPU	MBWERR16	MBWECLR16
EDMAC	MBWERR17	MBWECLR17
DRW	MBWERR18	MBWECLR18
CEU	MBWERR19	MBWECLR19
MIPIDSI	MBWERR20	MBWECLR20
MIPICSI	MBWERR21	MBWECLR21

15.3.31 MBWERRCLR : Master Bufferable Write Error Clear Register

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0x1B08

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	MBWE CLR21	MBWE CLR20	MBWE CLR19	MBWE CLR18	MBWE CLR17	MBWE CLR16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	MBWE CLR9	MBWE CLR8	—	—	—	—	—	—	MBWE CLR1	MBWE CLR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MBWECLR0	Master Bufferable Write Error Clear 0: No effect 1: Clear the MBWERRSTAT.MBWERR0 flag	W
1	MBWECLR1	Master Bufferable Write Error Clear 0: No effect 1: Clear the MBWERRSTAT.MBWERR1 flag	W
7:2	—	The write value should be 0.	W
8	MBWECLR8	Master Bufferable Write Error Clear 0: No effect 1: Clear the MBWERRSTAT.MBWERR8 flag	W
9	MBWECLR9	Master Bufferable Write Error Clear 0: No effect 1: Clear the MBWERRSTAT.MBWERR9 flag	W
15:10	—	The write value should be 0.	W
16	MBWECLR16	Master Bufferable Write Error Clear 0: No effect 1: Clears the MBWERRSTAT.MBWERR16 flag	W

Bit	Symbol	Function	R/W
17	MBWECLR17	Master Bufferable Write Error Clear 0: No effect 1: Clear the MBWERRSTAT.MBWERR17 flag	W
18	MBWECLR18	Master Bufferable Write Error Clear 0: No effect 1: Clear the MBWERRSTAT.MBWERR18 flag	W
19	MBWECLR19	Master Bufferable Write Error Clear Writing 1 to this bit clears the MBWERRSTAT.MBWERR19 flag. 0: No effect 1: Clear the MBWERRSTAT.MBWERR19 flag	W
20	MBWECLR20	Master Bufferable Write Error Clear 0: No effect 1: Clear the MBWERRSTAT.MBWERR20 flag	W
21	MBWECLR21	Master Bufferable Write Error Clear 0: No effect 1: Clear the MBWERRSTAT.MBWERR21 flag	W
31:22	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-2

15.3.32 BUSIRQEN<Master Name> : BUS Error Interrupt Enable Register

(Master Name: CPU0, CPU1, DMAC0, DMAC1, NPU, EDMAC, GLCDC, TDRW, MIPI0, MIPI1, CEU)

Base address: BUS = 0x4000_3000
BUS_NS = 0x5000_3000

Offset address: 0x1A08 (BUSIRQENCPU0), 0x1A18 (BUSIRQENCPU1), 0x1A28 (BUSIRQENDMAC0), 0x1A38 (BUSIRQENDMAC1),
0x1A48 (BUSIRQENNPU), 0x1A58 (BUSIQENEDMAC), 0x1A68 (BUSIRQENGLCDC), 0x1A78 (BUSIRQENTDRW),
0x1A88 (BUSIRQENMIPI0), 0x1A98 (BUSIRQENMIPI1), 0x1AA8 (BUSIRQENCEU)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	EN
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EN	Interrupt enable after bus error detected Bus interrupt request permission setting to ICU when a bus error (except Master Bufferable Write Error) occurs. 0: Disable 1: Enable	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

When writing 1 to BUSIRQEN<Master Name>, stop the bus access that causes an error in the corresponding bus master.

EN bit (Interrupt enable after bus error detected)

To output a bus error interrupt request to the ICU due to an error, select the interrupt in the OAD setting corresponding to each error, and enable the interrupt request with this bit. The setting of this bit does not depend on the output a bus error interrupt request due to Master Bufferable Write Error.

15.4 Endian and Data Alignment

The external bus has a data alignment function to control which byte of the data bus (D31 to D24, D23 to D16, D15 to D08, D07 to D00) is used when accessing the external address space (the CS and SDRAM areas). Alignment is based on the bus specifications of the area to be accessed (8-bit, 16-bit, or 32-bit bus space), the data size, and the endian order.

15.4.1 Data Alignment Control for the CS Area

15.4.1.1 32-Bit Bus Space

When a 32-bit width is selected for a bus space by the BSIZE[1:0] bits in CSnCR, the address buses A23 to A2 are enabled to output address signals in units of 32 bits, and the address buses A1 and A0 are disabled (always output low).

When byte strobe mode is selected (the WRMOD bit = 0 in CSnMOD), the WR0 to WR3 pins are enabled. The BC0 to BC3 pins are not used.

When single-write strobe mode is selected (the WRMOD bit = 1 in CSnMOD), only the WR0 pin is enabled and always outputs the low level during write access, regardless of the data size. The WR1 to WR3 pins are invalid (always output high). The valid byte position is indicated by the BC0 to BC3 pins.

In 32-bit bus space, the valid positions of data external to the MCU and of control signals differ according to whether the endian is big or little.

Table 15.18 Data alignment (little-endian) in 32-bit bus space

Data size	Access address	Number of access	Bus cycle Unit of data Address			WR3/BC3	WR2/BC2	WR1/BC1	WR0/BC0								
						RD											
						Data bus											
						D31	D24	D23	D16	D15	D8	D7	D0				
8 bits	4n	One	First	8 bits	4n							7		0			
	4n + 1	One			4n					7		0					
	4n + 2	One	First	8 bits	4n				7		0						
	4n + 3	One			4n	7		0									
16 bits	4n	One	First	16 bits	4n					15		8	7	0			
	4n + 2	One	First	16 bits	4n	15		8	7		0						
32 bits	4n	One	First	32 bits	4n	31		24	23		16	15		8	7		0
64 bits	4n	Two	First	32 bits	4n	31		24	23		16	15		8	7		0
			Second	32 bits	4n + 4	63		56	55		48	47		40	39		32

Table 15.19 Data alignment (big-endian) in 32-bit bus space

Data size	Access address	Number of access											
					WR3/BC3	WR2/BC2		WR1/BC1		WR0/BC0			
			RD										
			Data bus										
			Bus cycle	Unit of data	Address	D31	D24	D23	D16	D15	D8	D7	D0
8 bits	4n	One	First	8 bits	4n	7	0						
	4n + 1	One	4n			7	0						
	4n + 2	One	First	8 bits	4n				7	0			
	4n + 3	One	4n									7	0
16 bits	4n	One	First	16 bits	4n	15	8	7	0				
	4n + 2	One	First	16 bits	4n					15	8	7	0
32 bits	4n	One	First	32 bits	4n	31	24	23	16	15	8	7	0

15.4.1.2 16-Bit Bus Space

When a 16-bit bus space is selected in the BSIZE[1:0] bits in CSnCR, address buses A23 to A01 are enabled to output address signals in 16-bit units, and the address bus A00 is disabled (always outputs low).

When byte strobe mode is selected (WRMOD = 0 in CSnMOD), the WR0 and WR1 pins are enabled, and the WR2 and WR3 pins are disabled (fixed high). The BC0 to BC3 pins are not used.

When single-write strobe mode is selected (WRMOD = 1 in CSnMOD), only the WR0 pin is enabled, and it always outputs low during write access, regardless of the data size. The WR1 to WR3 pins are invalid (always outputs high). The valid byte position is indicated by the BC0 and BC1 pins.

The valid positions of control signals and data external to the MCU differ depending on the endian order. See [Table 15.20](#) and [Table 15.21](#).

Page access can occur for accesses to data in 32-bit units. Page access can only occur when an access does not extend over a 32-bit boundary and causes no change in the BC0 and BC1 signals. The conditions in which page access occurs are indicated by the letter (p) in [Table 15.20](#) and [Table 15.21](#).

Table 15.20 Data alignment (little-endian) in 16-bit bus space

Data size	Accessed address	Number of accesses	Bus cycle Unit of data Address			WR1/BC1		WR0/BC0	
						RD			
						Data bus			
						D15	D08	D07	D00
8 bits	4n	One	First	8 bits	4n			7	0
	4n + 1	One	First	8 bits	4n	7		0	
	4n + 2	One	First	8 bits	4n + 2			7	0
	4n + 3	One	First	8 bits	4n + 2	7		0	
16 bits	4n	One	First	16 bits	4n	15	8	7	0
	4n+2	One	First	16 bits	4n + 2	15	8	7	0
32 bits	4n	Two	First	16 bits	4n	15	8	7	0
			Second	16 bits	4n + 2 (p)	31	24	23	16
64 bits	4n	Four	First	16 bits	4n	15	8	7	0
			Second	16 bits	4n + 2 (p)	31	24	23	16
			Third	16 bits	4n + 4	47	40	39	32
			Forth	16 bits	4n + 6 (p)	63	56	55	48

(p):Page access (only when page access is enabled in the PRENB and PWENB bits in CSnMOD)

Table 15.21 Data alignment (big-endian) in 16-bit bus space (1 of 2)

Data size	Accessed address	Number of accesses	Bus cycle Unit of data Address			WR1/BC1	WR0/BC0				
						RD					
						Data bus					
						D15	D08	D07	D00		
8 bits	4n	One	First	8 bits	4n	7	0				
	4n + 1	One	First	8 bits	4n	7			0		
	4n + 2	One	First	8 bits	4n + 2	7	0				
	4n + 3	One	First	8 bits	4n + 2	7			0		

Table 15.21 Data alignment (big-endian) in 16-bit bus space (2 of 2)

Data size	Accessed address	Number of accesses	Bus cycle	Unit of data	Address	WR1/BC1		WR0/BC0	
						RD			
						Data bus			
						D15	D08	D07	D00
16 bits	4n	One	First	16 bits	4n	15	8	7	0
	4n+2	One	First	16 bits	4n + 2	15	8	7	0
32 bits	4n	Two	First	16 bits	4n	31	24	23	16
			Second	16 bits	4n + 2	(p) 15	8	7	0

(p):Page access (only when page access is enabled in the PRENB and PWENB bits in CSnMOD)

15.4.1.3 8-Bit Bus Space

When an 8-bit bus space is selected by the BSIZE[1:0] bits in CSnCR, the address buses A23 to A0 are enabled to output address signals in byte units.

In 8-bit bus space, only the WR0# pin is valid regardless of write access mode, and always outputs the low level during write access. The WR1# to WR3# pins and the BC0# to BC3# pins are not used.

Page access can occur in access to data in 16-, 32- or 64-bit units. Specifically, page access can occur when an access does not spread over a 32-bit boundary. The conditions in which page access occurs are indicated by the letter (p) in [Table 15.22](#) and [Table 15.23](#).

In 8-bit bus space, the valid positions of data external to the MCU are D7 to D0 and WR0# is used as the control signal, regardless of the endian mode.

Table 15.22 Data alignment (little-endian) in 8-bit bus space (1 of 2)

Data size	Accessed address	Number of accesses	Bus cycle Unit of data Address			WR1/BC1		WR0/BC0	
						RD			
						Data bus			
						D15	D08	D07	D00
8 bits	4n	One	First	8 bits	4n	70			
	4n + 1	One	First	8 bits	4n + 1	70			
	4n + 2	One	First	8 bits	4n + 2	70			
	4n + 3	One	First	8 bits	4n + 3	70			

Table 15.22 Data alignment (little-endian) in 8-bit bus space (2 of 2)

Data size	Accessed address	Number of accesses	Bus cycle Unit of data Address			WR1/BC1	WR0/BC0		
						RD			
						Data bus			
						D15	D08	D07	D00
16 bits	4n	Two	First	8 bits	4n		7	0	
			Second	8 bits	4n + 1 (p)		15	8	
	4n + 2	Two	First	8 bits	4n + 2		7	0	
			Second	8 bits	4n + 3 (p)		15	8	
32 bits	4n	Four	First	8 bits	4n		7	0	
			Second	8 bits	4n + 1 (p)		15	8	
			Third	8 bits	4n + 2 (p)		23	16	
			Fourth	8 bits	4n + 3 (p)		31	24	
64 bits	4n	Eight	First	8 bits	4n		7	0	
			Second	8 bits	4n + 1 (p)		15	8	
			Third	8 bits	4n + 2 (p)		23	16	
			Fourth	8 bits	4n + 3 (p)		31	24	
			Fifth	8 bits	4n + 4		39	32	
			Sixth	8 bits	4n + 5 (p)		47	40	
			Seventh	8 bits	4n + 6 (p)		55	48	
			Eighth	8 bits	4n + 7 (p)		63	56	

(p):Page access (only when page access is enabled in the PRENB and PWENB bits in CSnMOD)

Table 15.23 Data alignment (big-endian) in 8-bit bus space

Data size	Accessed address	Number of accesses	Bus cycle	Unit of data	Address	WR1/BC1	WR0/BC0		
						RD			
						Data bus			
						D15	D08	D07	D00
8 bits	4n	One	First	8 bits	4n			7	0
	4n + 1	One	First	8 bits	4n + 1			7	0
	4n + 2	One	First	8 bits	4n + 2			7	0
	4n + 3	One	First	8 bits	4n + 3			7	0
16 bits	4n	Two	First	8 bits	4n			15	8
			Second	8 bits	4n + 1 (p)		7	0	
	4n + 2	Two	First	8 bits	4n + 2			15	8
			Second	8 bits	4n + 3 (p)		7	0	
32 bits	4n	Four	First	8 bits	4n			31	24
			Second	8 bits	4n + 1 (p)		23	16	
			Third	8 bits	4n + 2 (p)		15	8	
			Fourth	8 bits	4n + 3 (p)		7	0	

(p):Page access (only when page access is enabled in the PRENB and PWENB bits in CSnMOD)

15.4.2 Data Alignment Control for the SDRAM Area

15.4.2.1 32-Bit Bus Space

When a 32-bit width is selected for a bus space by the BSIZE[1:0] bits in SDCCR, the address buses A26 to A02 are enabled to output address signals in units of 32 bits, and the address buses A01 and A00 are disabled (always output low). The valid byte position is indicated by DQM0 to DQM3 signals.

The external data is accessed using the D31 to D24, D23 to D16, D15 to D8, and D7 to D0 pins. Data can be accessed in either 8-bit, 16-bit, or 32-bit units at a time.

In 32-bit bus space, the valid positions of data external to the chip and of SDRAM control signals (DQM0 to DQM3) differ according to whether the endian is big or little. [Table 15.24](#) and [Table 15.25](#) show data alignment control when the endian is little and big, respectively.

When the SDAMOD.BE bit is set to 1, consecutive access may occur under the following conditions.

- When access to the same row address is accessed consecutively.
- Access with a data size of 64 bits that does not cross 64-bit address boundaries.

The conditions in which consecutive access occurs are indicated by the letter “(r1)” in [Table 15.24](#) and [Table 15.25](#).

Table 15.24 Data alignment (little-endian) in 32-bit bus space

Data size	Access address	Number of access			Address	DQM3		DQM2		DQM1		DQM0					
						WE											
						Data bus											
			Bus cycle	Unit of data		D31	D24	D23	D16	D15	D8	D7	D0				
8 bits	4n	One	First	8 bits	4n (r1)									7	0		
	4n + 1	One			4n (r1)									7	0		
	4n + 2	One	First	8 bits	4n (r1)									7	0		
	4n + 3	One			4n (r1)	7								0			
16 bits	4n	One	First	16 bits	4n (r1)									15	8	7	0
	4n + 2	One	First	16 bits	4n (r1)	15	8	7						0			
32 bits	4n	One	First	32 bits	4n (r1)	31	24	23	16	15	8	7	0				
64 bits	4n	Two	First	32 bits	4n (r1)	31	24	23	16	15	8	7	0				
			Second	32 bits	4n + 4 (r1)	63	56	55	48	47	40	39	32				

Table 15.25 Data alignment (big-endian) in 32-bit bus space (1 of 2)

Data size	Access address	Number of access	Bus cycleUnit of dataAddress			DQM3		DQM2		DQM1		DQM0	
						WE							
						Data bus							
						D31	D24	D23	D16	D15	D8	D7	D0
8 bits	4n	One	First	8 bits	4n (r1)	70							
	4n + 1	One			4n (r1)			70					
	4n + 2	One	First	8 bits	4n (r1)					70			
	4n + 3	One			4n (r1)							70	

Table 15.25 Data alignment (big-endian) in 32-bit bus space (2 of 2)

Data size	Access address	Number of access	Bus cycle Unit of data Address			DQM3		DQM2		DQM1		DQM0	
						WE							
						Data bus							
						D31	D24	D23	D16	D15	D8	D7	D0
16 bits	4n	One	First	16 bits	4n (r1)	15 8		7 0					
	4n + 2	One	First	16 bits	4n (r1)					15 8		7 0	
32 bits	4n	One	First	32 bits	4n (r1)	31 24		23 16		15 8		7 0	

15.4.2.2 16-Bit Bus Space

When a 16-bit bus space is selected in the BSIZE[1:0] bits in SDCCR, address buses A26 to A01 are enabled to output address signals in 16-bit units, and the address bus A00 is disabled (always outputs low). The valid byte position is indicated by the DQM0 and DQM1 signals.

External data is accessed using the DQ15 to DQ08 and DQ07 to DQ00 pins and DQM0 and DQM1 control signals. Data can be accessed in either 8-bit or 16-bit units at a time.

When the SDAMOD.BE bit is set to 1, consecutive access may occur under the following conditions.

- When access to the same row address is accessed consecutively.
- Access with a data size of 32 bits
- Access with a data size of 64 bits that does not cross 64-bit address boundaries.
- Access with a data size of 64 bits that does not cross 32-bit address boundaries.
- The conditions in which consecutive access occurs are indicated by the letter “(r1)” in [Table 15.26](#) and [Table 15.27](#).

The valid positions of control signals and data external to the MCU differ depending on the endian order. See [Table 15.26](#) and [Table 15.27](#).

Table 15.26 Data alignment in 16-bit bus space with little-endian order for SDRAM area (1 of 2)

							DQM1		DQM0					
											WE			
											Data bus			
Data size	Accessed address	Number of accesses	Bus cycle	Unit of data	Address									
8 bits	4n	One	First	8 bits	4n	(r1)			7	0				
	4n + 1	One	First	8 bits	4n	(r1)	7	0						
	4n + 2	One	First	8 bits	4n + 2	(r1)			7	0				
	4n + 3	One	First	8 bits	4n + 2	(r1)	7	0						

Table 15.26 Data alignment in 16-bit bus space with little-endian order for SDRAM area (2 of 2)

							DQM1		DQM0					
											WE			
													Data bus	
Data size	Accessed address	Number of accesses	Bus cycle	Unit of data	Address		DQ15	DQ08	DQ07	DQ00				
16 bits	4n	One	First	16 bits	4n	(r1)	15	8	7	0				
	4n+2	One	First	16 bits	4n + 2	(r1)	15	8	7	0				
32 bits	4n	Two	First	16 bits	4n	(r1)	15	8	7	0				
			Second	16 bits	4n + 2	(r1)	31	24	23	16				
64 bits	4n	Four	First	16 bits	4n	(r1)	15	8	7	0				
			Second	16 bits	4n + 2	(r1)	31	24	23	16				
			Third	16 bits	4n + 4	(r1)	47	40	39	32				
			Fourth	16 bits	4n + 6	(r1)	63	56	55	48				

Table 15.27 Data alignment in 16-bit bus space with big-endian order for SDRAM area

Data size	Accessed address	Number of accesses	Bus cycle Unit of data Address				DQM1		DQM0	
							WE			
							Data bus			
						DQ15	DQ08	DQ07	DQ00	
8 bits	4n	One	First	8 bits	4n	(r1)	7			

15.4.2.3 8-Bit Bus Space

When an 8-bit width is selected in the BSIZE[1:0] bits in SDCCR, address buses A26 to A00 are enabled to output address signals in 8-bit units.

External data is accessed using the DQ07 to DQ00 pins and DQM0 control signal. Eight-bit data is accessed in single 8-bit access, 16-bit data with two 8-bit accesses, and 32-bit data with four 8-bit accesses.

When the SDAMOD.BE bit is set to 1, consecutive access may occur under the following conditions.

- When access to the same row address is accessed consecutively.
- Access with a data size of 16 / 32bit.
- Access with a data size of 64 bits that does not cross 64-bit address boundaries.
- Access with a data size of 64 bits that does not cross 32-bit address boundaries.

Table 15.28 and Table 15.29 show the case where continuous access occurs (r1).

The valid positions of control signals and data external to the MCU differ depending on the endian order. See Table 15.28 and Table 15.29.

Table 15.28 Data alignment in 8-bit bus space with little-endian order for SDRAM area (1 of 2)

Data size	Accessed address	Number of accesses	Bus cycle Unit of data Address				DQM1		DQM0					
							Data bus							
											WE			
											DQ15	DQ08	DQ07	DQ00
8 bits	4n	One	First	8 bits	4n	(r1)		7	0					
	4n + 1	One	First	8 bits	4n + 1	(r1)		7	0					
	4n + 2	One	First	8 bits	4n + 2	(r1)		7	0					
	4n + 3	One	First	8 bits	4n + 3	(r1)		7	0					
16 bits	4n	Two	First	8 bits	4n	(r1)		7	0					
			Second	8 bits	4n + 1	(r1)		15	8					
	4n + 2	Two	First	8 bits	4n + 2	(r1)		7	0					
			Second	8 bits	4n + 3	(r1)		15	8					
32 bits	4n	Four	First	8 bits	4n	(r1)		7	0					
			Second	8 bits	4n + 1	(r1)		15	8					
			Third	8 bits	4n + 2	(r1)		23	16					
			Fourth	8 bits	4n + 3	(r1)		31	24					

Table 15.28 Data alignment in 8-bit bus space with little-endian order for SDRAM area (2 of 2)

Data size	Accessed address	Number of accesses	Bus cycle Unit of data Address				DQM1		DQM0			
							Data bus				WE	
											DQ15	DQ08
64 bits	4n	Eight	First	8 bits	4n	(r1)			7	0		
			Second	8 bits	4n + 1	(r1)			15	8		
			Third	8 bits	4n + 2	(r1)			23	16		
			Fourth	8 bits	4n + 3	(r1)			31	24		
			Fifth	8 bits	4n + 4	(r1)			39	32		
			Sixth	8 bits	4n + 5	(r1)			47	40		
			Seventh	8 bits	4n + 6	(r1)			55	48		
			Eighth	8 bits	4n + 7	(r1)			63	56		

Table 15.29 Data alignment in 8-bit bus space with big-endian order for SDRAM area (1 of 2)

Data size	Accessed address	Number of accesses	Bus cycle Unit of data Address				DQM1		DQM0	
							WE			
				DQ15	DQ08	DQ07	DQ00			
8 bits	4n	One	First	8 bits	4n	(r1)		7	0	
	4n + 1	One	First	8 bits	4n + 1	(r1)		7	0	
	4n + 2	One	First	8 bits	4n + 2	(r1)		7	0	
	4n + 3	One	First	8 bits	4n + 3	(r1)		7	0	
16 bits	4n	Two	First	8 bits	4n	(r1)		15	8	
			Second	8 bits	4n + 1	(r1)		7	0	
	4n + 2	Two	First	8 bits	4n + 2	(r1)		15	8	
			Second	8 bits	4n + 3	(r1)		7	0	

Table 15.29 Data alignment in 8-bit bus space with big-endian order for SDRAM area (2 of 2)

Data size	Accessed address	Number of accesses	Bus cycle Unit of data Address				DQM1		DQM0	
							WE			
				DQ15	DQ08	DQ07	DQ00			
32 bits	4n	Four	First	8 bits	4n	(r1)		31	24	
			Second	8 bits	4n + 1	(r1)		23	16	
			Third	8 bits	4n + 2	(r1)		15	8	
			Fourth	8 bits	4n + 3	(r1)		7	0	

15.5 Operation of CS Area Controller

15.5.1 Separate Bus

This section describes the periods shown in the timing diagrams. The CS area controller (CSC) operates in synchronization with the external bus clock, BCLK. Operation cycles, such as wait cycles, specified in the CSC register, are counted on BCLK. In the following description, the frequencies of BCLK and EBCLK pin output are the same, unless otherwise noted. Access through the external bus starts at the same point as the output of a rising edge on the EBCLK pin. However, if the external bus clock, BCLK, and the output on the EBCLK pin are at different frequencies, the wait settings can cause the start of access for the second and subsequent rounds to coincide with the falling edge of the output on the EBCLK pin. See [Figure 15.11](#) to [Figure 15.15](#). If recovery cycles are inserted for bus access, the setting for the number of recovery cycles can also cause the start of access for the second and subsequent rounds to coincide with the falling edge of the output on the EBCLK pin. See [Figure 15.33](#).

Tw1 to Twn (Clock Cycles of Wait for a Normal Read Cycle or Normal Write Cycle)

The period from Tw1 to Twn is the number of clock cycles from the start of access through the external bus clock to 1 cycle before the strobe signal is valid. The number of cycles is selectable from 0 to 31. Within this period, the timing of CSn, RD, and WRn assertion (driving the signals low) is determined by the respective wait settings. The wait periods are controlled by the following bits in the CSn Wait Control Register 2 (CSnWCR2).

- CS Assert Wait Select bits (CSON)
- RD Assert Wait Select bits (RDON)
- WR Assert Wait Select bits (WRON)
- Write-Data Output Wait Select bits (WDON)

The number of clock cycles for each of these wait periods is selectable as a value from 0 to 7, counted from the start of external bus access. The selectable number of cycles is also within the overall number of clock cycles required for waiting to read or write.

Tend (Clock Cycle where the Strobe Signal is Valid)

Tend is the next clock cycle after completion of the wait period for a normal cycle of read or write, or for a cycle of page read or page write. If the wait select bit for these cycles is 0, bus access starts on the clock cycle where the strobe signal is valid. The RD and WRn signals are negated in the next clock cycle. For a read access, the clock cycle where the strobe signal is valid is where the data to be read is sampled. If an external wait is enabled, the wait signal is sampled on the cycle where the strobe signal is valid. The bus cycle is extended if the wait signal is low. The bus cycle completes in the next clock cycle if the wait signal is high. Tend indicates the cycle where sampling of the wait signal starts.

After the first cycle where the strobe signal is valid during page access, second and subsequent page access operations (see [Tpwl to TpwN \(Page-Read Cycle Wait or Page-Write Cycle Wait\)](#)) start in the next cycle, except during write access with a setting other than 0 for write-data output extension clock cycles (see [Tdw1 to Tdwn \(Write-Data Output Extension Clock](#)

Cycles)). If the setting for the RD or WR assertion wait is any value other than 0, the RD and WRn signals are negated in the next clock cycle. If the setting is 0, assertion continues. Additionally, the CSn signal continues to be asserted rather than negated.

Tn1 to Tnm (Clock Cycles of CS Extension)

For normal access, Tn1 to Tnm represent the clock cycles of the period following the cycle where the strobe signal is valid (Tend) up to negation of the CSn signal. For read or write access, the negation timing can be controlled by the read-access CS Extension Cycle Select bits (CSROFF) and the write-access CS Extension Cycle Select bits (CSWOFF) in the CSn Wait Control Register 2 (CSnWCR2). The number of cycles is counted from the cycle following the cycle where the strobe signal is valid.

For page access, Tn1 to Tnm represent the clock cycles of the period following the last cycle where the strobe signal is valid up to negation of the CSn signal.

For write access, setting the Write Data Output Extension Cycle Select bits (WDOFF) controls extension of the period where the address and output data is valid.

Tdw1 to Tdwn (Write-Data Output Extension Clock Cycles)

For write access, if the wait setting for the write-data output extension is any value other than 0, the specified clock cycles are inserted from the cycle following the cycle where the strobe signal is valid (Tend).

For normal access, this period is inserted within the clock cycle period for CS extension (Tn1 to Tnm).

For page access, this period is inserted within the clock cycle period where the strobe signal is valid and subsequent page accesses, or within the clock cycle period for the CS extension (Tn1 to Tnm). Valid address and data output are extended over this period, and the WRn signal is negated.

Tpw1 to Tpwn (Page-Read Cycle Wait or Page-Write Cycle Wait)

For the second and subsequent bus cycles during page access, the values for a page-read cycle wait or page-write cycle wait are used instead of the settings for a normal read or write cycle wait. The settings in the WR Assert Wait Select bits become enabled in the same way as for the first access. The RD assertion control operation depends on the page read access mode setting (the PRMOD bit in CSnMOD) as follows:

- CSnMOD.PRMOD = 0: A wait for RD assertion is inserted in the same way as for the first access, and the RD signal is negated
- CSnMOD.PRMOD = 1: Although a wait for RD assertion is inserted in the same way as for normal-access compatibility mode, the RD signal continues to be asserted over this period.

Tr1 to Trn (Recovery Cycles)

Recovery cycles can be inserted from the point where a bus cycle is complete (CSn signal negation). The number of recovery cycles can be controlled by setting the Read Recovery (RRCV) or Write Recovery (WRCV) bits in the CSn Recovery Cycle Register (CSnREC). Both numbers of recovery cycles are counted from the end of a bus cycle (CSn negation) and can be selected from 0 to 15 cycles. For more information, see [section 15.5.4. Insertion of Recovery Cycles](#).

(1) Normal Access

When the PRENB and PWENB bits in CSnMOD are set to 0 to disable page read and page write access, all bus accesses take the form of normal read and write operations. Even when these bits are set to 1 to enable page read and page write access, bus access other than page access takes the form of normal read and write operations. [Figure 15.4](#) to [Figure 15.6](#) show the normal access operations.

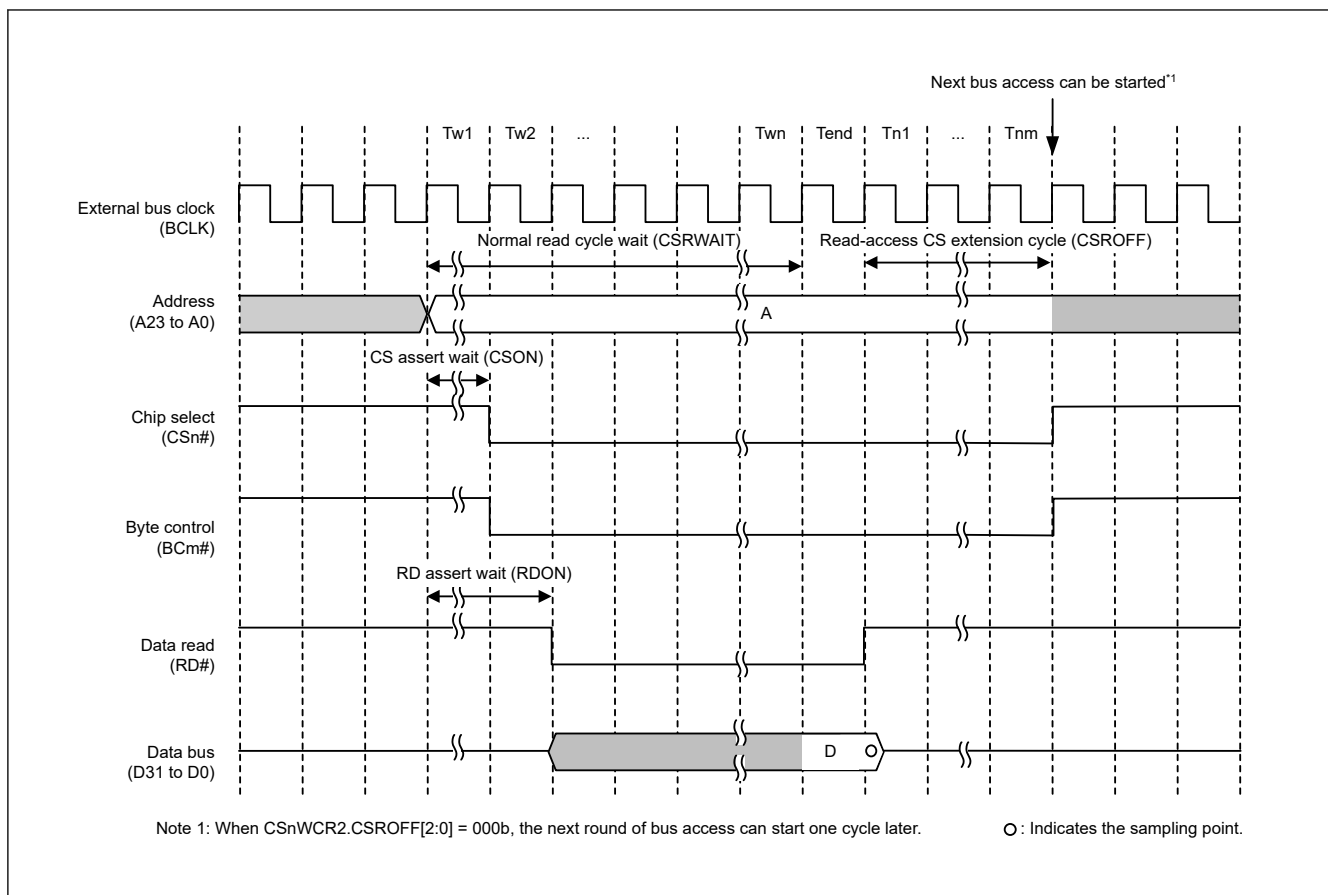


Figure 15.4 Bus timing for normal read operation (n = 0 to 7, m = 0 to 3)

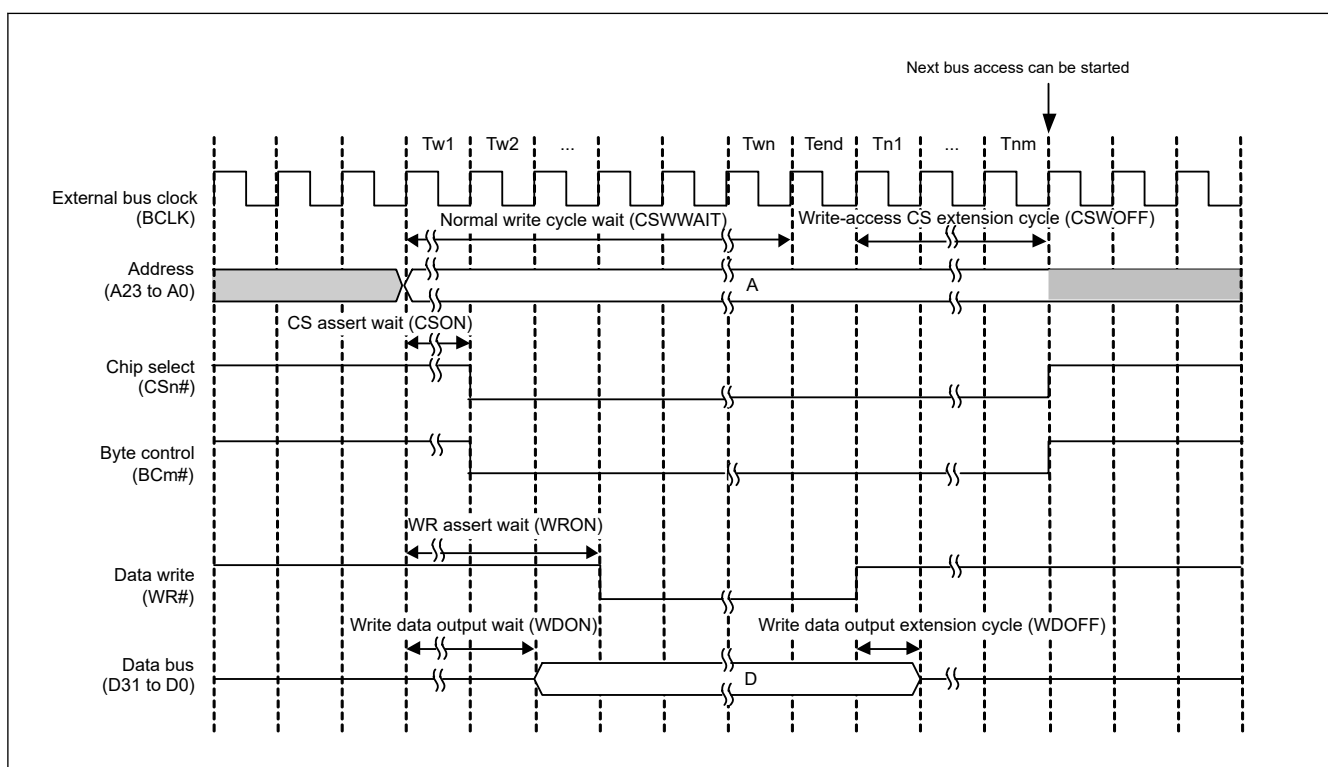


Figure 15.5 Bus timing for normal write operation in single-write strobe mode (n = 0 to 7, m = 0 to 3)

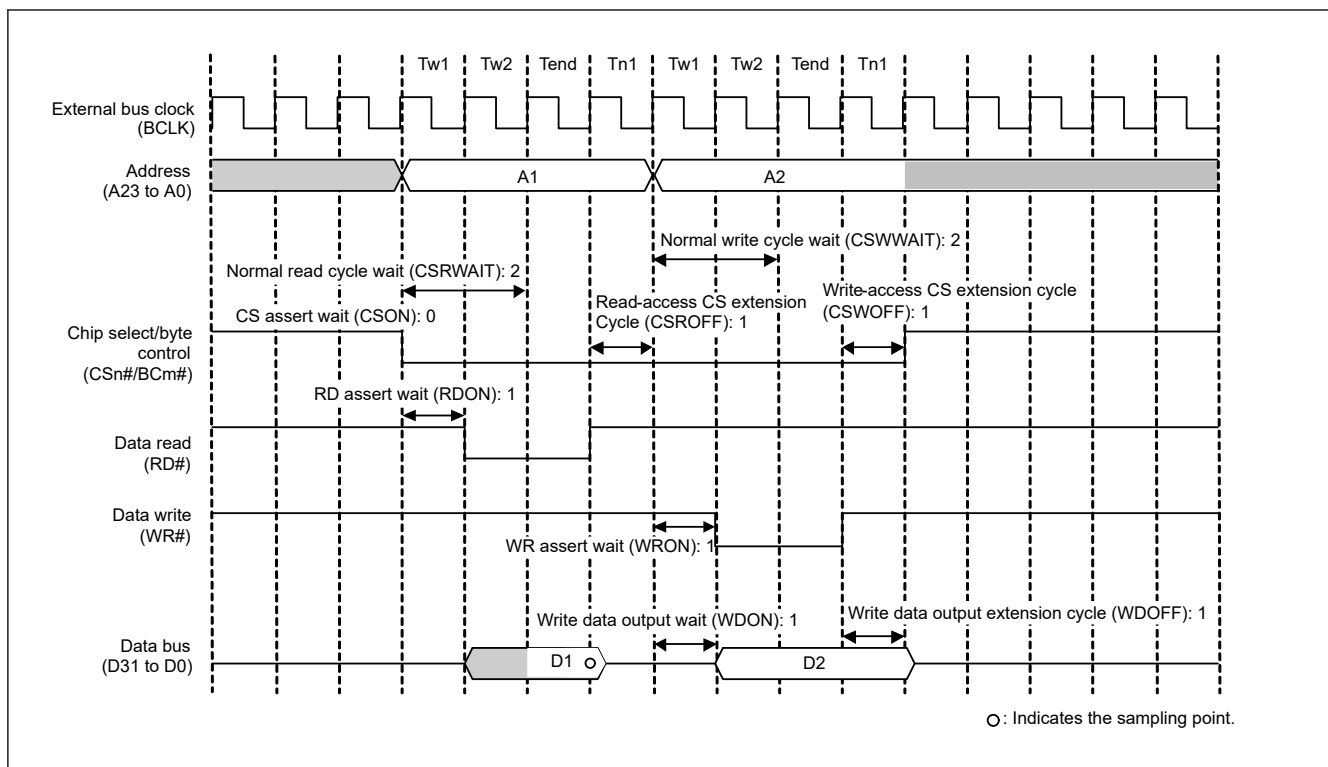


Figure 15.6 Example of normal access operation for read and write (n = 0 to 7, m = 0 to 3)

When two or more rounds of external bus access are required in response to a single request for transfer from a bus master, normal access operations are repeated. See [Tw1 to Twn \(Clock Cycles of Wait for a Normal Read Cycle or Normal Write Cycle\)](#) to [Tdw1 to Tdwn \(Write-Data Output Extension Clock Cycles\)](#). [Figure 15.7](#) and [Figure 15.8](#) show examples of operations when two rounds of bus access are generated in response to a single transfer request. If the recovery cycle insertion condition is satisfied, recovery cycles [Tr1 to Trn \(Recovery Cycles\)](#) are also inserted in the second and subsequent external bus accesses. See [Figure 15.31](#).

The values in the wait control registers shown in the figures are example settings. In your application, set the registers appropriately for the specifications of connected devices.

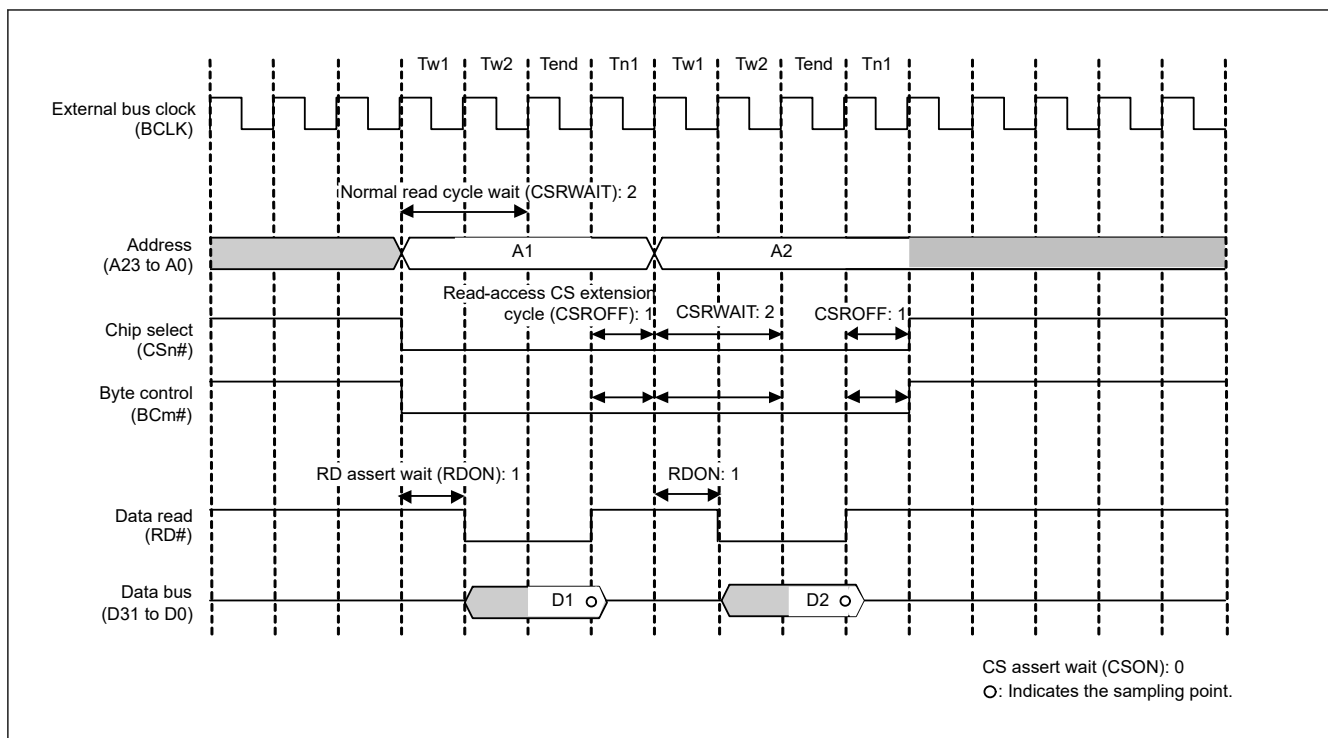


Figure 15.7 Example of normal read operation when two rounds of bus access are generated in response to a single transfer request ($n = 0$ to 7, $m = 0$ to 3)

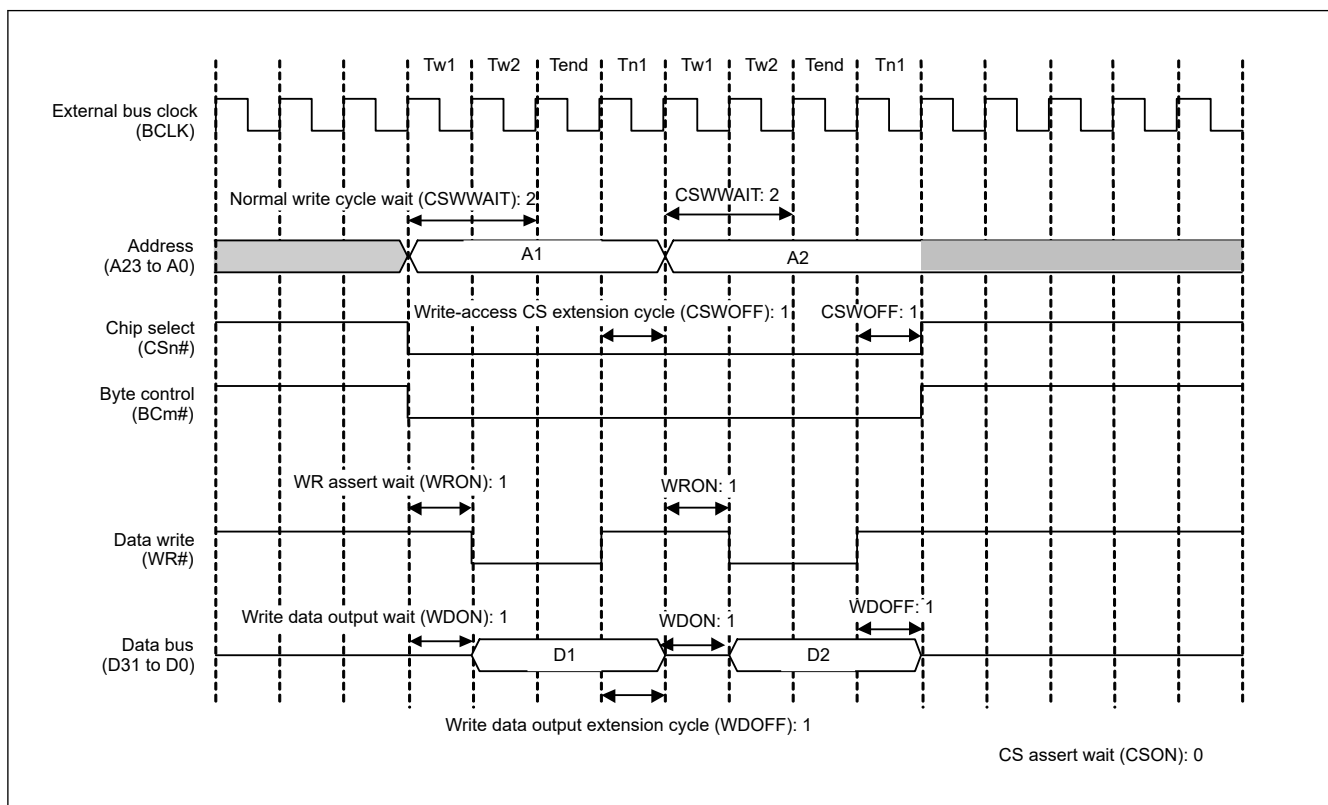


Figure 15.8 Example of normal write operation when two rounds of bus access are generated in response to a single transfer request in single-write strobe mode ($n = 0$ to 7, $m = 0$ to 3)

Figure 15.9 and Figure 15.10 show examples of normal read and write accesses to a 32-bit bus space in 16 bits.

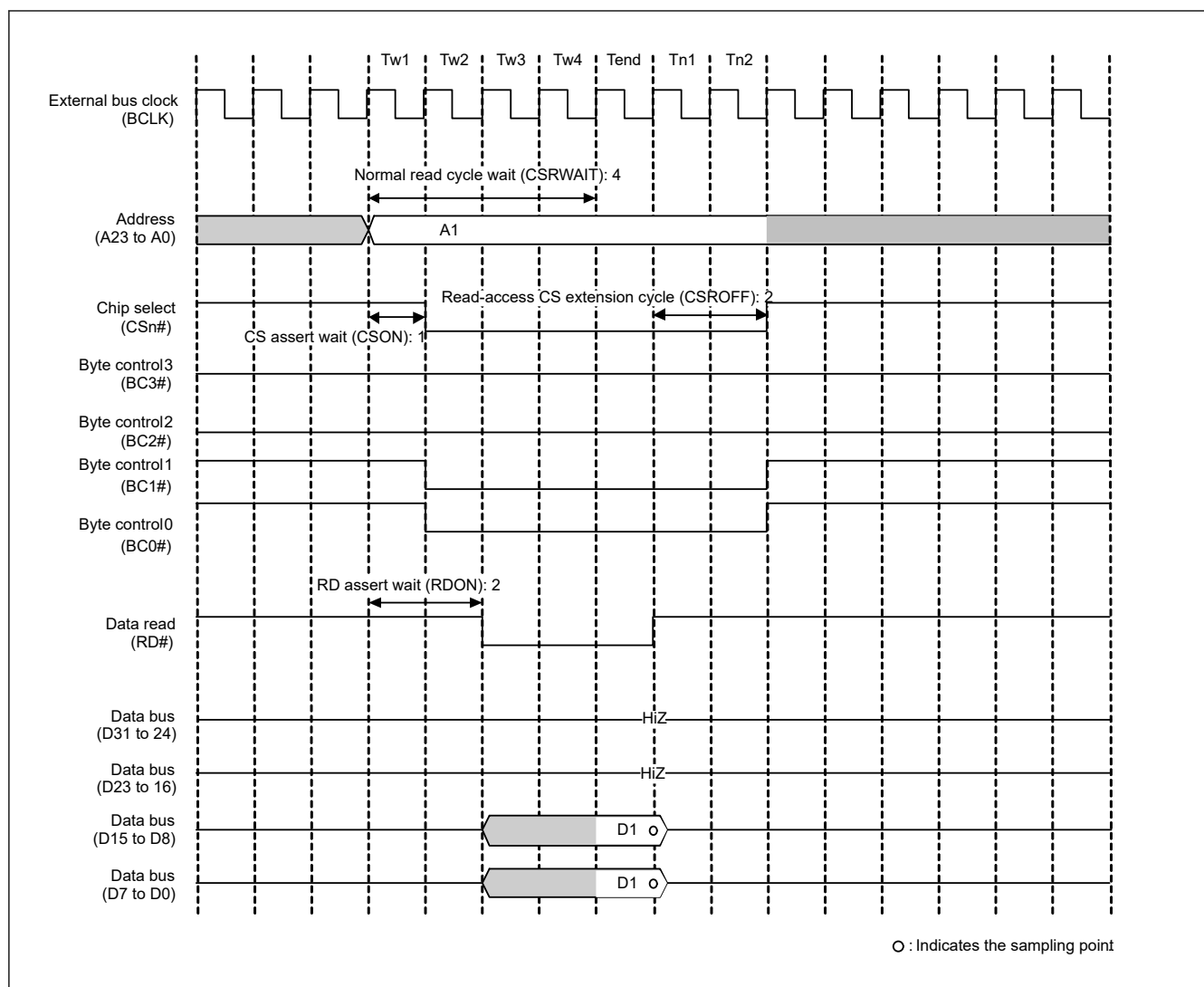


Figure 15.9 Example of normal read operation when 32-bit bus space is accessed in 16 bits (n = 0 to 7)

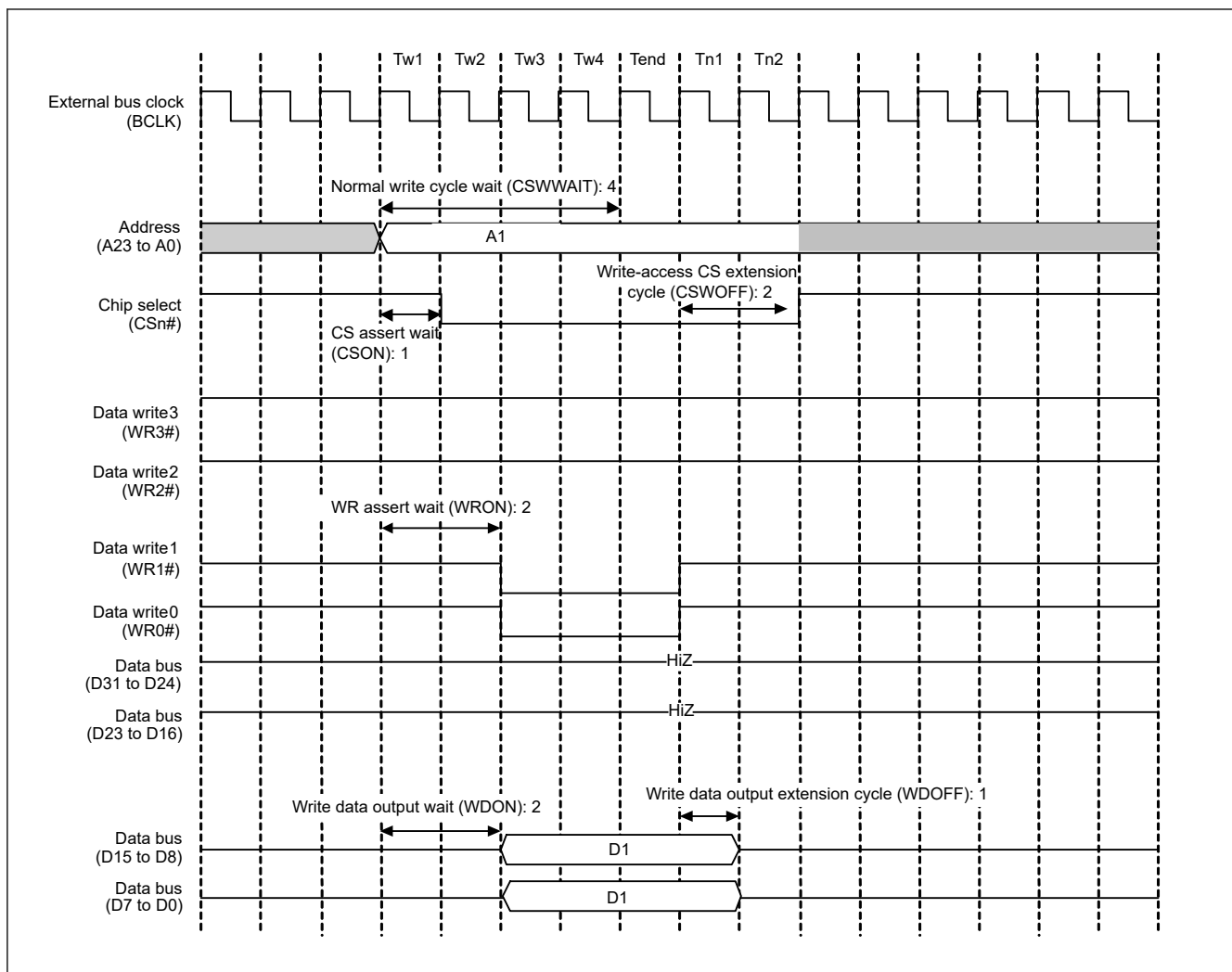


Figure 15.10 Example of normal write operation when 32-bit bus space is accessed in 16 bits, in byte strobe mode (n = 0 to 7)

Figure 15.11 to Figure 15.15 show examples of normal accesses made with the 1/2 BCLK selected with the EBCLK pin output select bit.

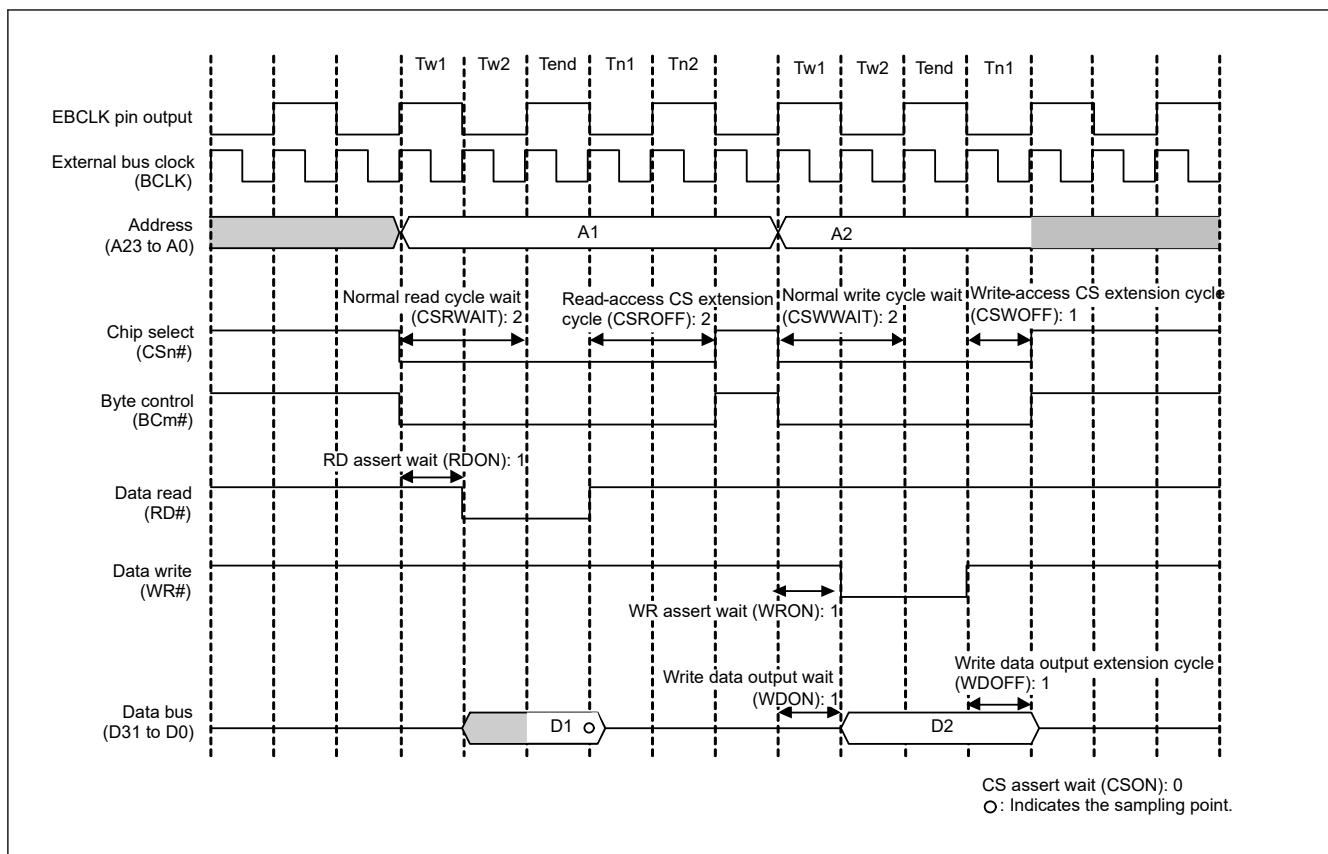


Figure 15.11 Example of normal access when BCLK/2 is selected in the EBCLK Pin Output Select bit ($n = 0$ to 7, $m = 0$ to 3)

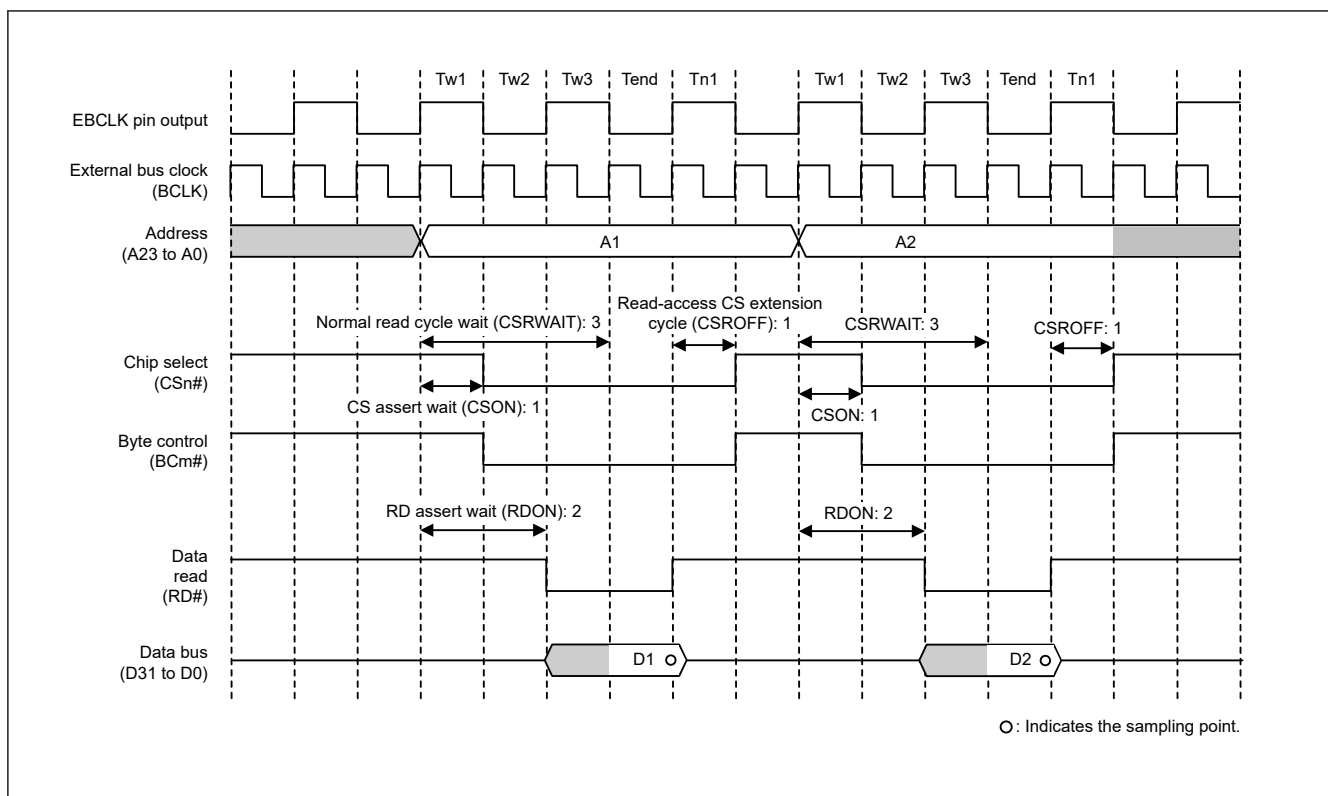


Figure 15.12 Example of normal read operation when BCLK/2 is selected in the EBCLK Pin Output Select bit ($n = 0$ to 7, $m = 0$ to 3)

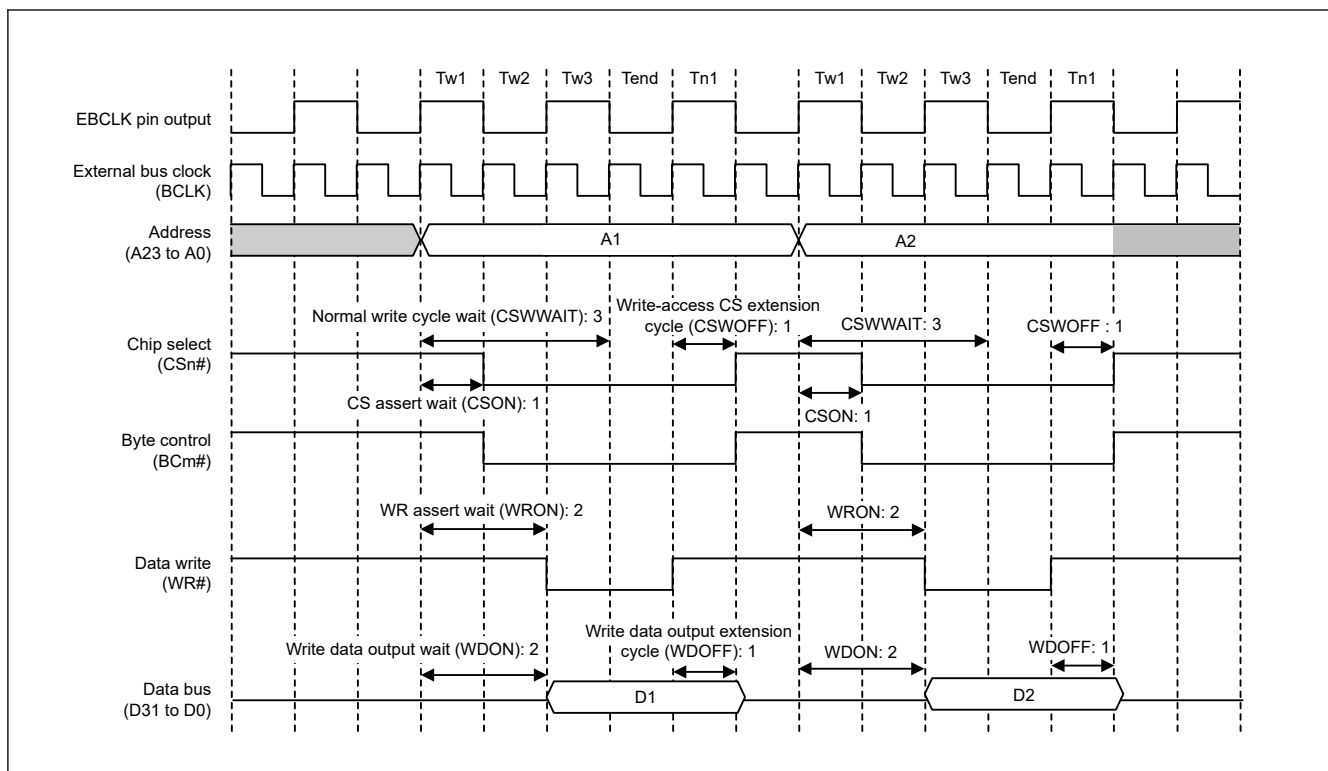


Figure 15.13 Example of normal write operation when BCLK/2 is selected in the EBCLK Pin Output Select bit ($n = 0$ to 7, $m = 0$ to 3)

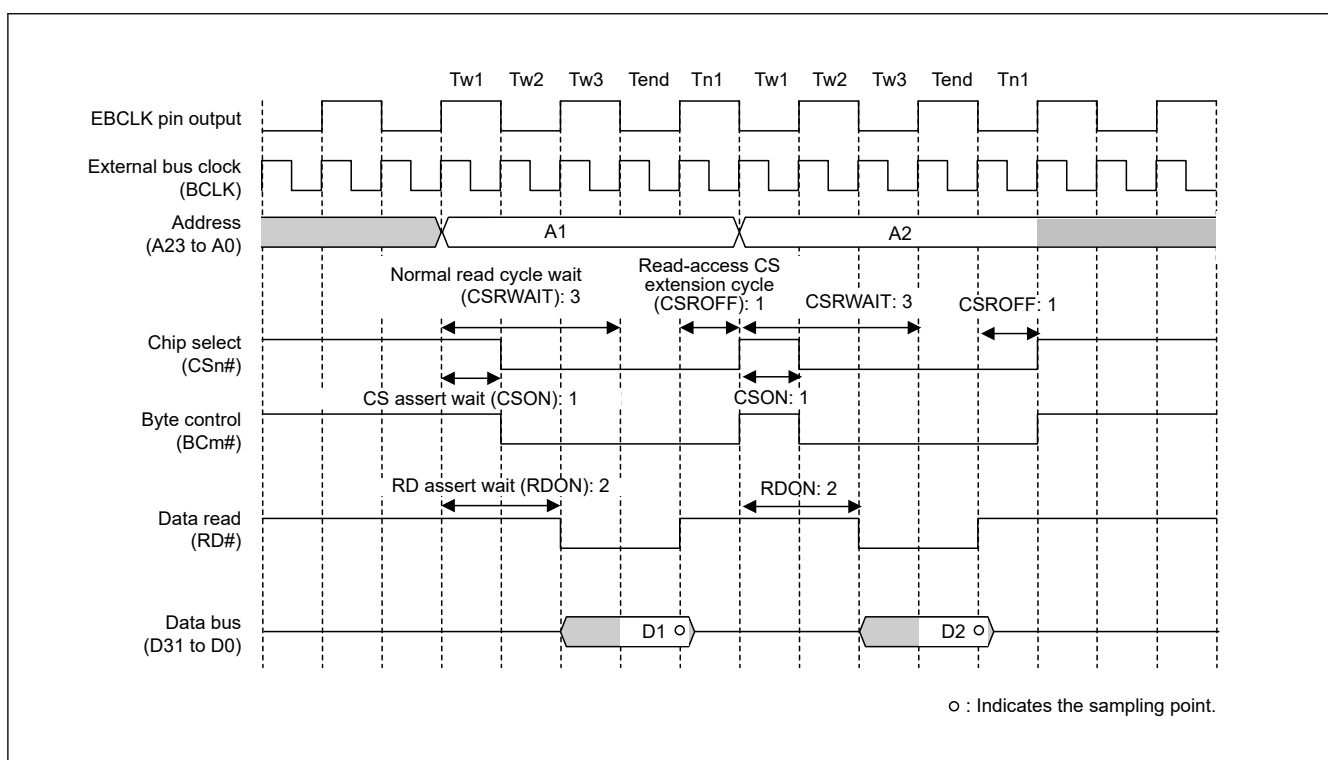


Figure 15.14 Example of normal read operation when BCLK/2 is selected in the EBCLK Pin Output Select bit and two rounds of bus access are generated in response to a single transfer request ($n = 0$ to 7, $m = 0$ to 3)

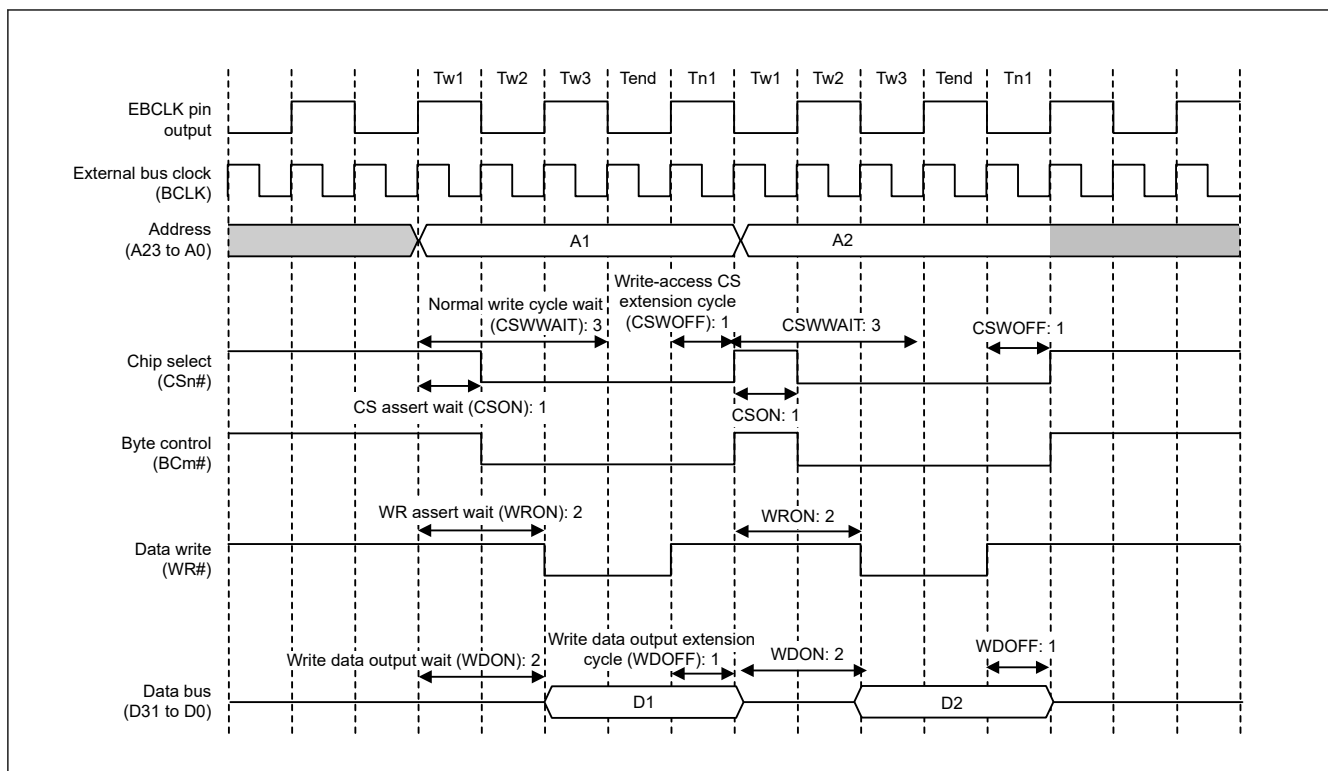


Figure 15.15 Example of normal write operation when BCLK/2 is selected in the EBCLK Pin Output Select bit and two rounds of bus access are generated in response to a single transfer request ($n = 0$ to 7, $m = 0$ to 3)

(2) Page Access

When the PRENB and PWENB bits in CSnMOD are set to 1 to enable page read and page write access, the bus access for page access operations becomes page read and write. Page access can only occur when two or more rounds of external bus access are required for a single transfer request from the bus master. However, normal access is made when split accesses are not aligned or access extends across the 32-bit boundary. See [Table 15.20](#) to [Table 15.23](#) for the conditions under which page access occurs.

[Figure 15.16](#) and [Figure 15.17](#) show examples of page access operations.

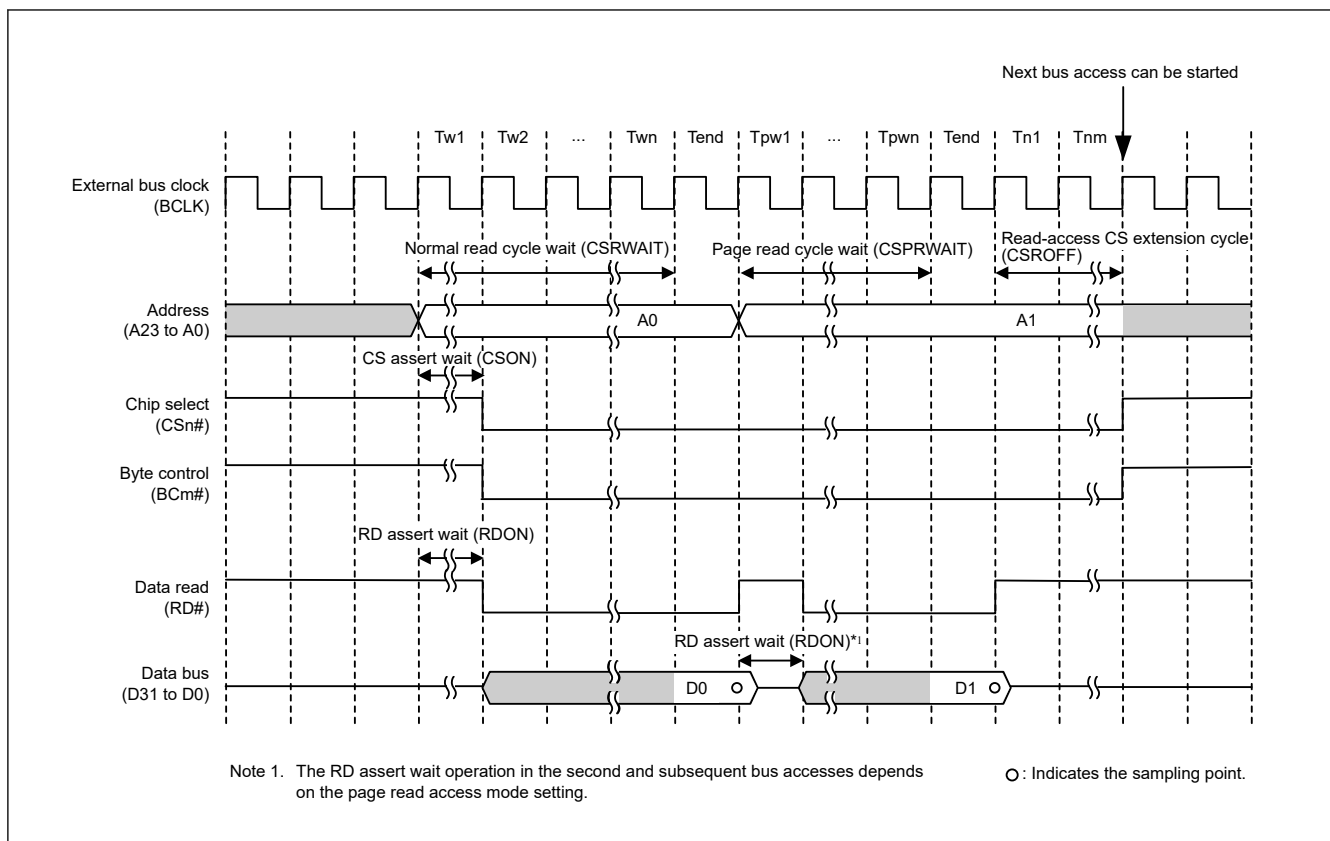


Figure 15.16 Page read access timing (n = 0 to 7, m = 0 to 3)

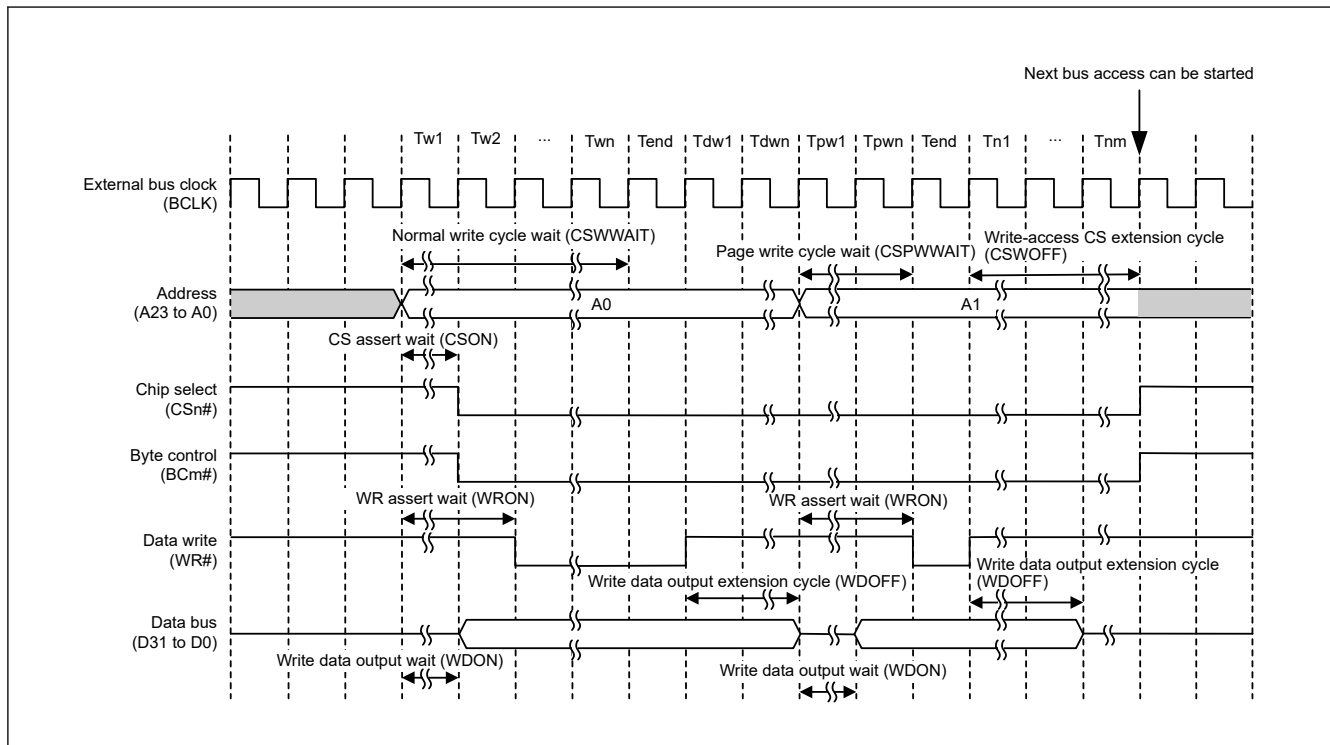


Figure 15.17 Page write access timing (n = 0 to 7, m = 0 to 3)

Figure 15.18 and Figure 15.19 show examples of operations for access to a 16-bit bus space in 32 bits. The values of the wait control registers shown in the figures are example settings. In your application, set the registers appropriately for the specifications of connected devices.

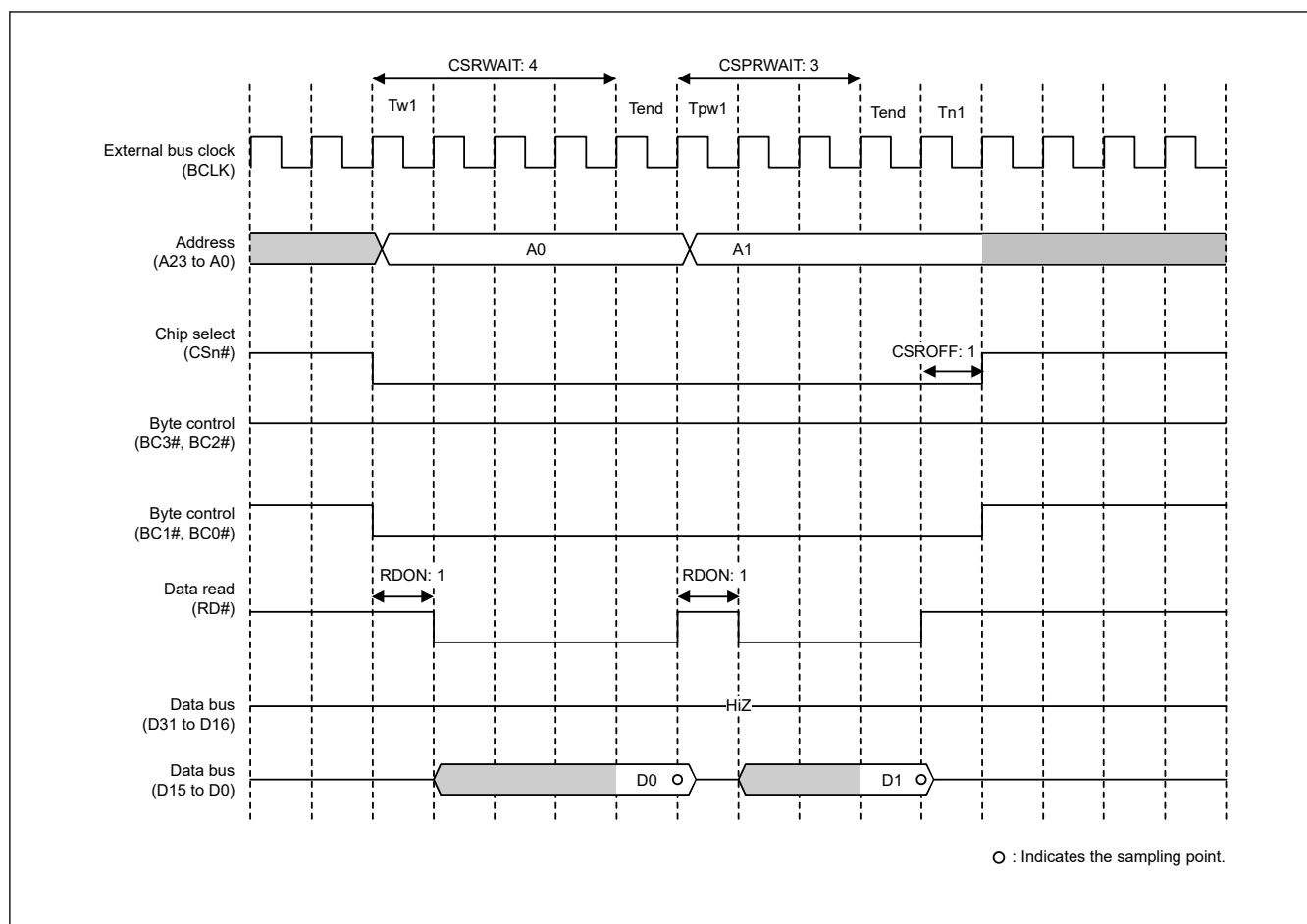


Figure 15.18 Example page read access operation when 16-bit bus space is accessed in 32 bits ($n = 0$ to 7)

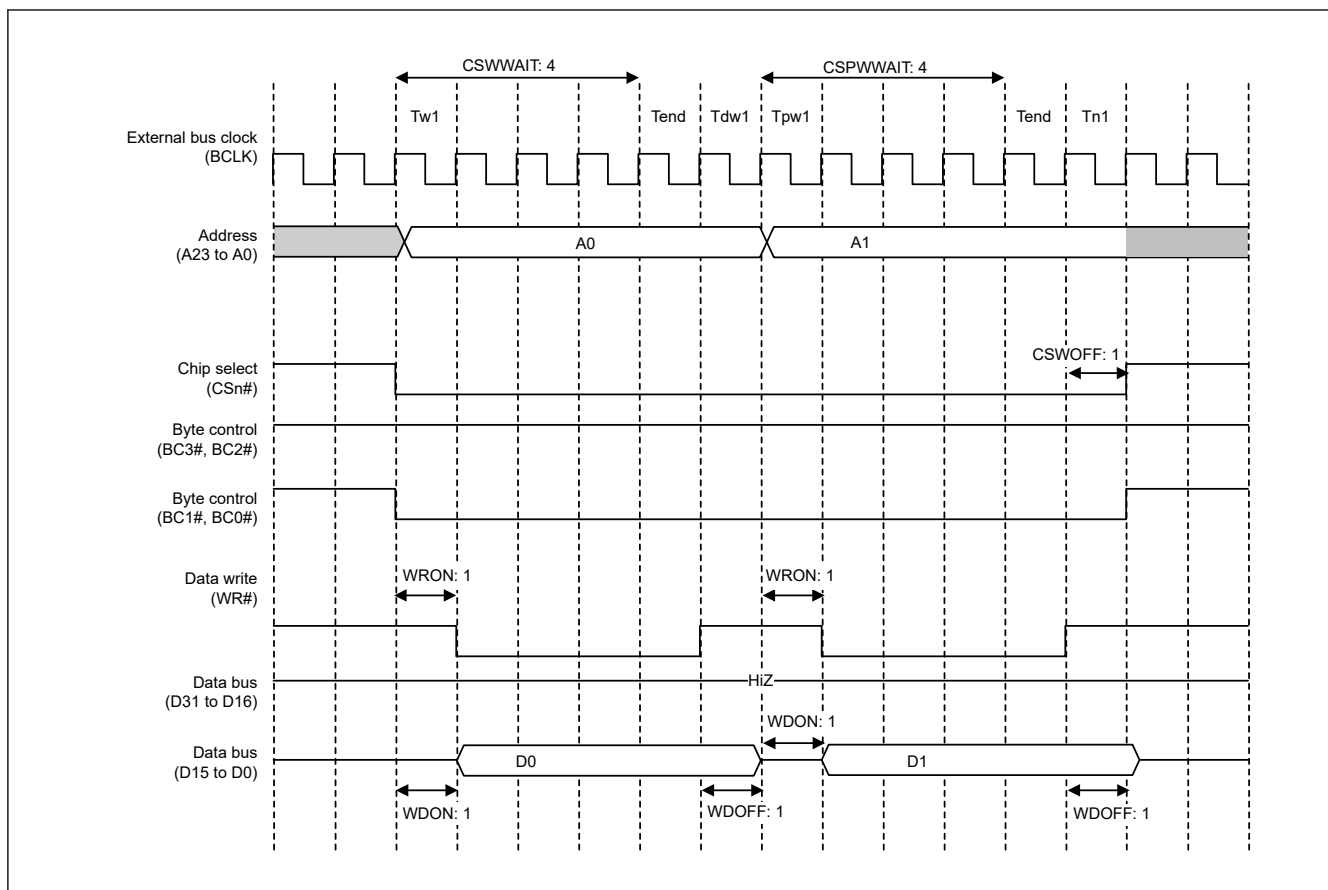


Figure 15.19 Example page write access operation when 16-bit bus space is accessed in 32 bits in single-write strobe mode ($n = 0$ to 7)

Figure 15.20 and Figure 15.21 show examples of page access operations when BCLK/2 is selected as the frequency division in the EBCLK Pin Output Select bit.

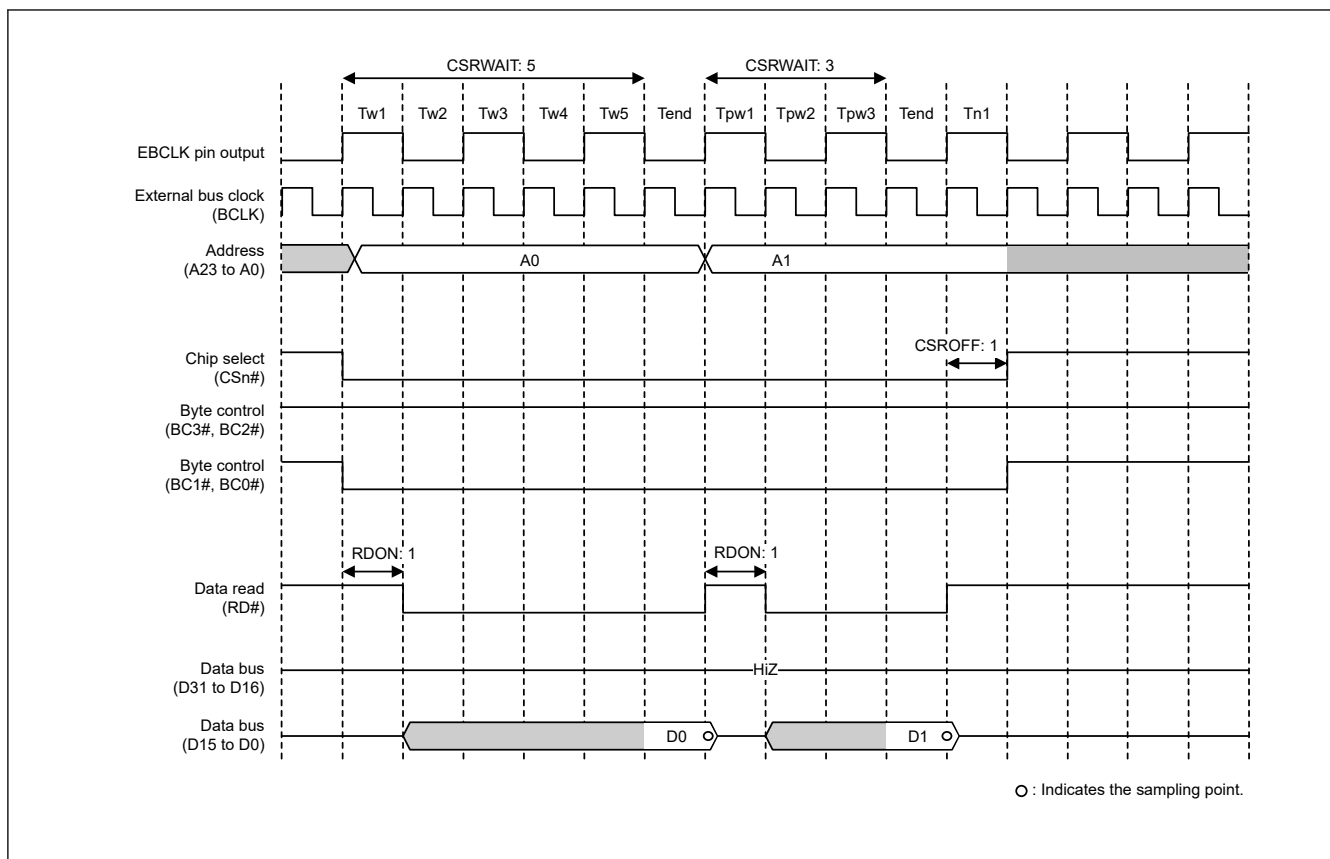


Figure 15.20 Example page read access operation when BCLK/2 is selected in the EBCLK Pin Output Select bit and two rounds of bus access are generated in response to a single transfer request ($n = 0$ to 7)

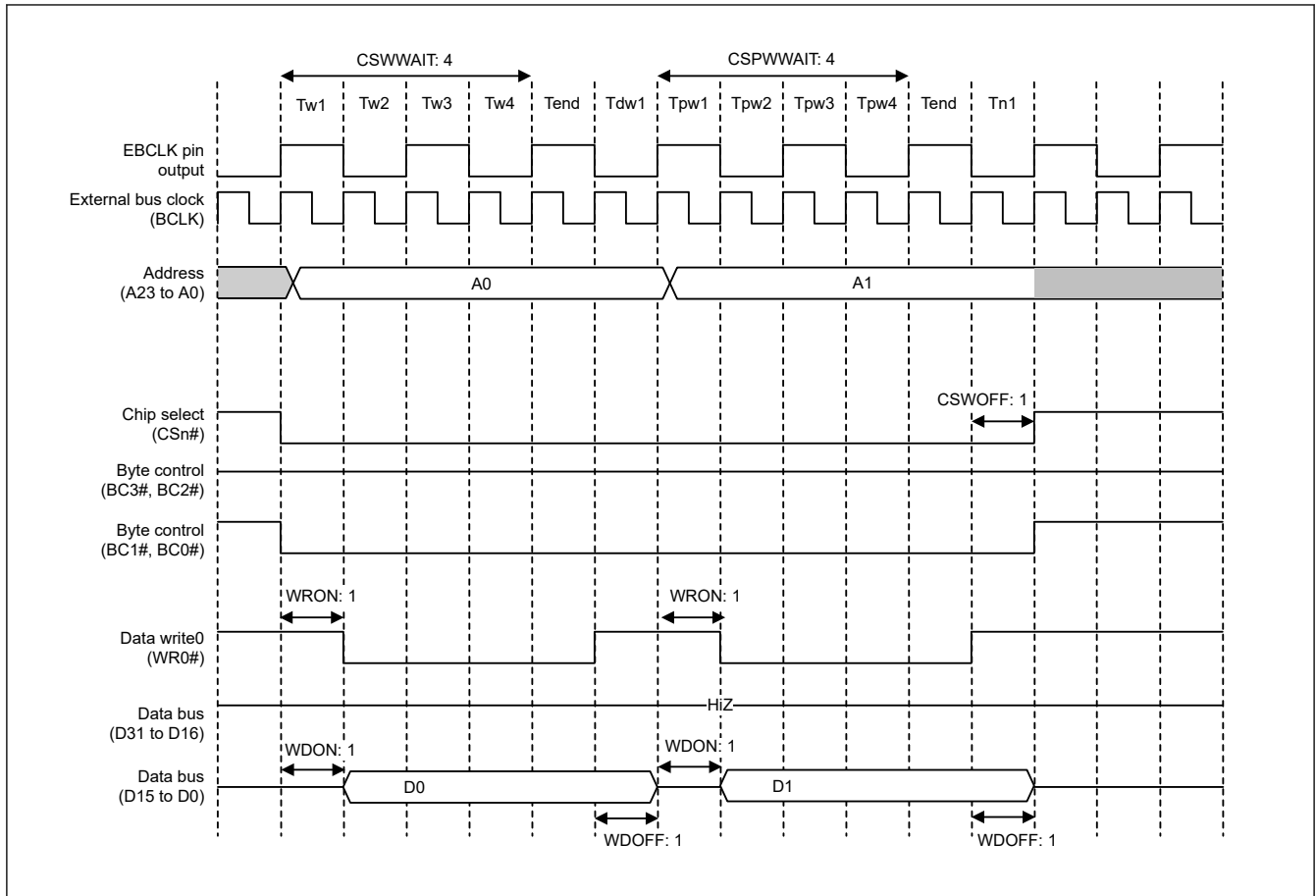


Figure 15.21 Example page write access operation when BCLK/2 is selected in the EBCLK Pin Output Select bit and two rounds of bus access are generated in response to a single transfer request, in single-write strobe mode ($n = 0$ to 7)

15.5.2 Address/Data Multiplexed Bus

When the address/data Multiplexed I/O Interface Select bit (MPXEN) in CSnCR is set to 1, addresses and data can be multiplexing input/output to/from the D15 to D00 pins in the corresponding area. Using this function enables direct connection of this MCU to peripherals of the MCU requiring address/data multiplexing. When 8-bit width is selected with the BSIZE[1:0] bits in CSnCR, D7 to D00 are multiplexed with A07 to A00. When 16-bit width is selected, D15 to D00 are multiplexed with A15 to A00. In the address/data multiplexed I/O space, accesses are controlled with the ALE, RD, WRn, and BCn signals.

Byte strobe mode or single-write strobe mode is selectable in the same way as for a separate bus. However, with regard to the BCn signals within the address cycle, the byte-control signal is output for the data being read or written.

During the address/data multiplexed I/O space access, after the number of wait cycles specified by the Address Cycle Wait Select bits (AWAIT[1:0]) in CSnWCR2 is inserted in the address output cycle, data access is performed.

Ta1 to Tan (Address Cycle Wait)

The period Ta1 to Tan is valid only when the address/data multiplexed I/O space is specified. This period is made up of the number of clock cycles between the start of external bus access and 1 cycle before the address latch (ALE) signal is negated. The number of cycles is selectable within the range from 0 to 3. Addresses are output until the next cycle of ALE signal negation (address cycle). The timing of ALE signal is the same as that of CS assertion. After the address cycle, a data cycle is started. CSnWCR1 and CSnWCR2 should be set so that an address cycle and a data cycle do not overlap.

Page access to the address/data multiplexed I/O space is invalid. When the PRENB or PWENB bit in CSnMOD is set to 1 to enable page-read or page-write access, these settings are ignored and normal read or write operation is performed.

Figure 15.22 to Figure 15.24 show examples of operations with the address/data multiplexed I/O interface

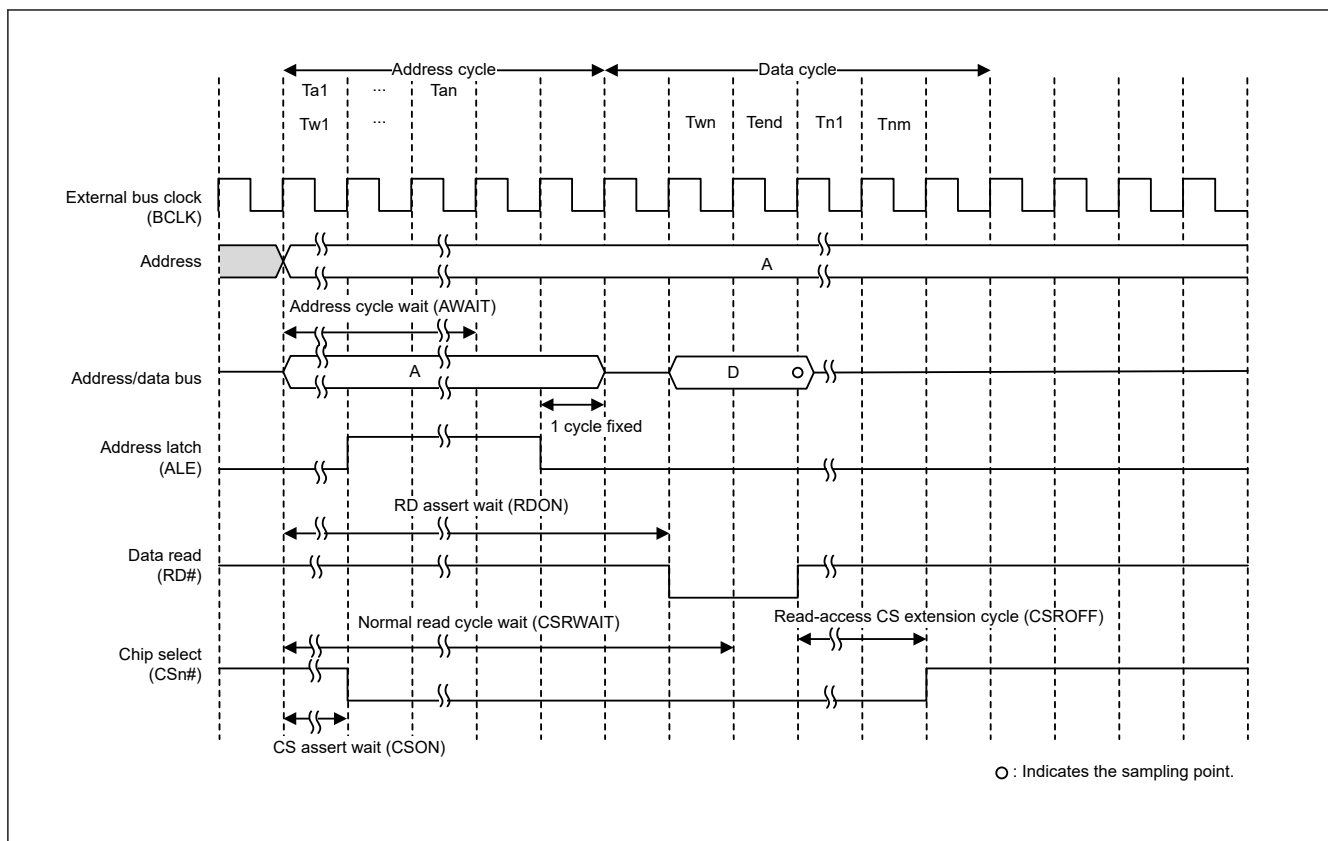


Figure 15.22 Example of read access operation with address/data multiplexed I/O interface (n = 0 to 7)

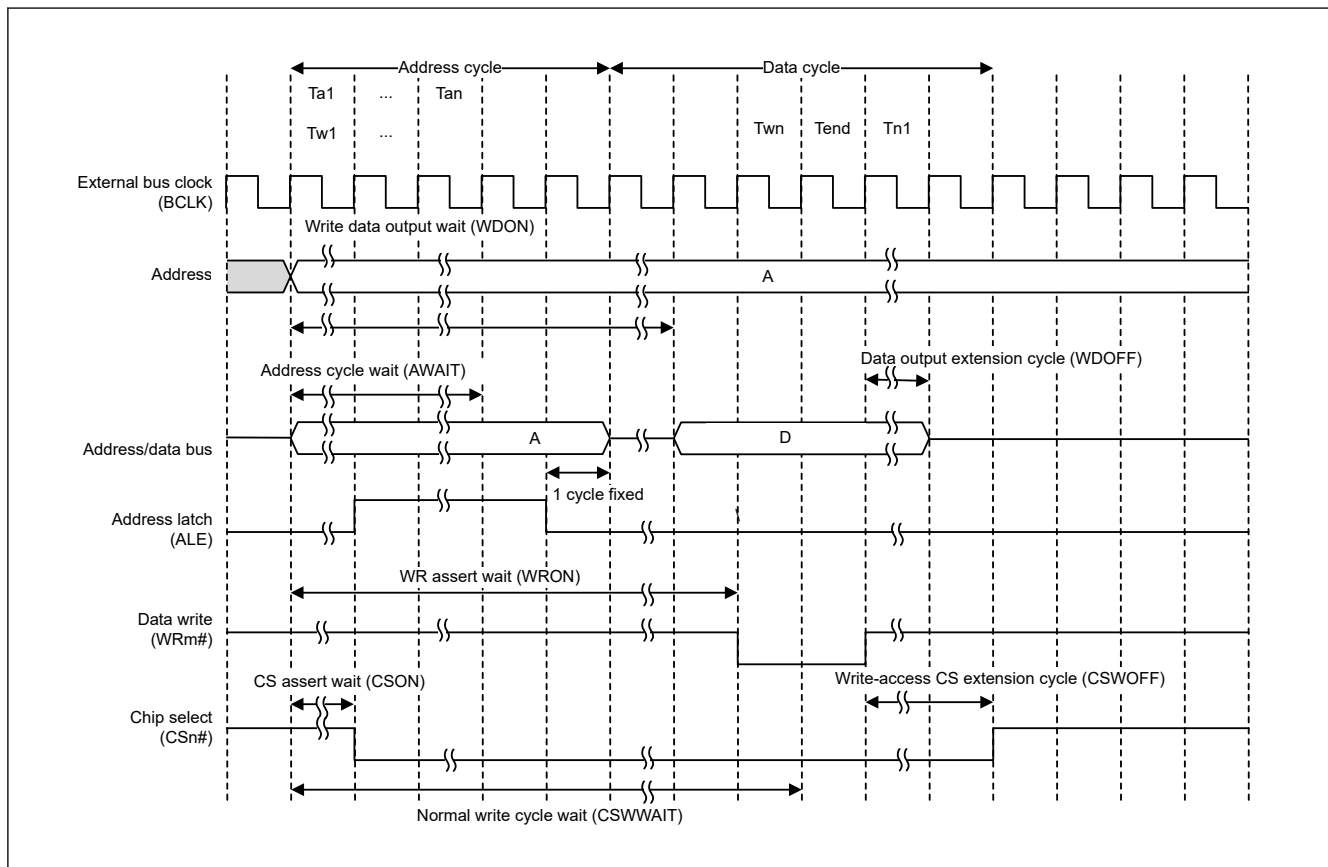


Figure 15.23 Example of write access operation with address/data multiplexed I/O interface (n = 0 to 7, m = 0, 1)

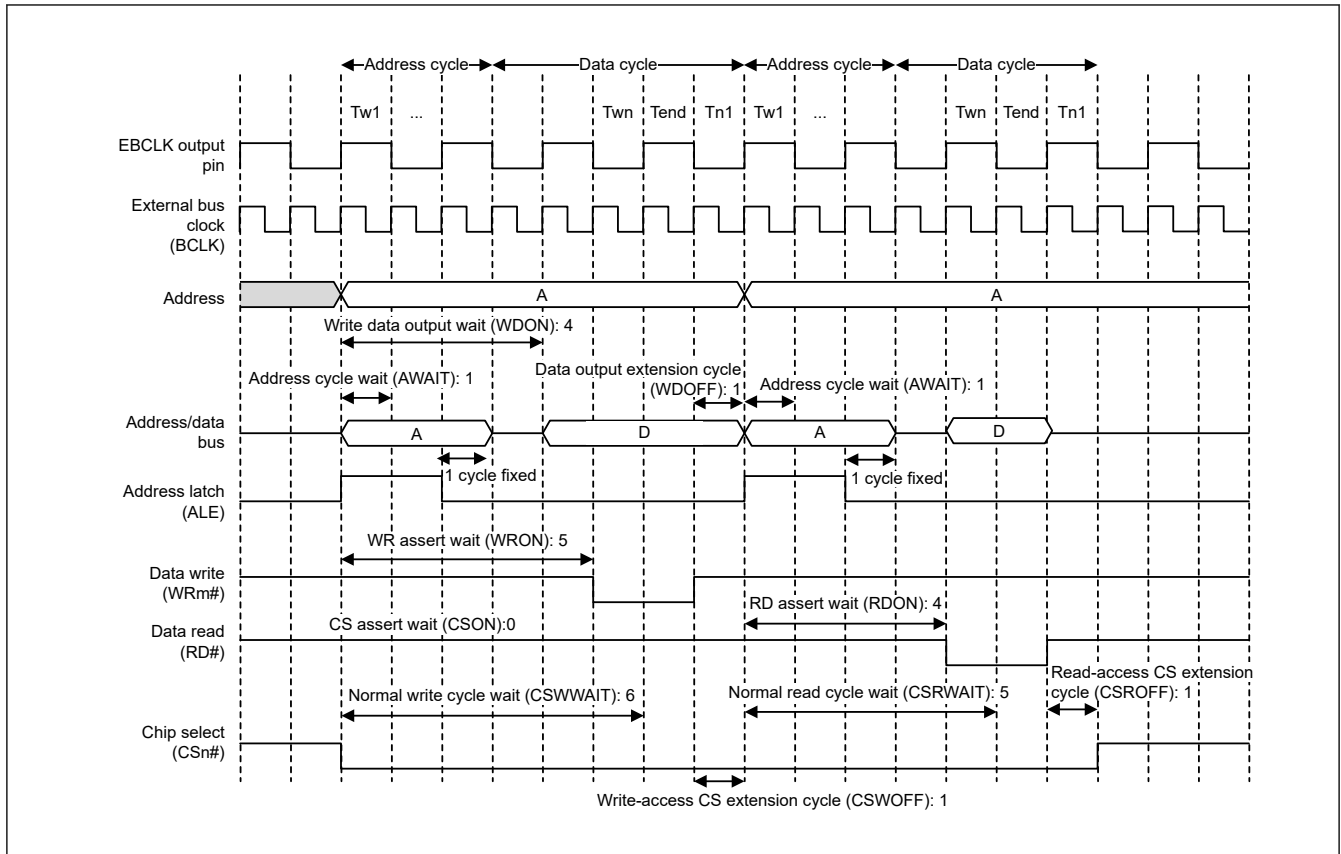


Figure 15.24 Example of bus timing with address/data multiplexed I/O interface (n = 0 to 7, m = 0, 1)

15.5.3 External Wait Function

Wait cycles can be extended by the WAIT signal beyond the length of the normal access cycle wait specified in the CSRWAIT[4:0] and CSWWAIT[4:0] bits in CSnWCR1, and the page access cycle wait specified in the CSPRWAIT[2:0] and CSPWWAIT[2:0] bits in CSnWCR1.

When external wait is enabled (EWENB = 1 in CSnMOD), wait cycles are inserted while the WAIT signal is held low. When external wait is disabled (EWENB = 0 in CSnMOD), the WAIT signal has no effect. All wait cycles specified in CSnWCR1 are inserted independently of the WAIT signal.

(1) Normal Access

Sampling of the WAIT signal begins on completion of the wait cycle (Tend) specified in CSnWCR1. The bus cycle is extended while the WAIT signal is held low. The wait cycle ends (Tend) at the next cycle after the WAIT signal goes high.

(2) Page Access

The first access operation is the same as the normal access operation. Sampling of the WAIT signal begins on completion of the wait cycle (Tend) specified in the CSnWCR1 register. The bus cycle is extended while the WAIT signal is held low. The wait cycle (Tend) ends at the next cycle after the WAIT signal goes high.

For the second and subsequent accesses, sampling of the WAIT signal begins on completion of the page access wait cycle (Tend). The page access wait cycle is extended while the WAIT signal is held low, and ends (Tend) at the next cycle after the WAIT signal goes high.

Figure 15.25 to Figure 15.28 show examples of external wait insertion timing with the separate bus interface.

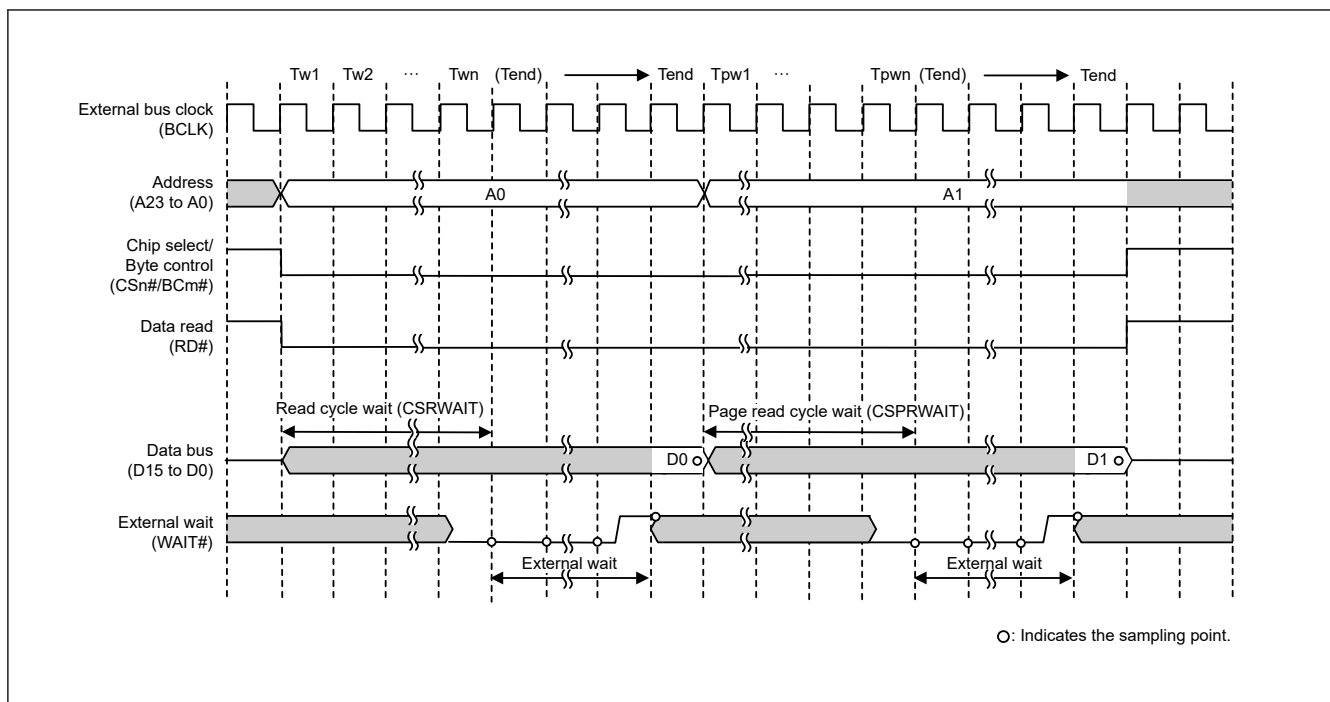


Figure 15.25 Example external wait timing for page read access to 16-bit bus space (when 1/1 BCLK is selected with the BCLK Pin Output Select bit) ($n = 0$ to 7, $m = 0, 1$)

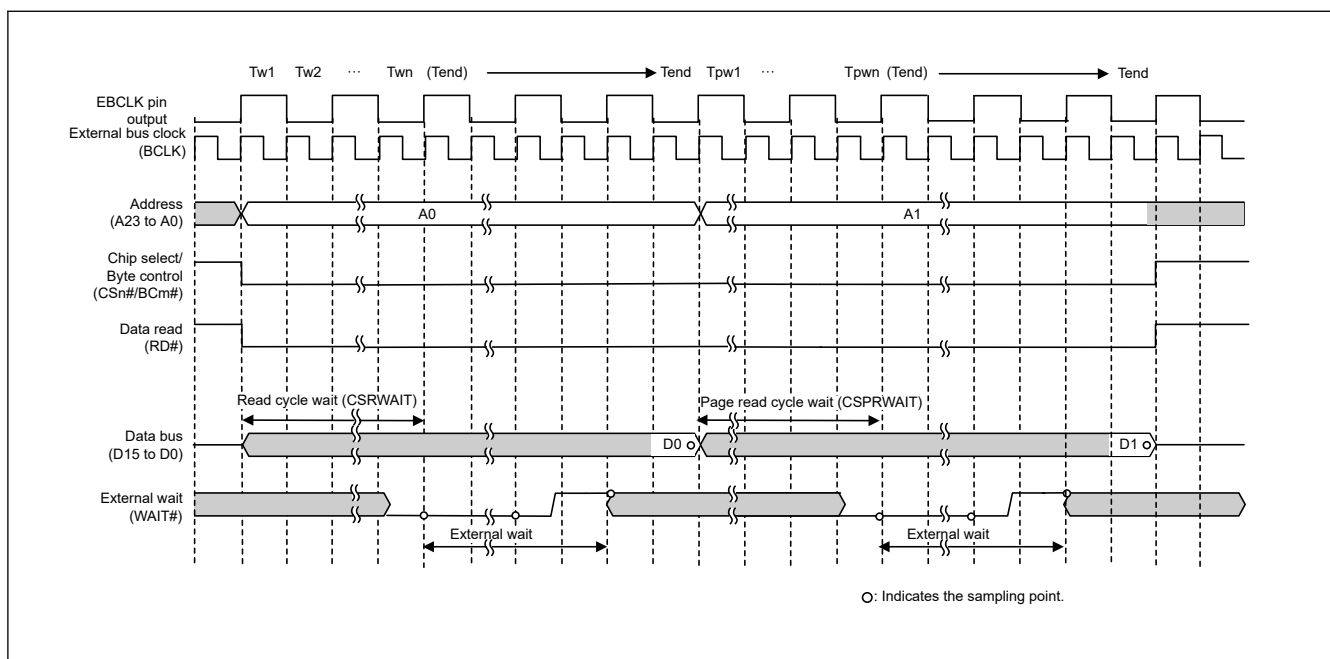


Figure 15.26 Example external wait timing for page read access to 16-bit bus space (when 1/2 BCLK is Selected with the EBCLK Pin Output Select bit) ($n = 0$ to 7, $m = 0, 1$)

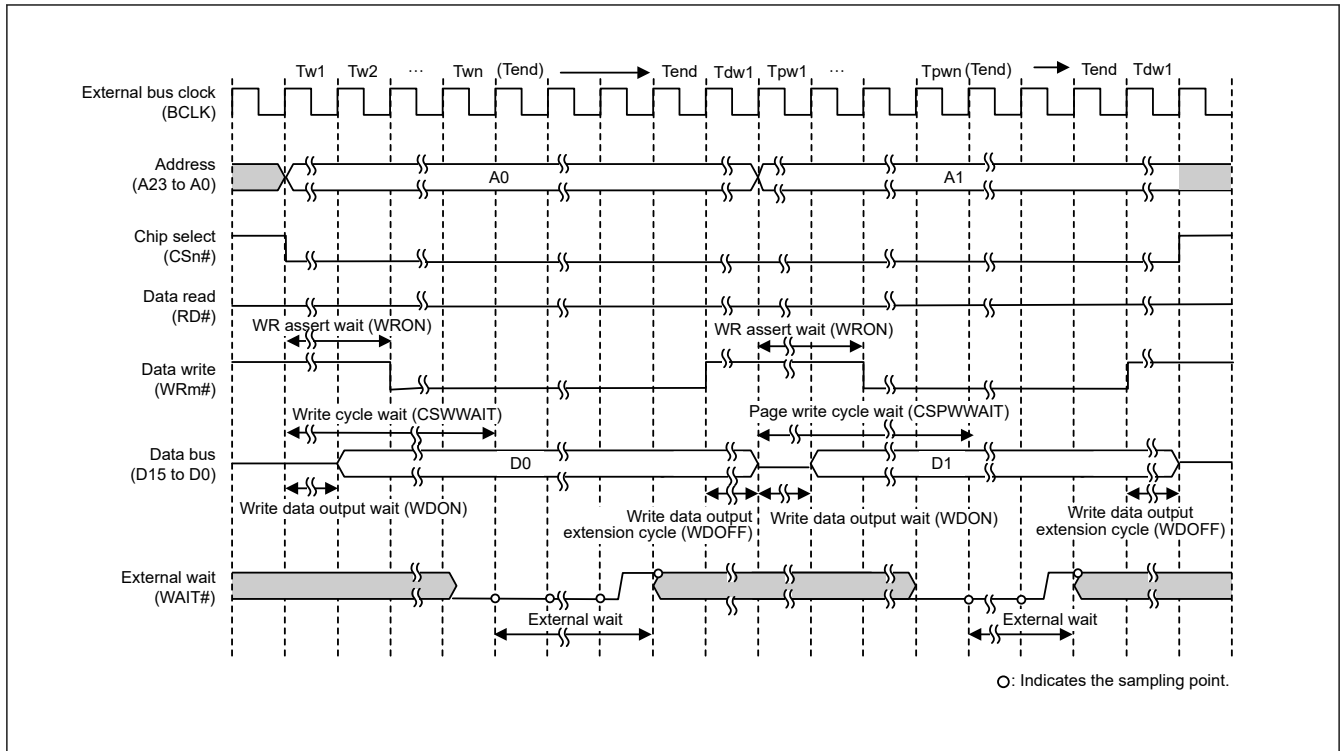


Figure 15.27 Example external wait timing for page write access to 16-bit bus space in byte strobe mode (when 1/1 BCLK is selected with the BCLK Pin Output Select bit) ($n = 0$ to 7, $m = 0, 1$)

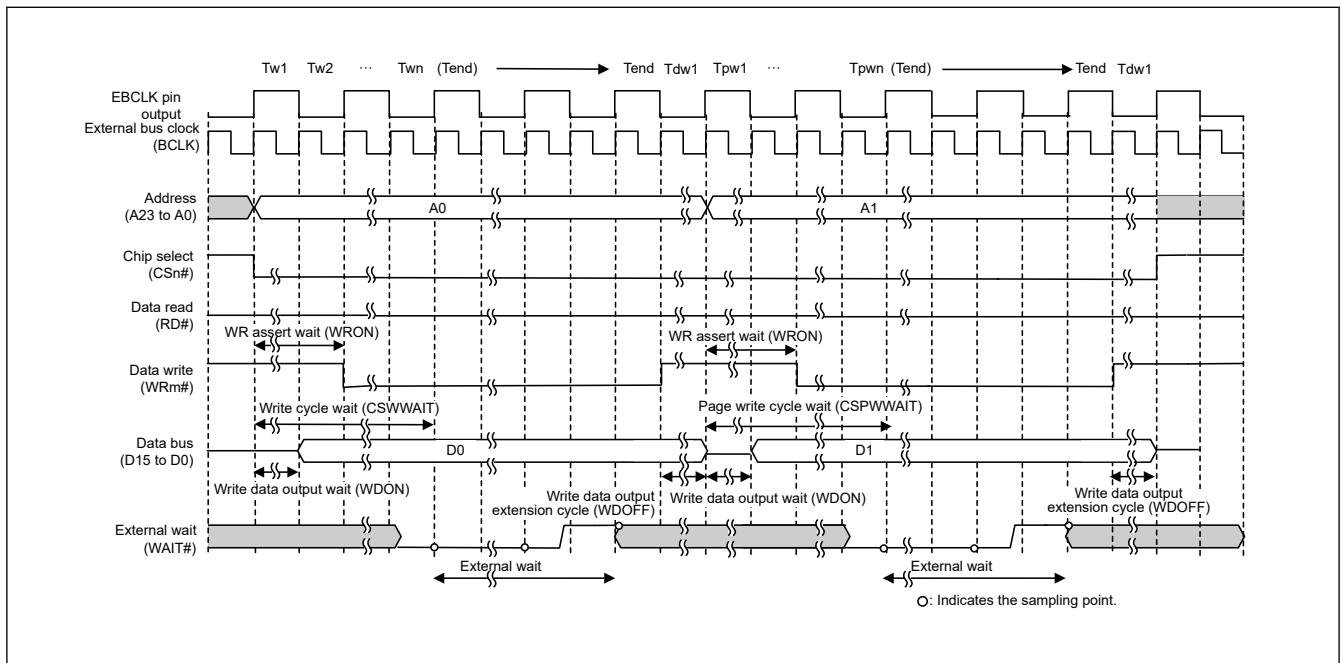


Figure 15.28 Example external wait timing for page write access to 16-bit bus space in byte strobe mode (when 1/2 BCLK is selected with the EBCLK Pin Output Select bit) ($n = 0$ to 7, $m = 0, 1$)

(3) Address/Data Multiplexed I/O Interface

In a data cycle with the address/data multiplexed I/O interface, programmed waits and pin waits using the WAIT pin can be inserted in the same way as that with the separate bus interface.

Address cycles are not affected by the wait control settings. Figure 15.29 shows an example of external wait insertion timing with the address/data multiplexed I/O interface.

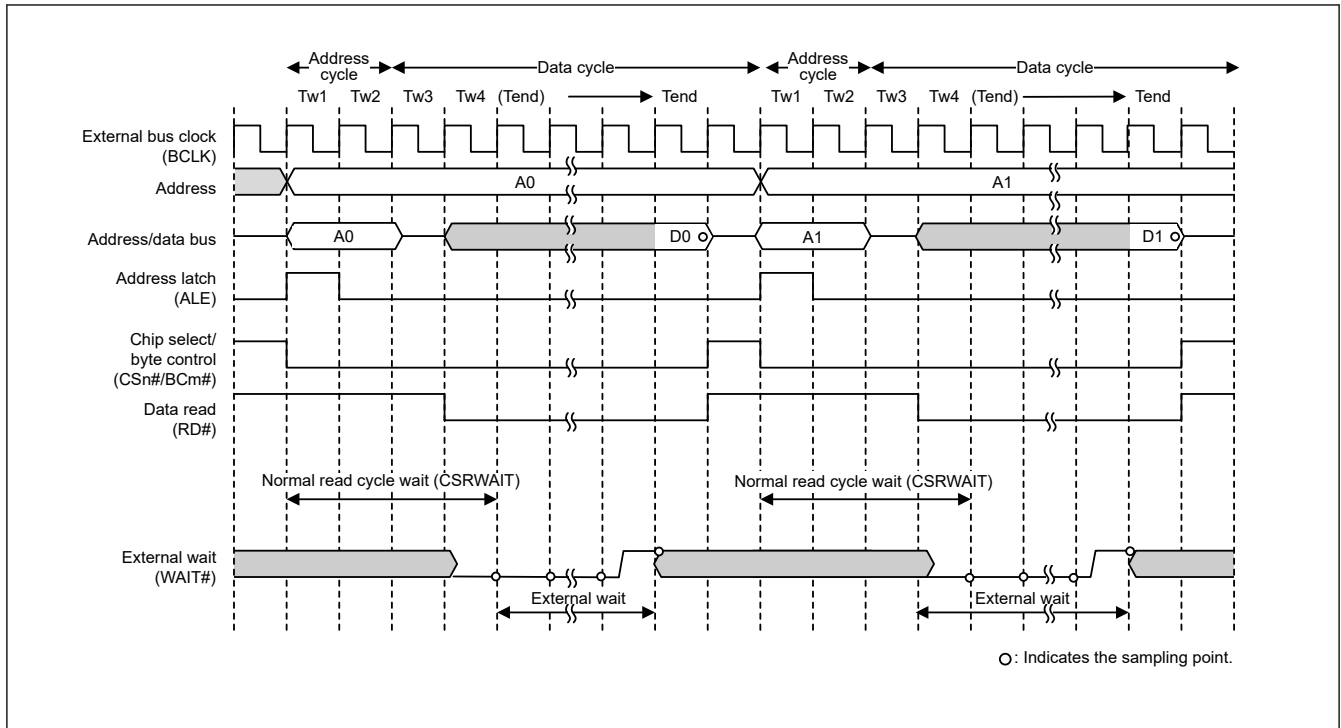


Figure 15.29 Example external wait insertion timing with address/data multiplexed I/O interface ($n = 0$ to 7 , $m = 0, 1$)

15.5.4 Insertion of Recovery Cycles

Recovery cycles can be inserted between consecutive rounds of external bus access by setting the Recovery Cycle Insertion Enable bit in CSRECEN to 1.

The number of recovery cycles to be inserted after read cycles and write cycles can be independently set for each area using CSnREC. When the preceding bus cycle is a write access, the number of write recovery cycles must be set with the WRCV[3:0] bits for the associated area. When the preceding bus cycle is a read access, the number of read recovery cycles must be set with the RRCV[3:0] bits for the associated area. For example, when a CS1 read access occurs after a CS0 read access, the number of recovery cycles to be inserted between them is set in the RRCV[3:0] bits in CS0REC.

Recovery cycle insertion can be enabled or disabled with RCVEN_i ($i = 0$ to 7) in CSRECEN when the preceding bus access is a separate bus access, and with RCVENM_j ($j = 0$ to 7) when the preceding bus access is an address/data multiplexed bus access.

Recovery cycles can be inserted on any of the following conditions:

- After a read access to the external bus, a read access is made to the external bus in the same area.
- After a read access to the external bus, a read access is made to the external bus in a different area.
- After a read access to the external bus, a write access is made to the external bus in the same area.
- After a read access to the external bus, a write access is made to the external bus in a different area.
- After a write access to the external bus, a read access is made to the external bus in the same area.
- After a write access to the external bus, a read access is made to the external bus in a different area.
- After a write access to the external bus, a write access is made to the external bus in the same area.
- After a write access to the external bus, a write access is made to the external bus in a different area.

The recovery cycle starts at the end of the preceding bus cycle, for example when the CS_n signal ($n = 0$ to 7) is negated. A high-level period of the CS_n signal is inserted for the specified recovery cycle period starting from this point.

In the fastest case, the CS_n signal for the next round of bus access is asserted immediately after the end of the recovery cycles. Even if the next request for access to an external address space is generated during the recovery period, the next access over the external bus starts immediately after the end of the recovery cycles.

When two or more external bus access cycles are required for a single transfer request from a bus master, and the recovery cycle insertion condition is satisfied, recovery cycles are also inserted between these bus access cycles. However, when page read access is enabled ($CSnMOD.PRENB = 1$) or page write access is enabled ($CSnMOD.PWENB = 1$), recovery cycles are not inserted except after the last bus access cycle of the transfer, even if the recovery cycle insertion condition is satisfied. See Figure 15.33.

Similarly, during normal access with page access enabled, recovery cycles are not inserted between bus access cycles but only after the last bus access cycle of the transfer. With the address/data multiplexed I/O interface, when the recovery cycle insertion condition is satisfied, recovery cycles are inserted between bus access cycles regardless of the page access enable setting.

Figure 15.30 to Figure 15.33 show examples of recovery cycle insertion with the separate bus interface.

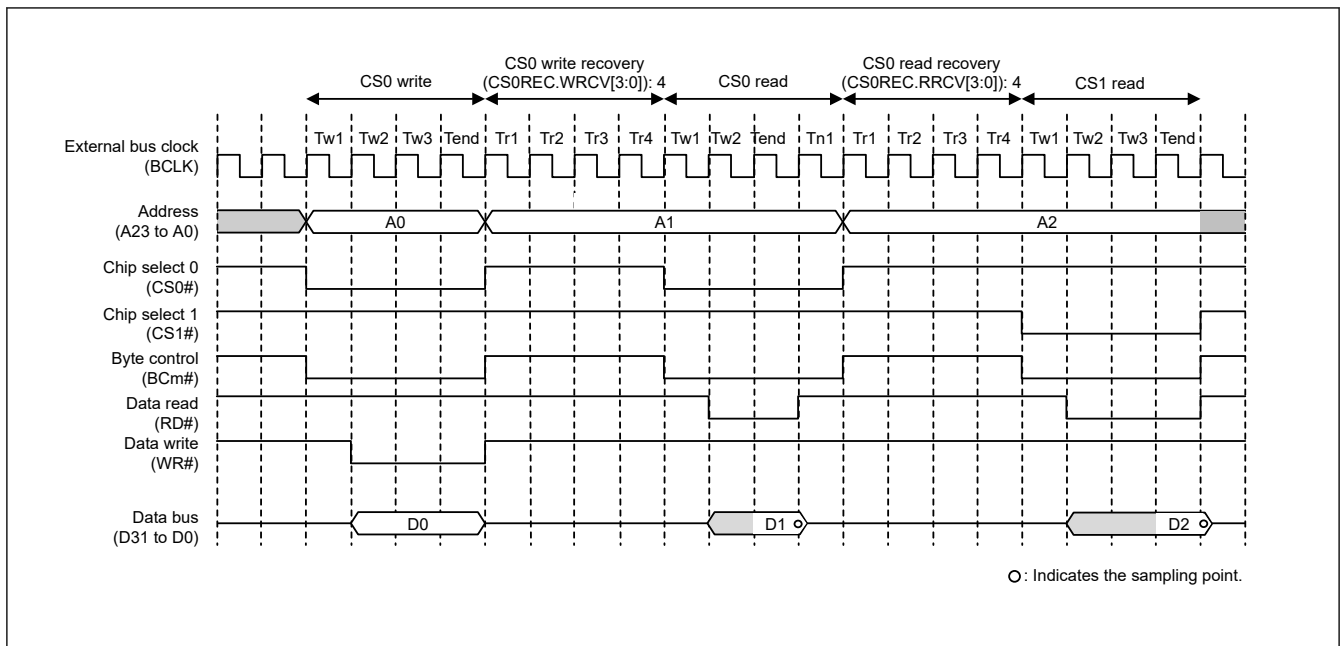


Figure 15.30 Example recovery cycle insertion with separate bus interface (m = 0 to 3)

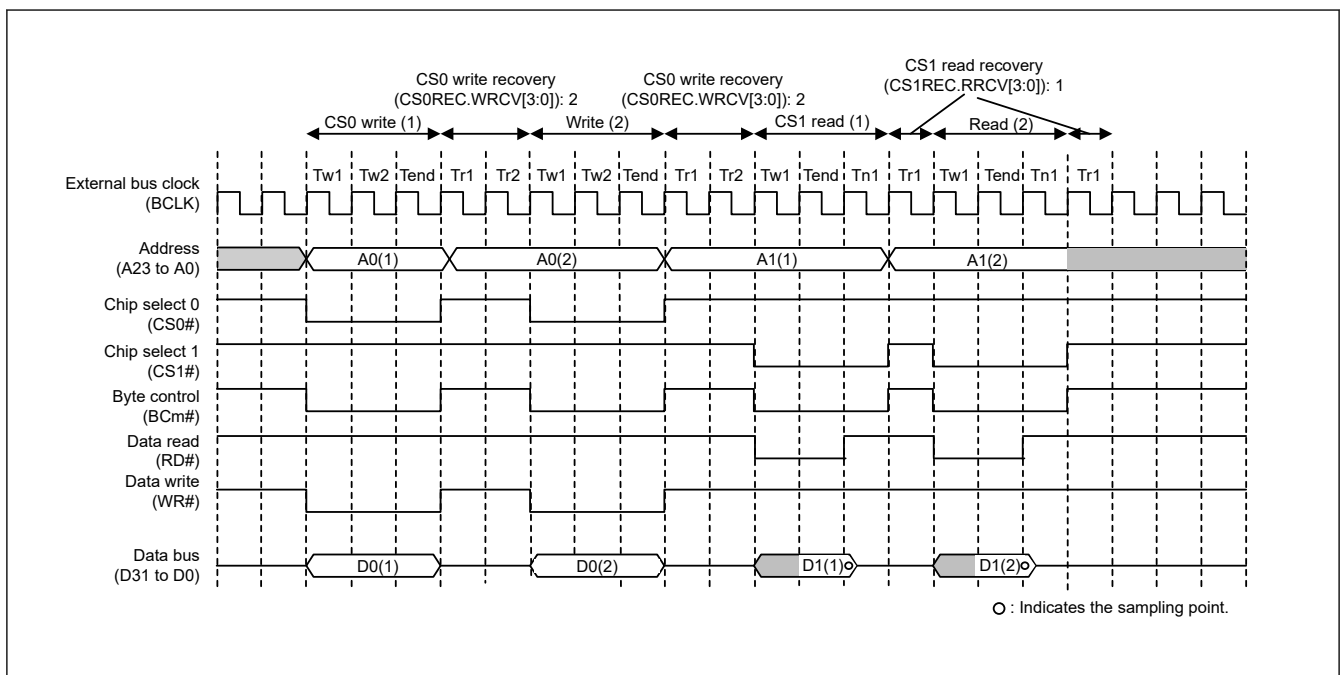


Figure 15.31 Example recovery cycle insertion when bus access is split, with separate bus interface and normal access (m = 0 to 3)

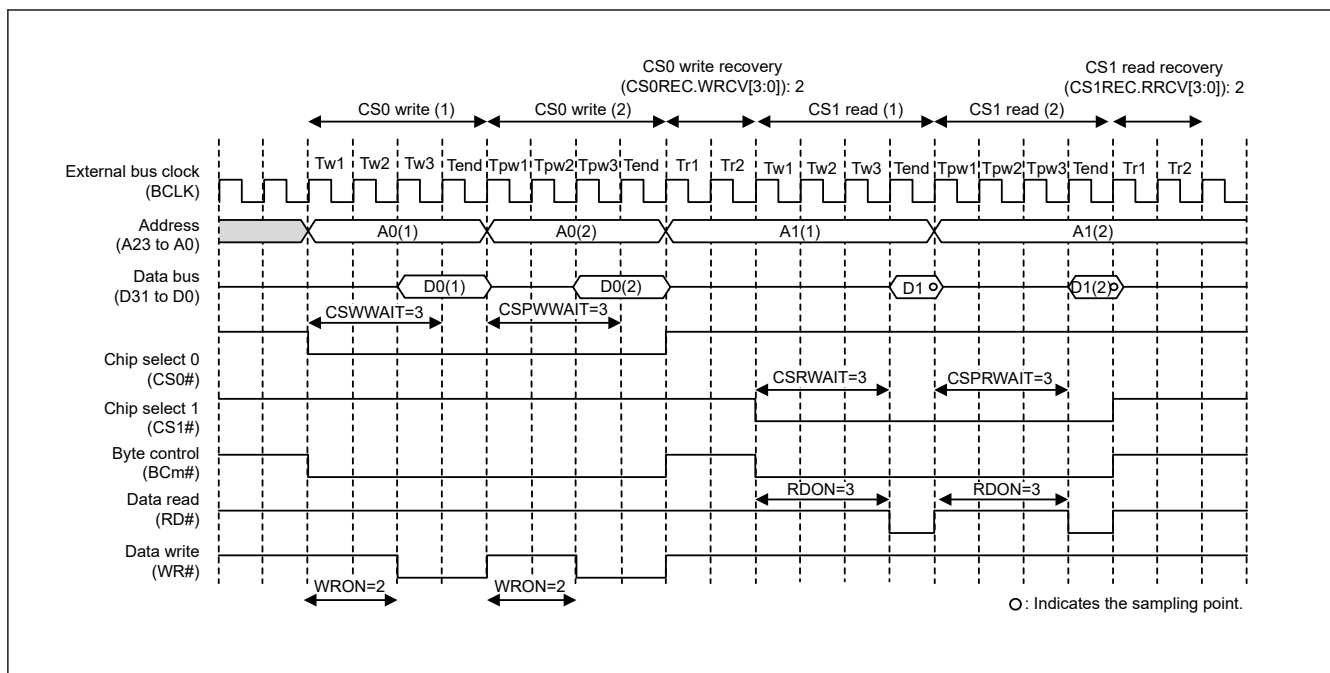


Figure 15.32 Example recovery cycle insertion when bus access is split, with separate bus interface and page access ($m = 0$ to 3)

Figure 15.33 shows an example operation when BCLK/2 is selected as the frequency division in the EBCLK Pin Output Select bit.

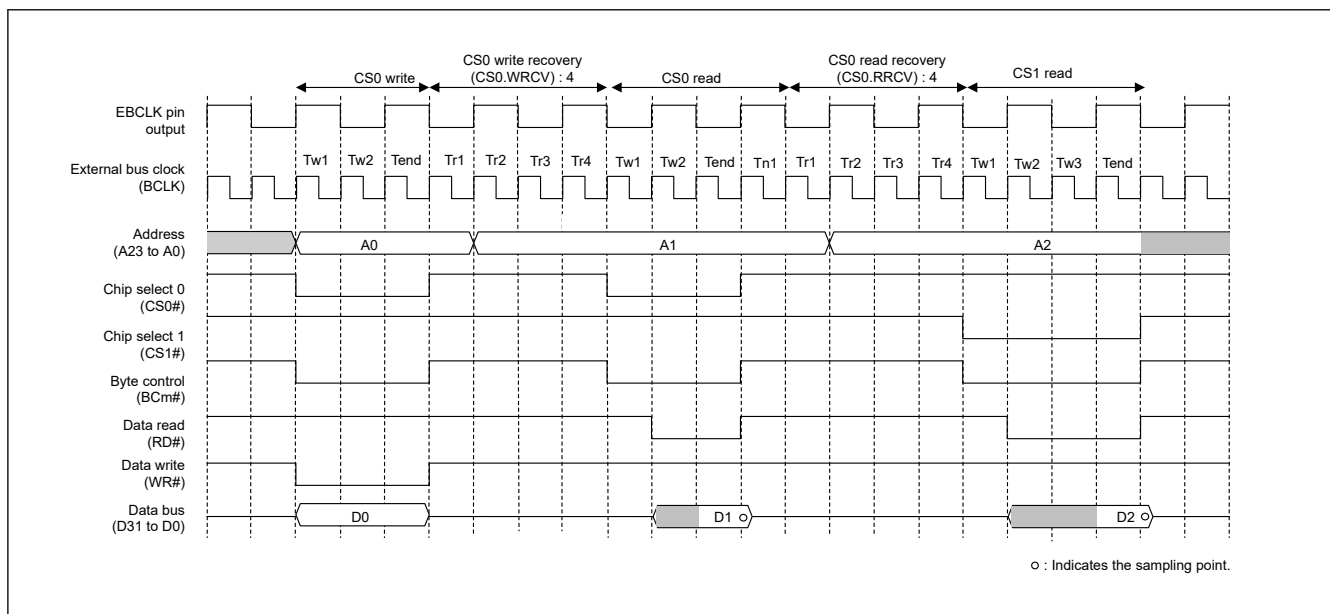


Figure 15.33 Example operation for recovery cycles when BCLK/2 is selected in the EBCLK Pin Output Select bit, with normal access through a separate bus interface ($m = 0$ to 3)

With the address/data multiplexed I/O interface, recovery cycles are inserted in the same way as that with the separate bus interface. Figure 15.34 and Figure 15.35 show examples of recovery cycle insertion with the address/data multiplexed I/O interface.

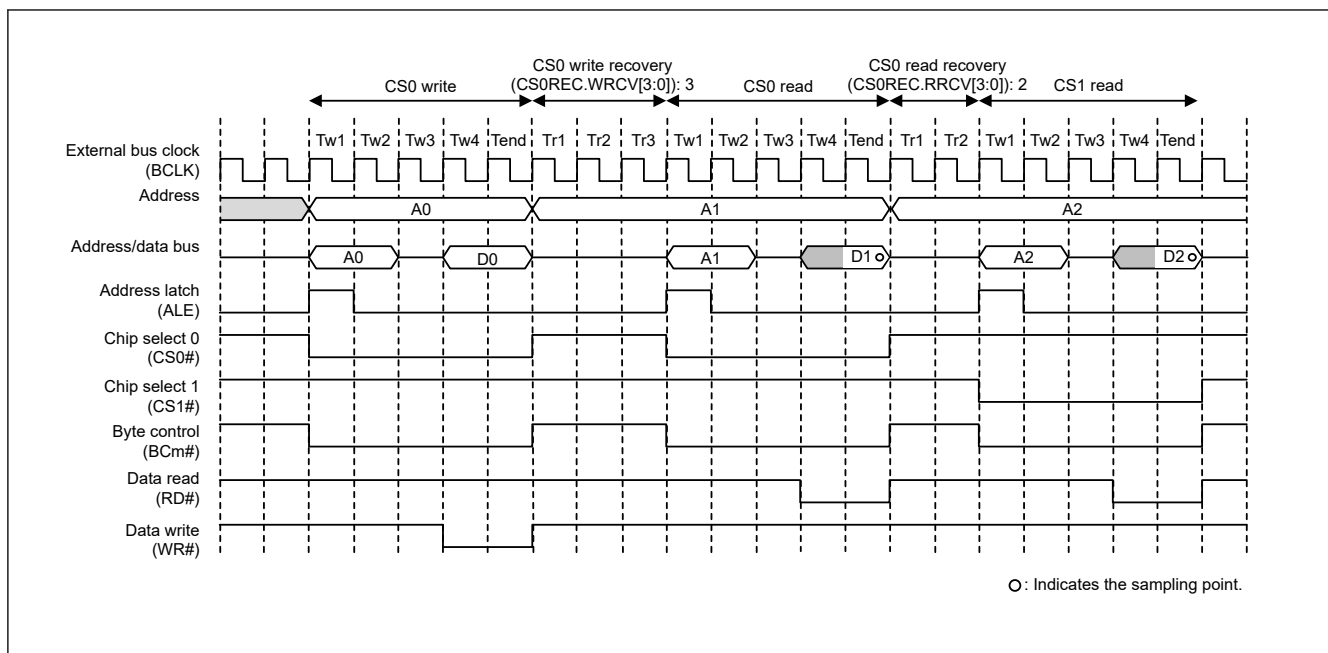


Figure 15.34 Example of recovery cycle insertion with address/data multiplexed I/O interface (m = 0, 1)

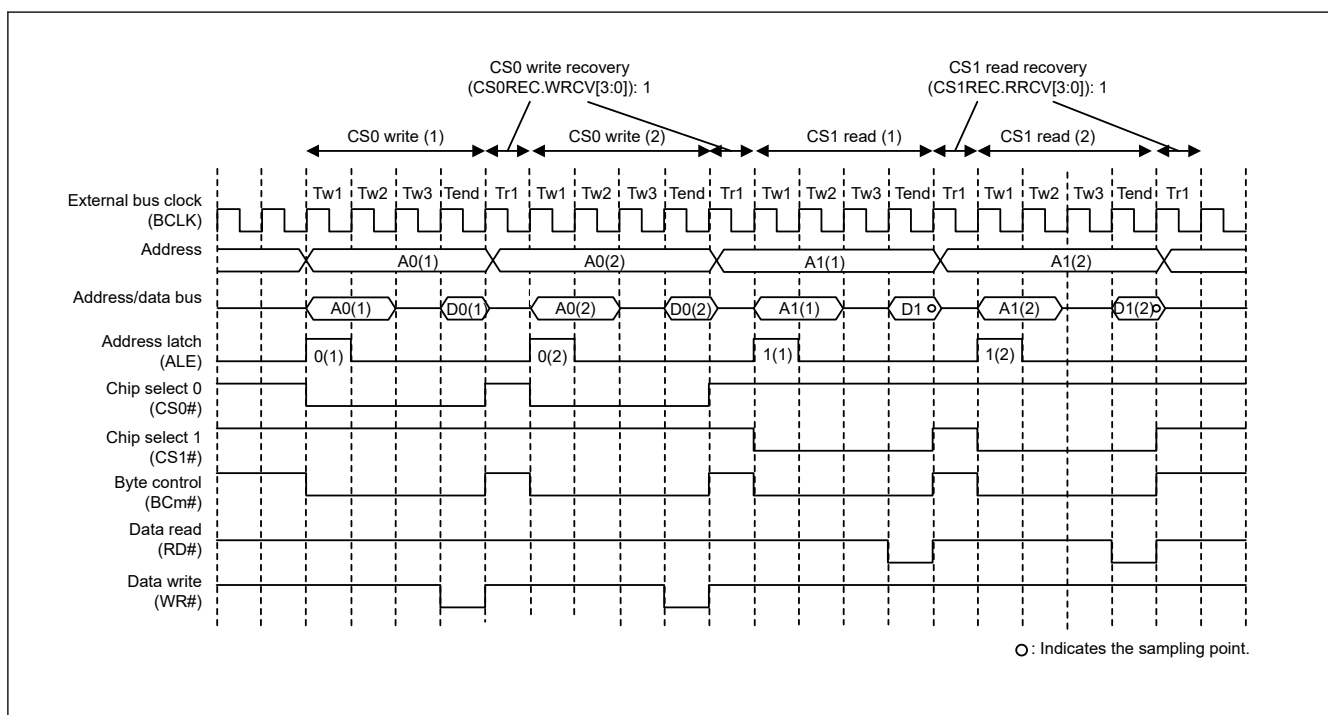


Figure 15.35 Example of recovery cycle insertion when a bus access is split with address/data multiplexed I/O interface (m = 0, 1)

15.5.5 No Access State

When no external address space is accessed, CSn, BCn, WRn, and RDn signals are high, ALE signal is low, and D31 to D00 are in the high-impedance state.

15.5.6 Limitations

(1) Constraints on using separate bus interface

Table 15.30 lists the constraints that apply to bits in the CSn Wait Control Register 1 (CSnWCR1) and CSn Wait Control Register 2 (CSnWCR2) when normal and page accesses occur.

Even if the Page Read Access Enable bit or Page Write Access Enable bit in the CSn Mode Register is set to enable (CSnMOD.PRENB = 1 or CSnMOD.PWENB = 1), the first page access or access that does not fall within the scope of a page access is a normal access operation. Because of this, constraints on normal access must be satisfied.

Table 15.30 Constraints on normal access and page access

Limitations at the time of normal access		Limitations at the time of page access	
Reading	Writing	Reading	Writing
$CSON[2:0] \leq CSRWAIT$ $RDON[2:0] \leq CSRWAIT$ $CSON[2:0] \leq RDON$	$1 \leq WDON[2:0]$ $CSON[2:0] \leq CSWWAIT$ $WRON[2:0] \leq CSWWAIT$ $WDON[2:0] \leq CSWWAIT$ $WDOFF[2:0] \leq CSWOFF$ $WDON[2:0] \leq WRON$ $CSON[2:0] \leq WRON$	$CSON[2:0] \leq CSPRWAIT$ $RDON[2:0] \leq CSPRWAIT$ $CSON[2:0] \leq RDON$	$1 \leq WDON[2:0]$ $CSON[2:0] \leq CSPWWAIT$ $WRON[2:0] \leq CSPWWAIT$ $WDON[2:0] \leq CSPWWAIT$ $WDOFF[2:0] \leq CSWOFF$ $WDON[2:0] \leq WRON$ $CSON[2:0] \leq WRON$

Note: When 2 or more external bus access cycles are required for a single transfer request from a bus master, and the recovery cycle insertion condition is satisfied, with page read access enabled (CSnMOD.PRENB = 1) or page write access enabled (CSnMOD.PWENB = 1), recovery cycles are not inserted between bus access cycles and are inserted only after the last bus access cycle of the transfer.

(2) Constraints on using address/data multiplexed bus interface

In the address/data multiplexed I/O space, page accesses are invalid. If a page access setting is specified, the setting is ignored and the normal read or write operation is performed. When the address/data multiplexed I/O interface is set, the BSIZE[1:0] bits in CSnCR should not be set to the 32-bit bus space. If set, the operation cannot be guaranteed.

Table 15.31 Constraints at the time of normal access

Reading	Writing
$CSON[2:0] \leq CSRWAIT$ $RDON[2:0] \leq CSRWAIT$ $CSON[2:0] \leq RDON$ $AWAIT[1:0] + 2 \leq RDON$ $CSON[2:0] \leq AWAIT$	$CSON[2:0] \leq CSWWAIT$ $WRON[2:0] \leq CSWWAIT$ $WDON[2:0] \leq CSWWAIT$ $WDOFF[2:0] \leq CSWOFF$ $WDON[2:0] \leq WRON$ $CSON[2:0] \leq WRON$ $AWAIT[1:0] + 2 \leq WRON$ $AWAIT[1:0] + 2 \leq WDON$ $CSON[2:0] \leq AWAIT$

(3) Constraint on pin multiplexing between the A00 and BC0 functions

Setting the single-write strobe mode is prohibited in the 8-bit bus space.

(4) Constraints when BCLK/2 is selected in the EBCLK Pin Output Select bit

When 1/2 cycle of BCLK is selected in the EBCLK Pin Output Select bit, the external bus access cycle starts on the rising edge of the EBCLK pin output. However, when 2 or more external bus access cycles are generated for a single transfer request from a bus master, the second or subsequent external bus access cycle can start on the falling edge of the EBCLK pin output, depending on the wait cycle settings. Set the registers appropriately for the specifications of connected devices.

(5) Instruction code constraints

You must fix the instruction code to little-endian order.

15.6 SDRAM Area Controller Operation

This section describes how the SDRAM area controller (SDRAMC) is enabled and the SDRAM bus width is set, followed by a description of the SDRAMC operations, including read, write, auto-refresh, self-refresh, initialization sequence, and mode register settings.

15.6.1 Enabling/Disabling SDRAM Access and Setting the SDRAM Bus Width

SDRAM access can be enabled or disabled using the SDC Control Register (SDCCR). The SDRAM bus width can also be set using SDCCR. The refresh operation is available even when the operation of the SDRAM address space is disabled, as long as self-refresh or auto-refresh is enabled.

15.6.2 No Access State

When no external address space is accessed, the SDCS, WE, RAS and CAS signals are high.

15.6.3 Insertion of Recovery Cycles

When access to the SDRAM area follows access to the CS area, data recovery cycles are inserted for the CS area controller (CSC). If the number of recovery cycles for the CSC is 0, the ACT command for the next SDRAM access is issued immediately after negation of CS_n signal at the earliest. If the number of recovery cycles are not 0, the ACT command is issued 2 cycles after the specified recovery cycle period elapsed after negation of CS_n signal at the earliest. Because no data conflicts can occur during access to the SDRAM area, there is no need to set data recovery cycles for the SDRAM (fixed to 0 cycle).

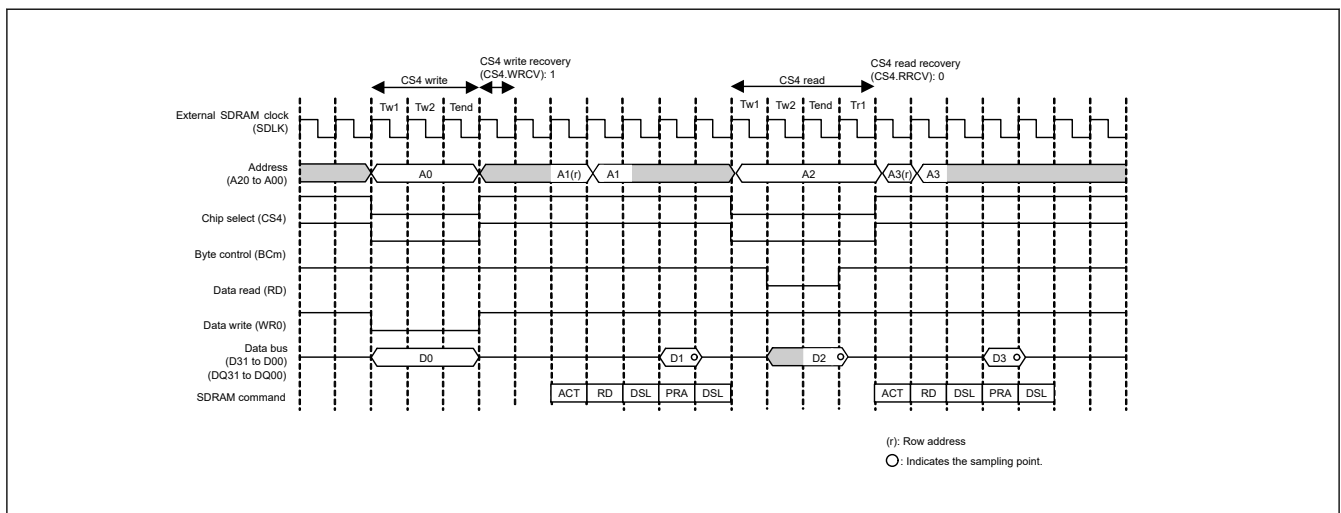


Figure 15.36 Example of recovery timing for SDRAM access

15.6.4 SDRAM Commands

To control the SDRAM, the SDRAMC issues a command for each bus cycle. Commands are defined by a combination of the SDCS, RAS, CAS, WE, CKE, and other signals. Table 15.32 lists the commands issued by the SDRAMC.

Table 15.32 SDRAMC commands

Name	Abbreviation	Command	SDCS	RAS	CAS	WE	CKE		BA1	BA0
							n-1	n		
DESL	DSL	Device deselect	H	x	x	x	H	x	x	x
ACTV	ACT	Bank active	L	L	H	H	H	x	V	V
READ	RD	Read	L	H	L	H	H	x	V	V
WRIT	WRI	Write	L	H	L	L	H	x	V	V
PRE	PRE	Precharge	L	L	H	L	H	x	V	V
PALL	PRA	All bank precharge	L	L	H	L	H	x	x	x
REF	RFA	Auto-refresh	L	L	L	H	H	x	x	x
MRS	MRS	Mode register set	L	L	L	L	H	x	L	L
SELF	RFS	Self-refresh entry	L	L	L	H	H	L	x	x
SELF	RFX	Self-refresh end	H	x	x	x	L	H	x	x

Note: H = High level, L = Low level, V = Valid, x = Don't care.

n = Command issue cycle, n - 1 = 1 cycle before the command is issued.

15.6.5 Conditions for Setting the SDRAMC Registers

The SDRAMC registers must only be modified when all the conditions shown in Table 15.33 are satisfied.

Table 15.33 Conditions for register modification

Function or operation	Registers	Conditions
Self-refresh	SDSELF ^{*1}	- SDRAM access is disabled (SDCCR.EXENB = 0^{*2}) - Auto-refresh operation is enabled (SDRFEN.RFEN = 1)
Auto-refresh	SDRFCR	Self-refresh operation is disabled (SDSELF.SFEN = 0)
	SDRFEN	- SDRAM access is disabled (SDCCR.EXENB = 0^{*2}) - Self-refresh operation is disabled (SDSELF.SFEN = 0)
Initialization sequence	SDIR ^{*1}	SDICR is not set yet, and the same conditions as for SDICR modification are satisfied
	SDICR ^{*1}	- Auto-refresh operation is disabled (SDRFEN.RFEN = 0) - Self-refresh operation is disabled (SDSELF.SFEN = 0)
Address register	SDADR	- SDRAM access is disabled (SDCCR.EXENB = 0^{*2}) - Auto-refresh operation is disabled (SDRFEN.RFEN = 0) - Self-refresh operation is disabled (SDSELF.SFEN = 0)
Timing register	SDTR	- Self-refresh operation is in progress (SDSELF.SFEN = 1) or - SDRAM access is disabled (SDCCR.EXENB = 0^{*2}) - Auto-refresh operation is disabled (SDRFEN.RFEN = 0) - Self-refresh operation is disabled (SDSELF.SFEN = 0)
Mode register	SDMOD ^{*1}	- SDRAM access is disabled (SDCCR.EXENB = 0^{*2}) - Self-refresh operation is disabled (SDSELF.SFEN = 0)
Access Mode register	SDAMOD	- SDRAM access is disabled (SDCCR.EXENB = 0^{*2}) - Auto-refresh operation is disabled (SDRFEN.RFEN = 0) - Self-refresh operation is disabled (SDSELF.SFEN = 0)

Note 1. Before modifying this register, confirm that all the status bits in SDSR are 0.

Note 2. After writing 0 to the EXENB bit, confirm that it is cleared to 0.

15.6.6 Self-Refresh

Transition to or recovery from self-refresh mode is controlled with the SDRAM Self-Refresh Control Register (SDSELF). Immediately before the transition to self-refresh mode, an auto-refresh operation is performed. In self-refresh mode, the CKE signal is low. Immediately after recovery from self-refresh mode, the auto-refresh cycle starts.

Figure 15.37 and Figure 15.38 show timing examples of the transition to and recovery from self-refresh mode.

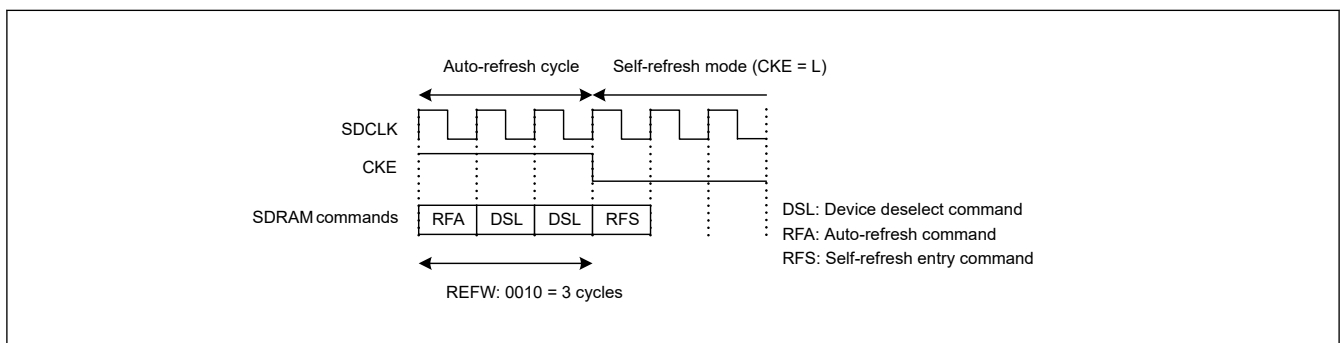


Figure 15.37 Example timing for transition to self-refresh mode when SDRFCR.REFW[3:0] = 0010 (3 cycles)

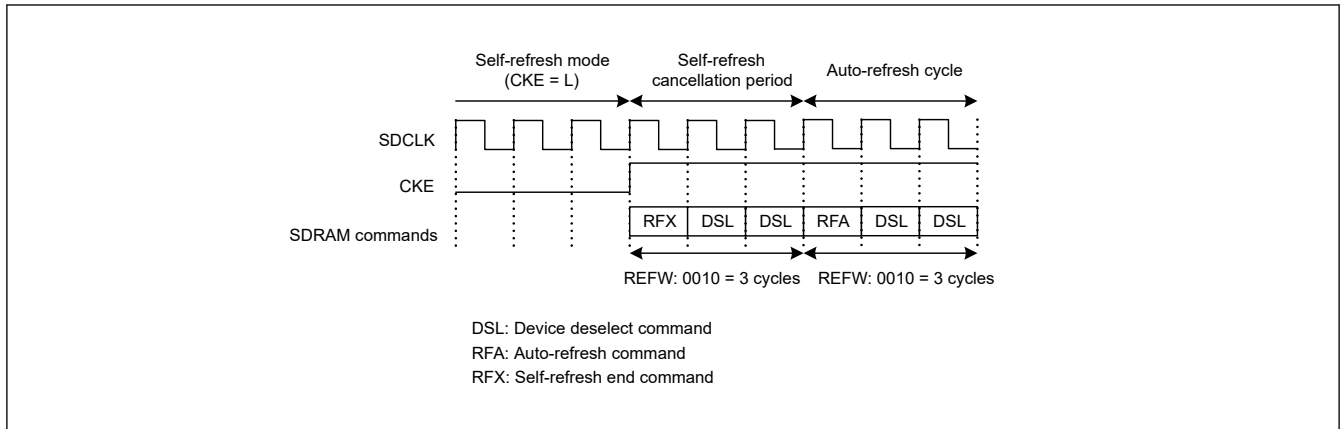


Figure 15.38 Example timing for recovery from self-refresh mode

(1) Self-refresh in Software Standby mode

When invoking self-refresh in Software Standby mode, first follow the procedure shown in [section 15.6.11.2. Procedure for Transitioning to and Recovering from Self-refresh Mode](#). Next, set up the transition to Software Standby mode. In this mode, set the Output Port Enable bit (OPE) in the Standby Control Register (SBYCR) to 1 to hold the output state of the address bus and bus control signals.

After canceling Software Standby mode, follow the procedure shown in [section 15.6.11.2. Procedure for Transitioning to and Recovering from Self-refresh Mode](#). For details on invoking and canceling Software Standby mode, see [section 11, Low Power Mode](#).

(2) Self-refresh in Deep Software Standby mode

Deep Software Standby mode is invoked from within Software Standby mode. On this transition, the pin states remain unchanged. Therefore, invoking of self-refresh in Deep Software Standby mode can be handled the same as for Software Standby mode with one additional setting. You must also set the I/O Port Keep bit (IOKEEP) in the Deep Software Standby mode Control Register (DPSBYCR) to 1.

Because the SDRAMC is reset internally when Deep Software Standby mode is canceled, the SDRAM control registers must be set again. After canceling Software Standby mode, follow the procedure in this section to cancel self-refresh.

[Figure 15.39](#) shows self-refresh timing in Deep Software Standby mode. For details on invoking and canceling Deep Software Standby mode, see [section 11, Low Power Mode](#).

To cancel self-refresh mode:

1. Set DPSBYCR.IOKEEP to 1 to keep the CKE signal output low in Deep Software Standby mode.
2. Start the clock supply to the SDRAMC.
3. Set the SDRAM control registers (SDCMOD, SDADR, and SDTR) again. These registers were initialized by an internal reset on entering Deep Software Standby mode.
4. Enable an auto-refresh operation by setting SDRFEN.RFEN to 1.
5. Check that all the status bits in SDSR are cleared to 0 and set SDSELF.SFEN to 1 to select self-refresh mode again.
6. Modify the port settings for the SDRAM interface.
7. Set SDCKOCR.SDCKOEN to 1 to start the clock supply to the SDRAM with the SDCLK pin.
8. Check that all the status bits in SDSR are cleared to 0 and set SDSELF.SFEN to 0 to cancel self-refresh mode.

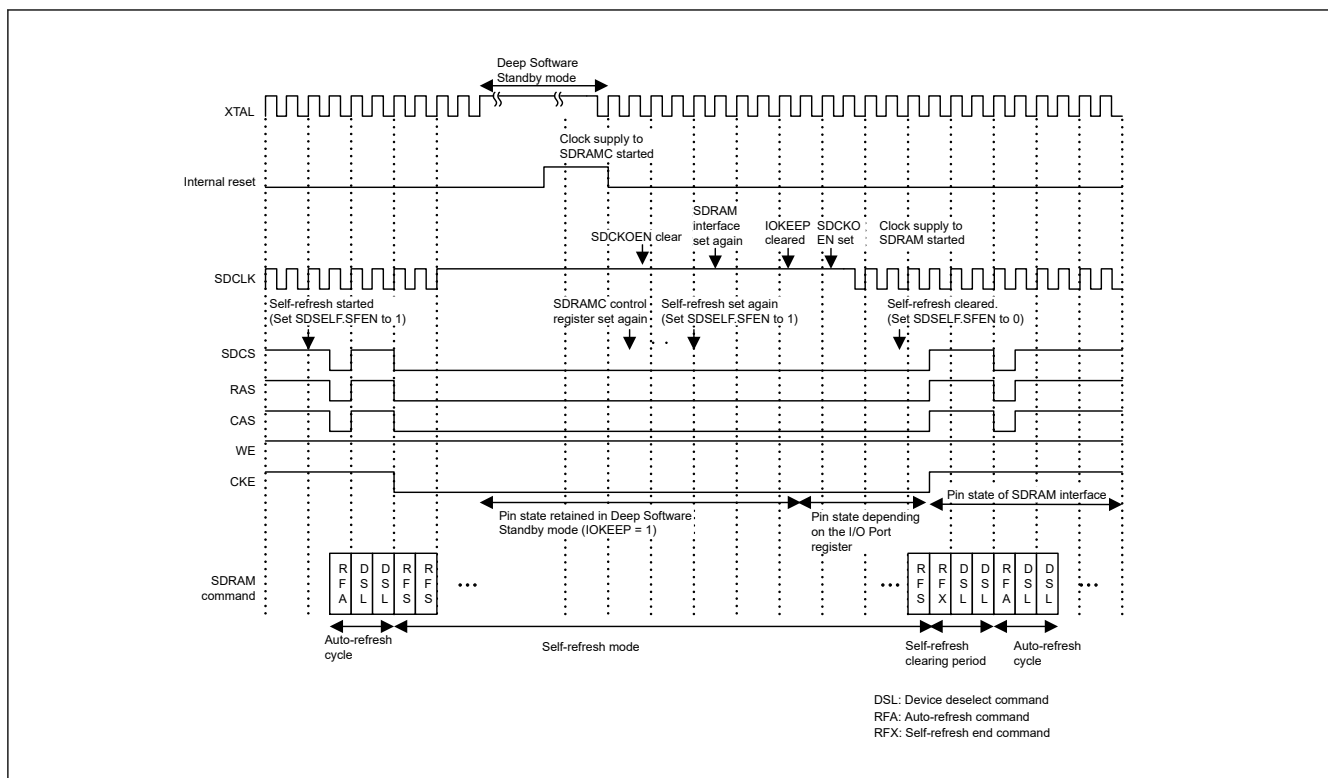


Figure 15.39 Example timing for self-refresh cycle in Deep Software Standby mode

15.6.7 Auto-Refresh

The auto-refresh cycle can be started by setting the Auto-Refresh Operation Enable bit (RFEN) in the SDRAM Auto-Refresh Control Register (SDRFEN) to 1. After the cycle starts, refresh requests are generated at fixed intervals determined by the refresh counter. However, because refresh requests are not accepted during read or write access, the auto-refresh cycle might be suspended.

If an SDRAM access and a refresh request are generated at the same time, the refresh request takes precedence. A CS area access and a refresh request can be made at the same if the SDCS, RAS, CAS, WE, and CKE signals, which are required for issuing the refresh command, are exclusively provided for SDRAM access.

The refresh counter is halted during a self-refresh operation. After recovery from the self-refresh mode, the auto-refresh cycle starts and the counter value is reset, resuming the counter operation.

Figure 15.40 shows a timing example of an auto-refresh cycle.

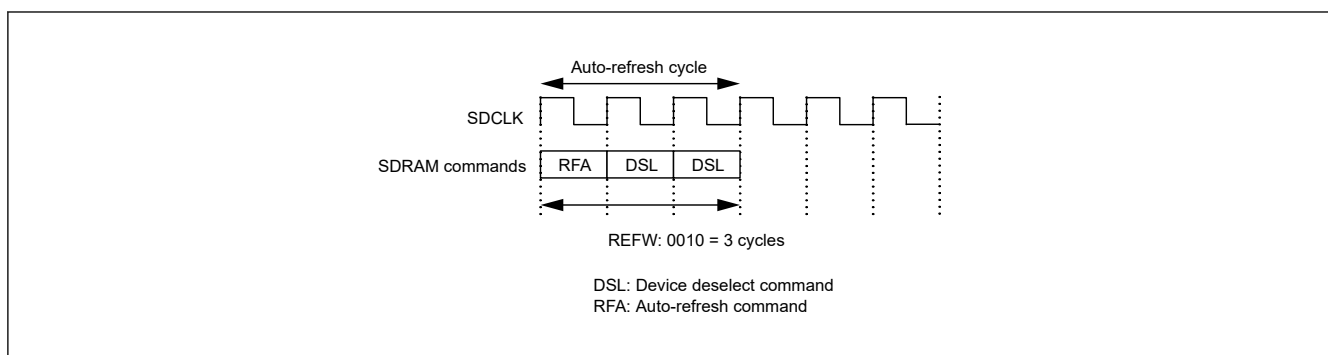


Figure 15.40 Example timing for auto-refresh cycle (1)

Figure 15.41 and Figure 15.42 show examples of operation when an auto-refresh request is generated during single access and continuous access, respectively.

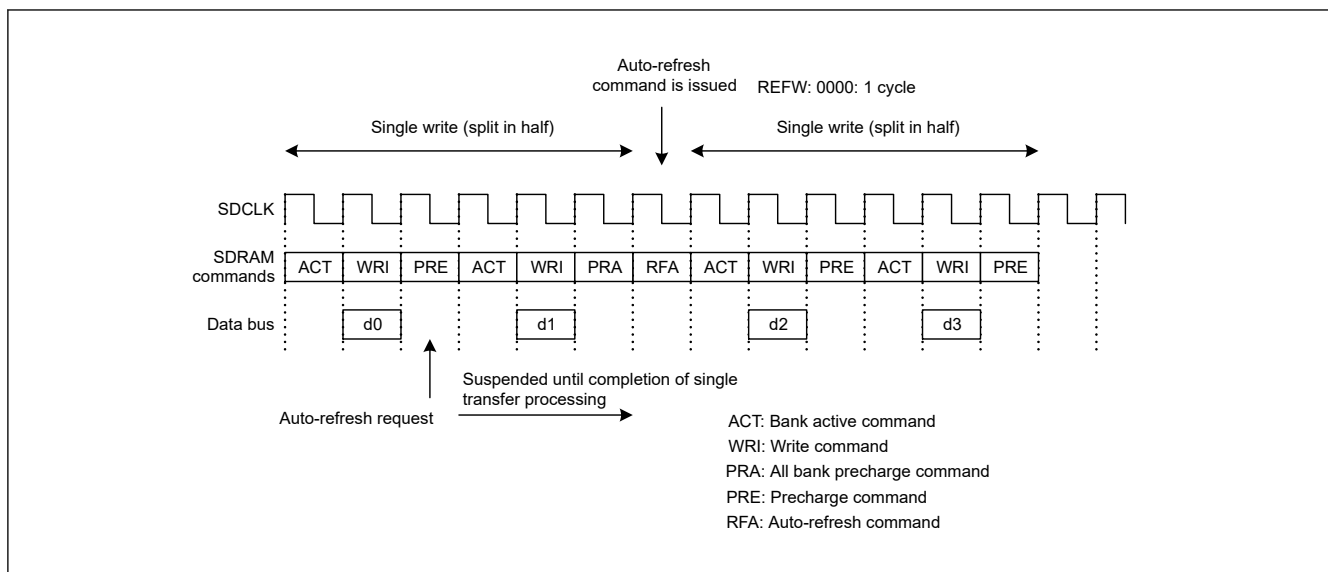


Figure 15.41 Example timing for auto-refresh cycle (2), when the auto-refresh request is made during single access

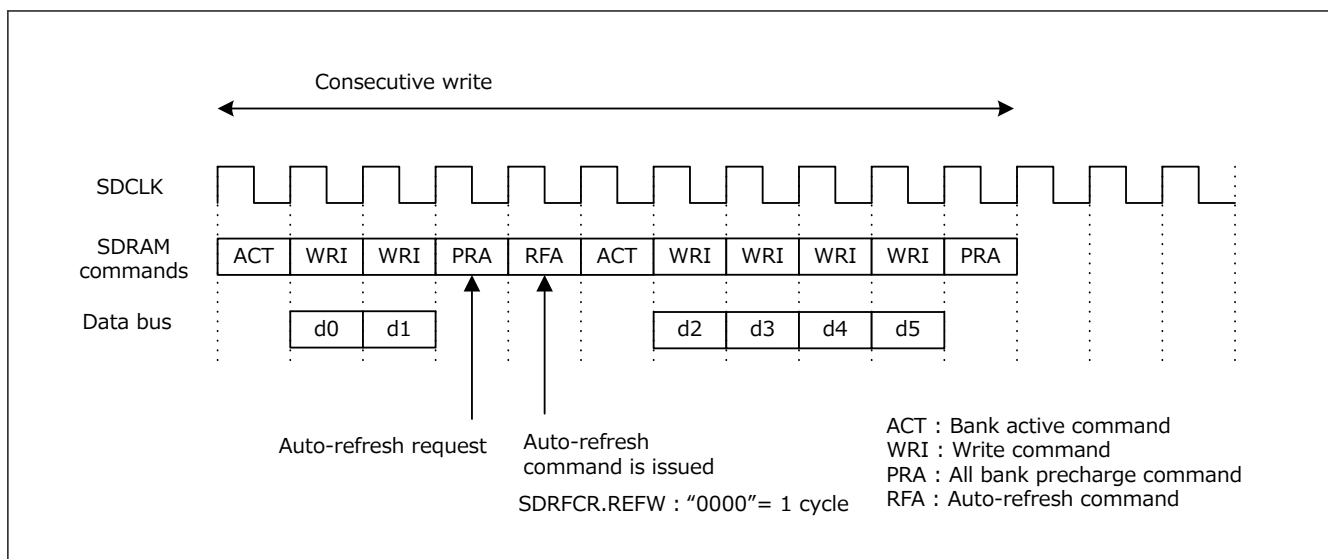


Figure 15.42 Example timing for auto-refresh Cycle (3) when auto-refresh request is made during continuous access

15.6.8 Initialization Sequencer

The SDRAMC has a sequencer to issue SDRAM initialization commands. After a reset, the initialization sequencer must be activated without fail. Operation is not guaranteed if the SDRAM is not initialized.

The SDRAM initialization sequencer issues an all-bank precharge command followed by auto-refresh commands n times, where $n = 1$ to 15. The SDRAM initialization sequence timing can be set using the SDRAM Initialization Register (SDIR). The SDRAM initialization sequence can be activated using the SDRAM Initialization Sequence Control Register (SDICR). These registers must be set only when the conditions listed in Table 15.33 are satisfied.

Figure 15.43 shows a timing example of the SDRAM initialization sequence. When the ARFC[3:0] bits in SDIR are set so that auto-refresh operation is performed two or more times, auto-refresh cycles are repeated in the initialization sequence accordingly.

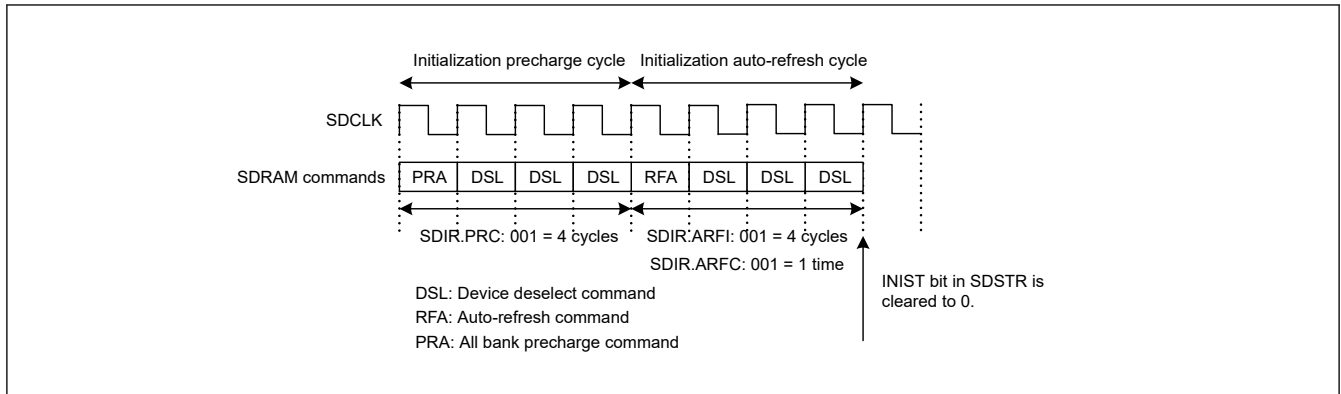


Figure 15.43 Example timing for SDRAM initialization sequence

15.6.9 Setting the Mode Register

Setting the SDRAM Mode Register (SDMOD) allows the mode register set command to be issued to the SDRAM and the value set in the MR[14:0] bits in SDMOD to be output to the lower bits of the address, specifically to A14 to A00 for 8-bit bus width, A15 to A01 for 16-bit bus width or A16 to A02 for 32-bit bus width. Before setting the mode register, set the SDRAM Bus Width Select bits in the SDC Control Register (SDCCR.BSIZE[1:0]) to determine the data bus width of the SDRAM.

Figure 15.44 shows the mode register setting timing.

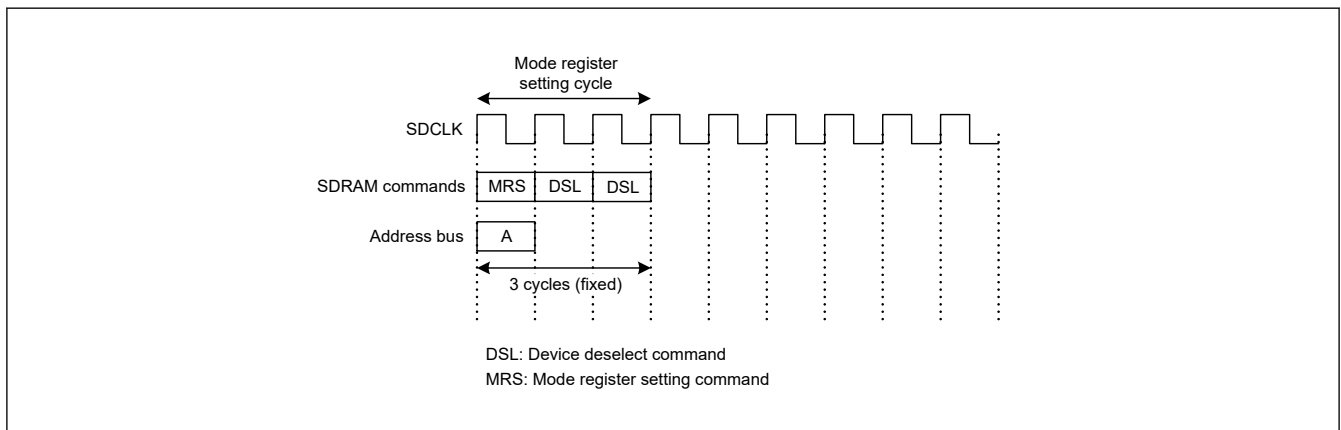


Figure 15.44 Mode register setting timing

15.6.10 Read/Write Access

There are two types of SDRAM read / write access:

- **Single access**
The row address is output each time the data is accessed to access the data.
- **Consecutive access**
When the same row address is accessed consecutively, after the row address is output, the data can be accessed at high speed simply by changing the column address.

When the continuous access enable bit (SDAMOD.BE) of the SDRAM access mode register is 1, consecutive access operation of SDRAM is enabled.

If the transfer from the bus master has a burst length of 2 or more and the transfer data size is less than or equal to the external bus width, consecutive access will occur.

(1) Single access

Figure 15.45 shows timing examples for single read and single write. The access timing changes depending on the setting of the SDRAM timing register (SDTR). For details, see [section 15.6.11.3. Timing Register Settings and Access Timing](#).

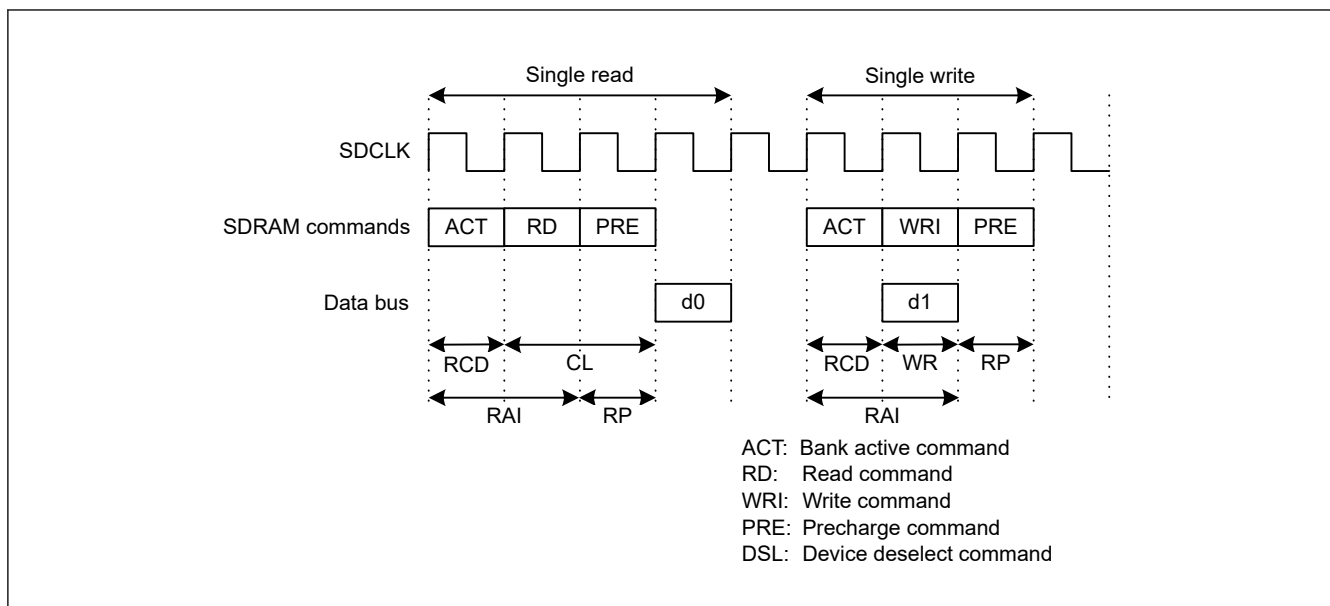


Figure 15.45 Example timing for single read/write

(2) Consecutive access

Figure 15.46 shows an example of the timing when performing consecutive read and continuous write of 4 data.

In addition, if the row address of SDRAM changes during transfer, the row is automatically deactivated and activated.

The access timing changes depending on the setting of the SDRAM timing register (SDTR). For details, see [section 15.6.11.3. Timing Register Settings and Access Timing](#).

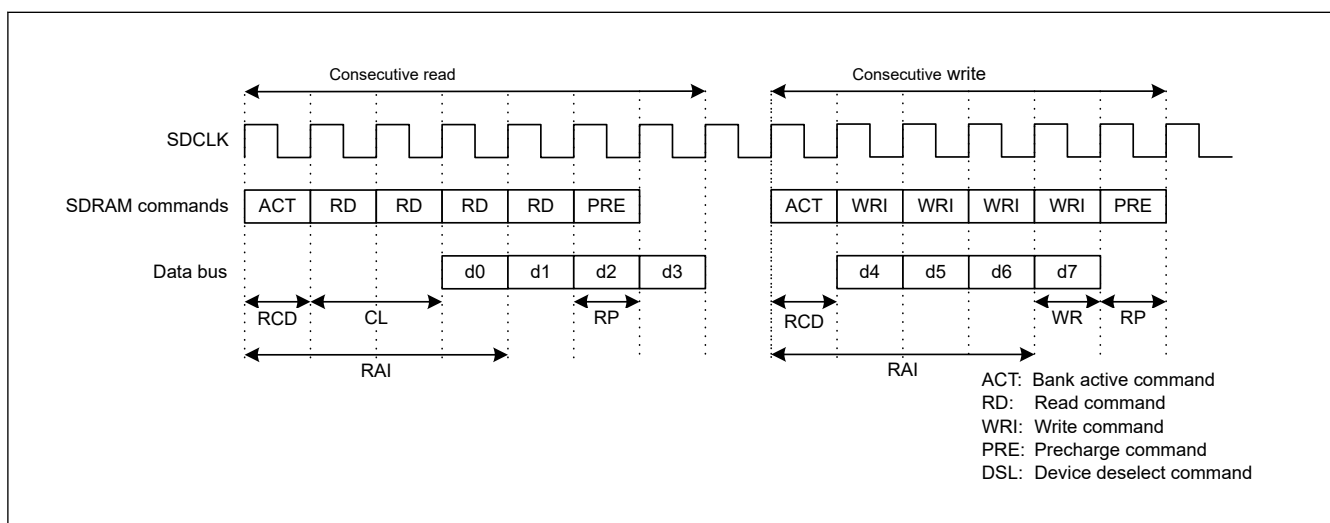


Figure 15.46 Example timing for Consecutive read/write

Figure 15.47 shows an example of the timing when the row address changes during consecutive writing.

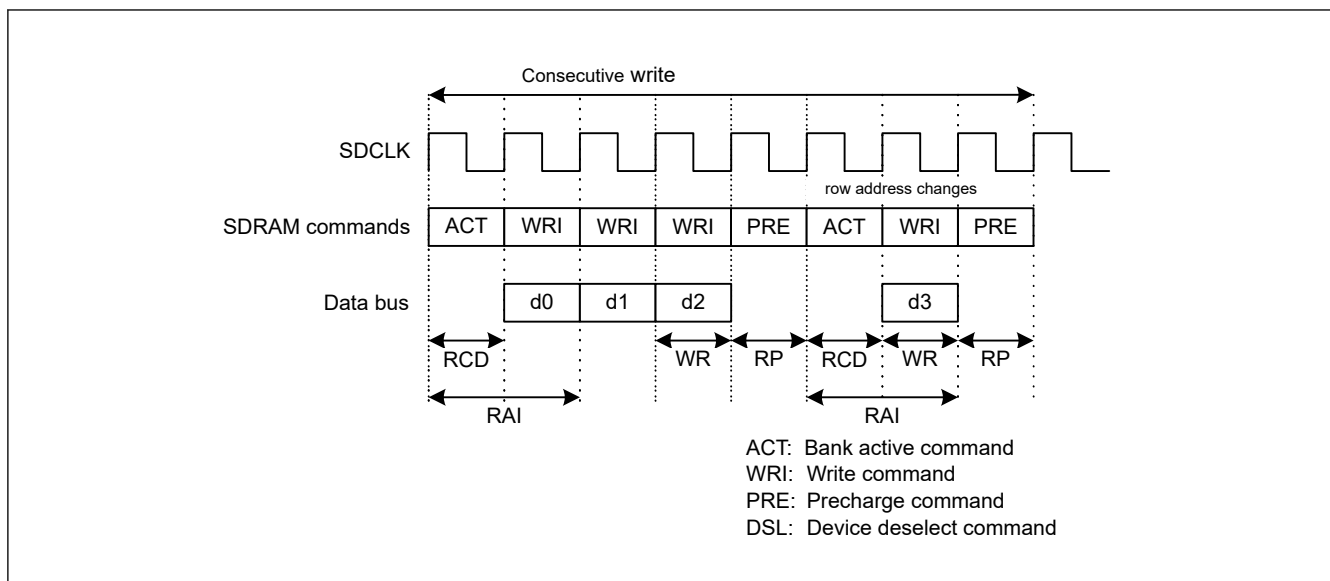


Figure 15.47 Example timing for Consecutive write when row address changes

Note: When consecutive burst write accesses are issued from bus masters to SDRAM area, a redundant write access may occur if the following conditions are all satisfied, the last write access to SDRAM in the preceding burst access is repeated with the same address and the same write data.

- The preceding burst write and the following burst write have the same transfer size, and the transfer size is 8, 16, or 32 bits.
- The address of the subsequent burst write is the last address of the preceding burst write plus the transfer size.
- The burst length of the preceding burst write is 2 or more and that of the subsequent burst write is 1.

15.6.11 SDRAMC Setting Examples

This section describes the following:

- SDRAMC setting procedure
- Timing register setting examples
- Procedure for transitioning to and recovering from self-refresh mode.

15.6.11.1 SDRAMC Access Procedure

[Figure 15.48](#) shows the SDRAMC setting procedure.

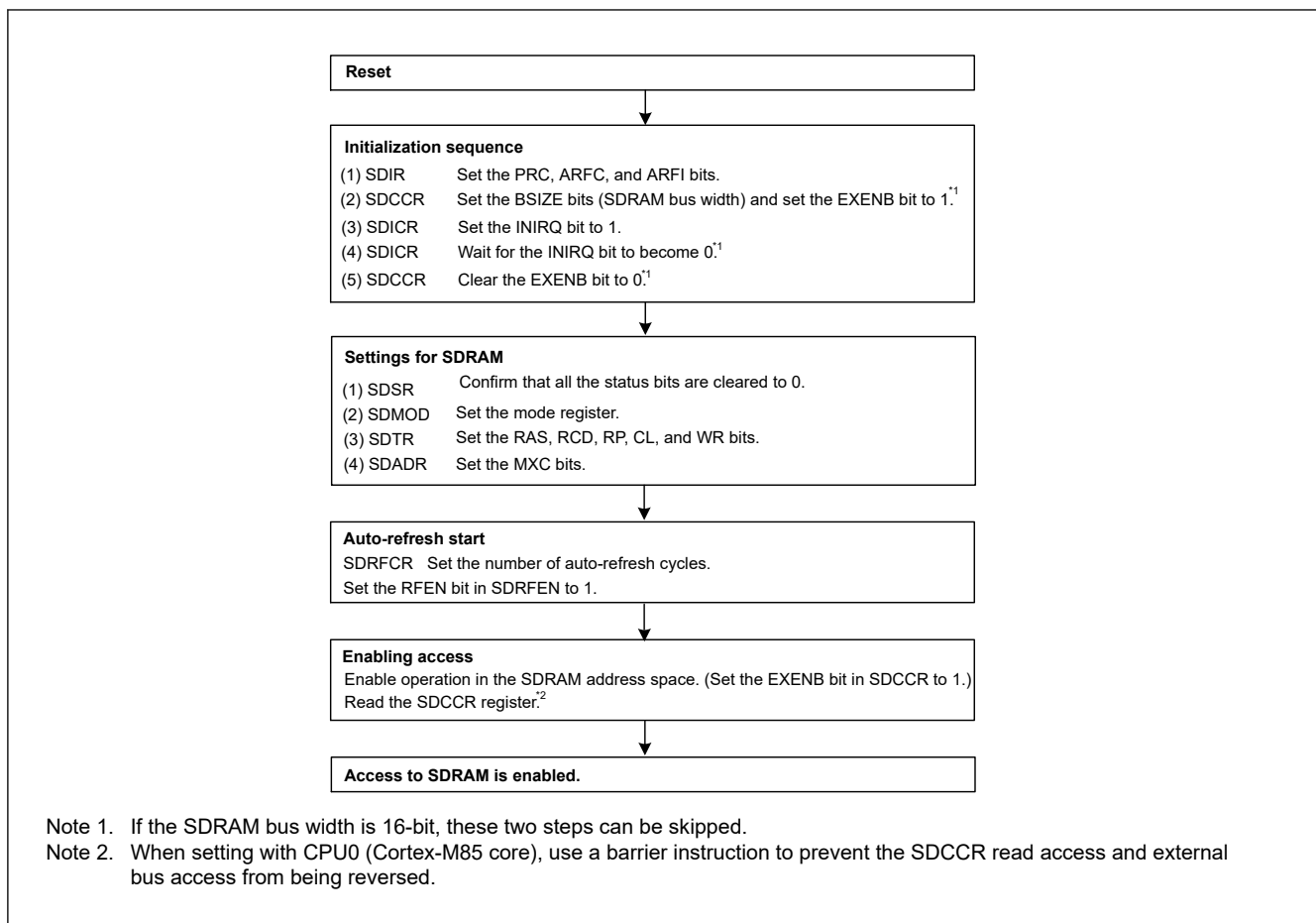


Figure 15.48 SDRAMC setting procedure

15.6.11.2 Procedure for Transitioning to and Recovering from Self-refresh Mode

Figure 15.49 shows the procedure for transitioning to and recovering from self-refresh mode.

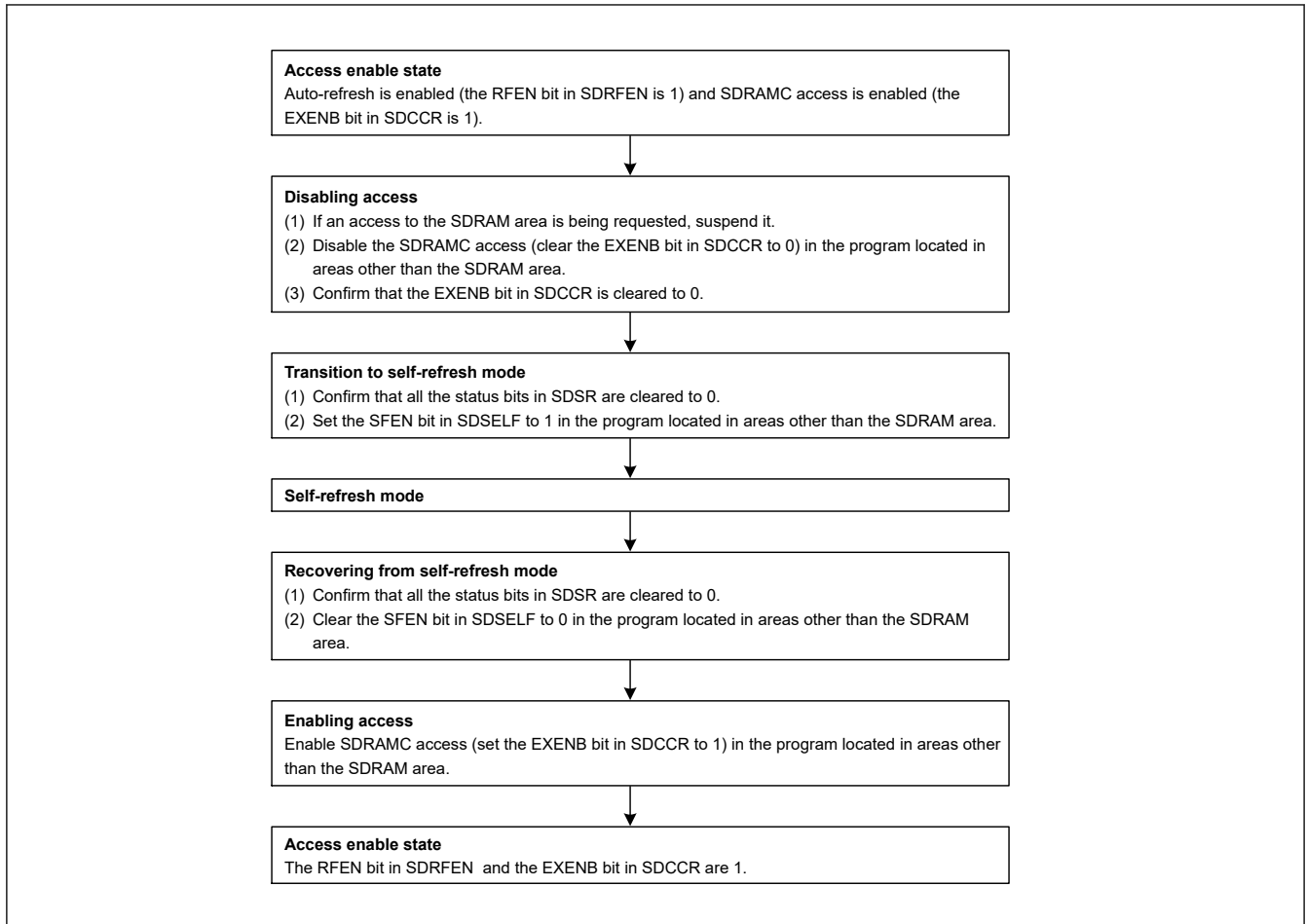


Figure 15.49 Procedure for transitioning to and recovering from self-refresh mode

Note: Self-refresh mode cannot be invoked during SDRAM access. SDRAM access must be disabled during both transition to and recovery from self-refresh mode. Follow the programming instructions shown in [Figure 15.50](#).

Before transitioning to self-refresh mode, disable access to the SDRAM area.

During transition to self-refresh mode, self-refresh operation, and recovery from self-refresh mode, do not allow any operand access or instruction fetch, including prefetch to the SDRAM area, to be generated.

[Figure 15.50](#) shows the procedure for transitioning to and recovering from self-refresh mode in Deep Software Standby mode.

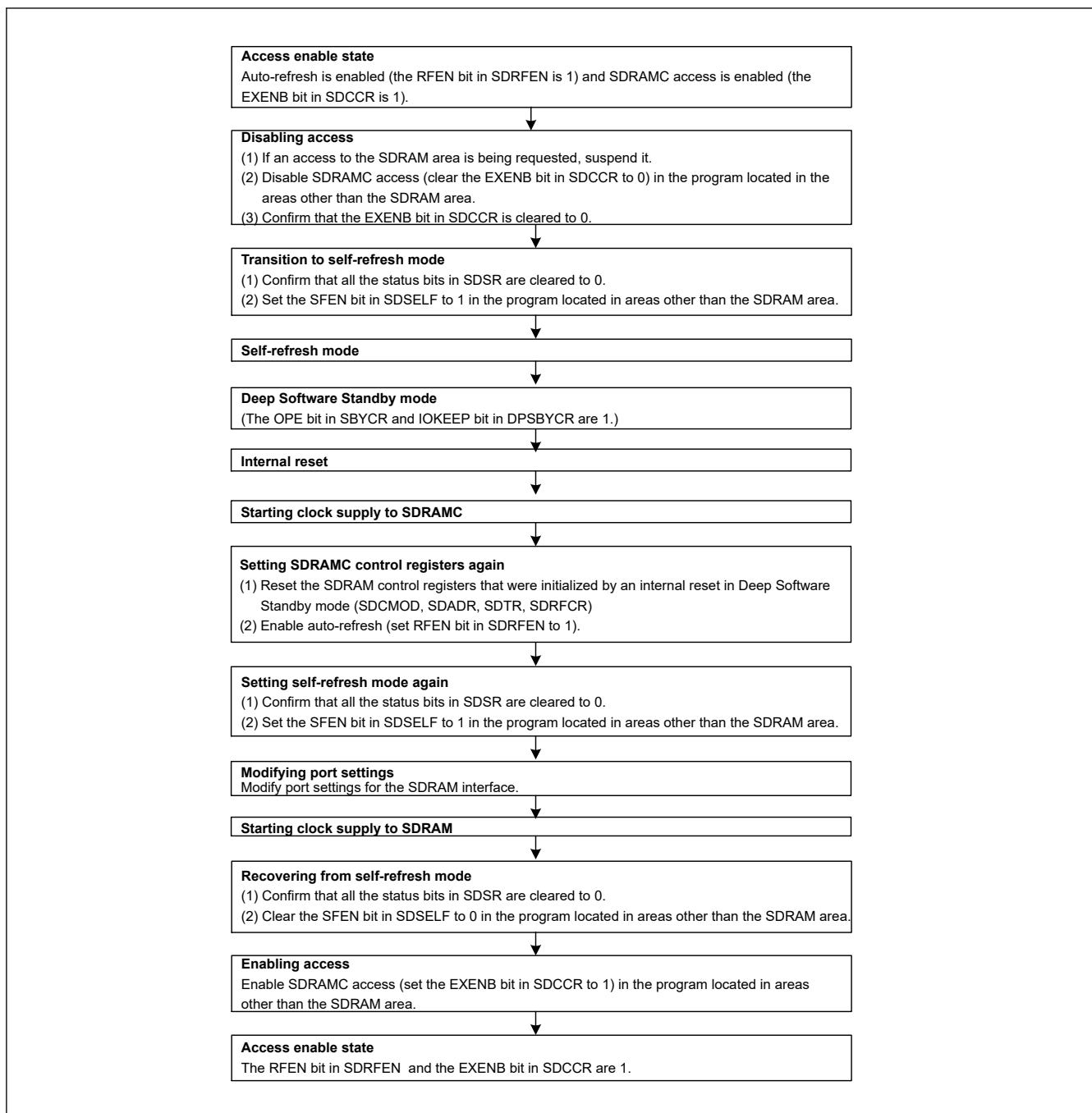


Figure 15.50 Procedure for transitioning to and recovering from self-refresh mode in Deep Software Standby mode

15.6.11.3 Timing Register Settings and Access Timing

This section describes the relationship between read and write timing and the settings in the SDRAM Timing Register (SDTR).

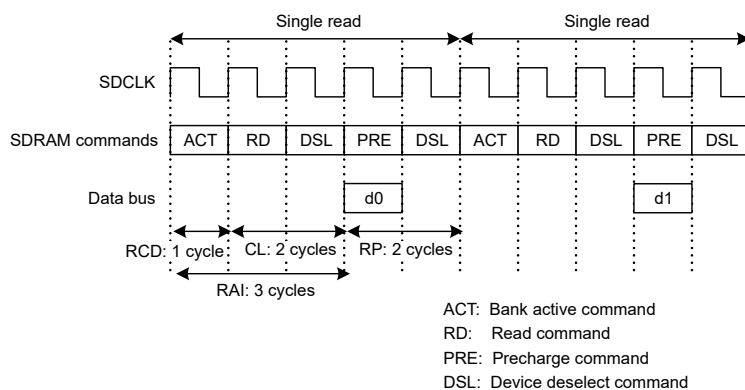
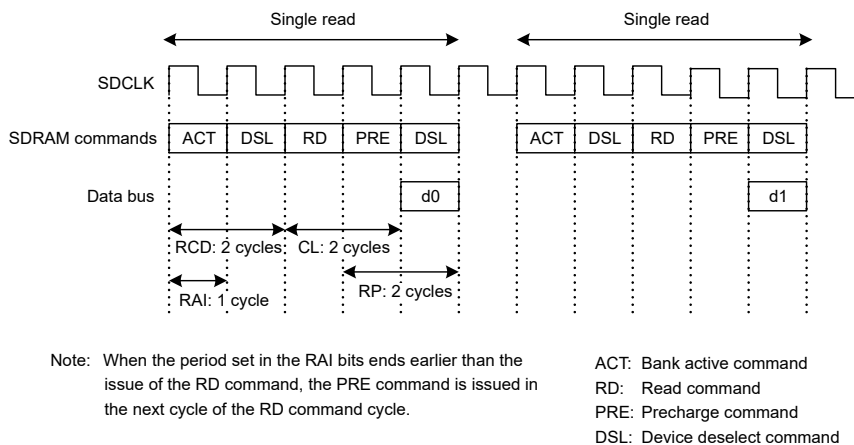
(1) Single read timing examples

Figure 15.51 to Figure 15.55 show the relationship between single read timing and the SDTR register settings. Table 15.34 shows the correspondence between the figures and the SDTR register settings.

During read access, the next bus access is enabled at the earliest 2 cycles after the read data becomes valid. However, if two or more accesses occur for one transfer request, the next bus access is enabled at the earliest 1 cycle after the read data becomes valid, as shown in Figure 15.55.

Table 15.34 Correspondence between timing figures and STDR register settings for single read timing

Figure number	RAI[2:0] settings	Number of cycles	RCD[1:0] settings	Number of cycles	RP[2:0] settings	Number of cycles	CL[2:0] settings	Number of cycles
Figure 15.51	010	3	00	1	001	2	010	2
Figure 15.52	000	1	01	2	001	2	010	2
Figure 15.53	000	1	01	2	001	2	011	3
Figure 15.54 Figure 15.55	010	3	00	1	000	1	010	2

**Figure 15.51 Example timing for single read (1)****Figure 15.52 Example timing for single read (2)**

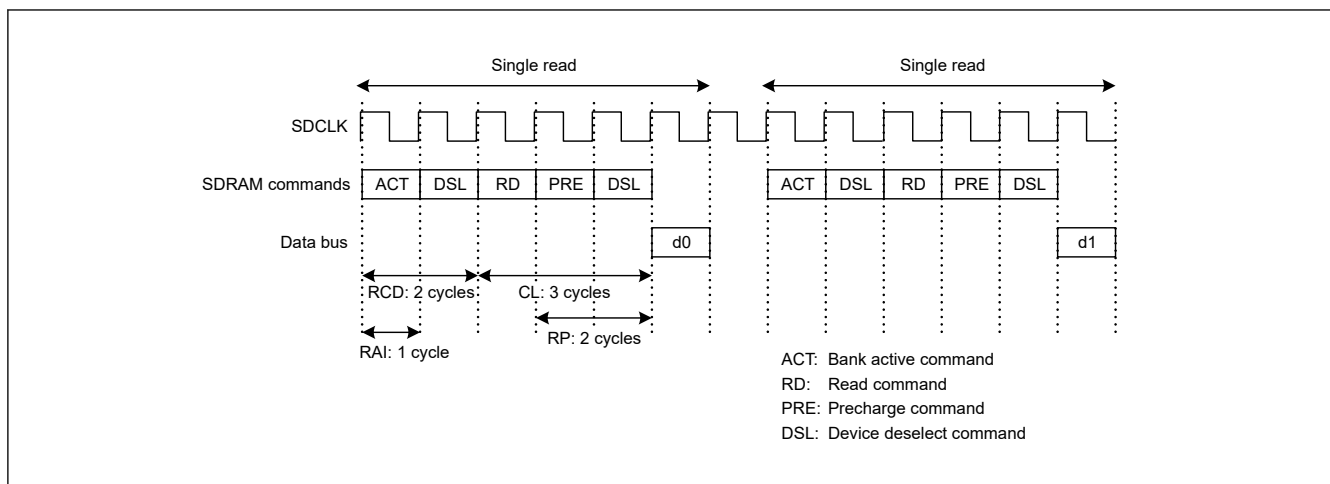


Figure 15.53 Example timing for single read (3)

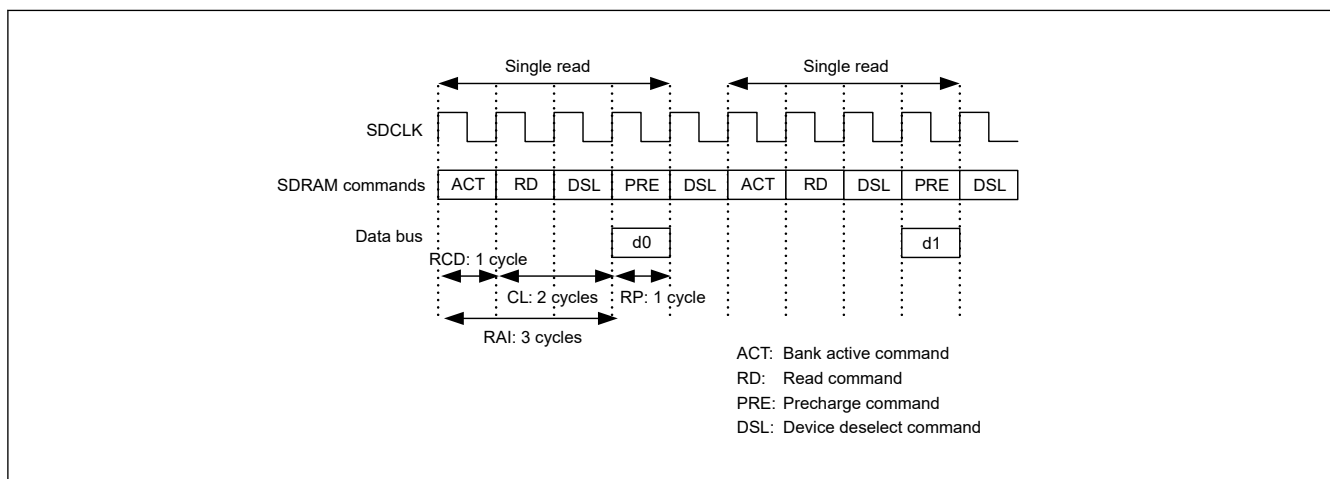


Figure 15.54 Example timing for single read (4)

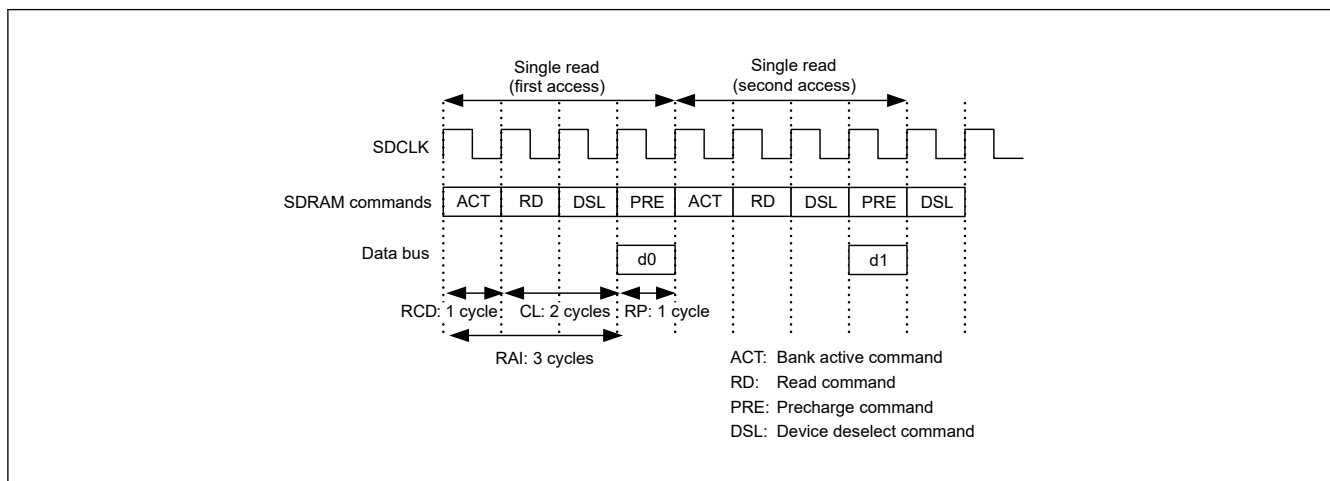


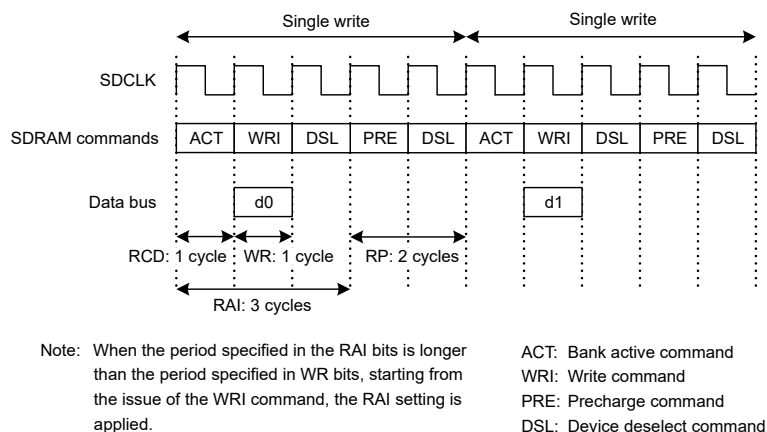
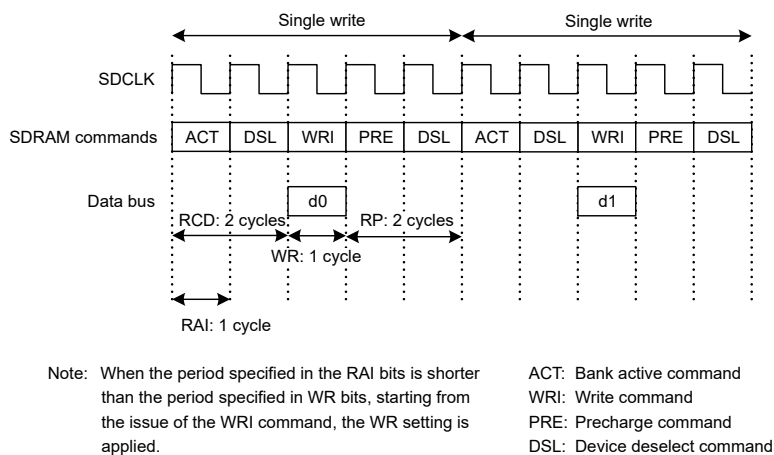
Figure 15.55 Example timing for single read (5), when two bus accesses occur for one transfer request

(2) Single write timing examples

Figure 15.56 to Figure 15.60 show the relationship between the single write timing and the SDTR register settings. Table 15.35 shows the association between the figures and the SDTR register settings. During write access, the next bus access is enabled at the earliest 2 cycles after a precharge command (PRE) is issued. However, if two or more accesses occur for one transfer request, the next bus access is enabled at the earliest 1 cycle after the PRE is issued, as shown in Figure 15.60.

Table 15.35 Association between timing figures and STDR register settings for single write timing

Figure number	RAI[2:0] settings	Number of cycles	RCD[1:0] settings	Number of cycles	RP[2:0] settings	Number of cycles	WR settings	Number of cycles
Figure 15.56	010	3	00	1	001	2	0	1
Figure 15.57	000	1	01	2	001	2	0	1
Figure 15.58	000	1	01	2	001	2	1	2
Figure 15.59 Figure 15.60	010	3	00	0	000	2	0	1

**Figure 15.56 Example timing for single write (1)****Figure 15.57 Example timing for single write (2)**

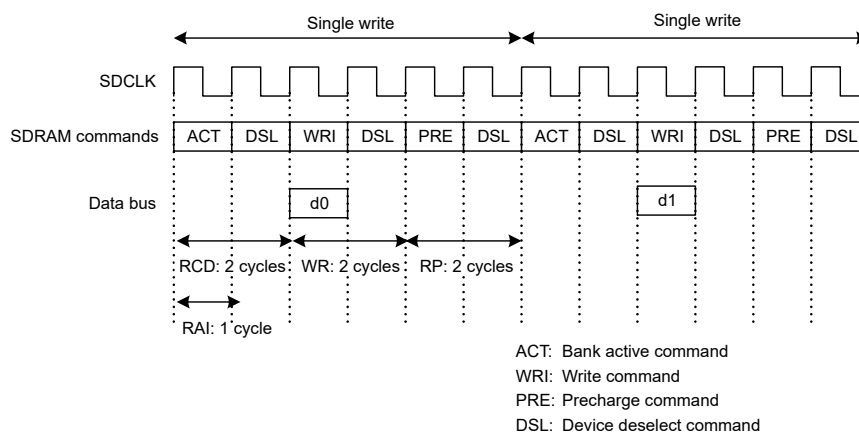


Figure 15.58 Example timing for single write (3)

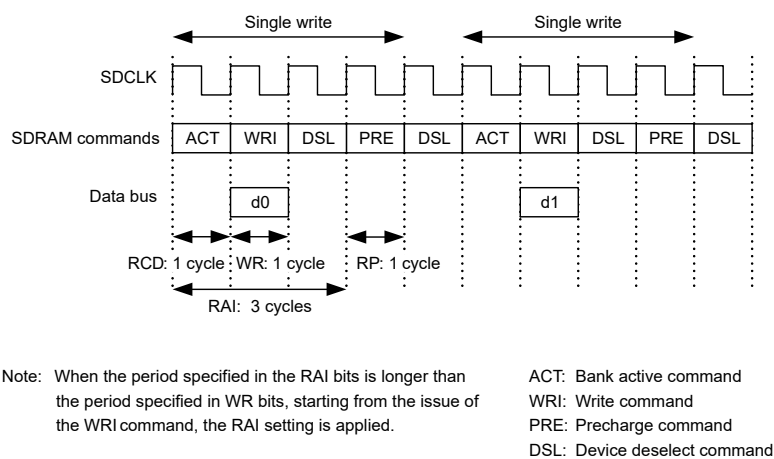


Figure 15.59 Example timing for single write (4)

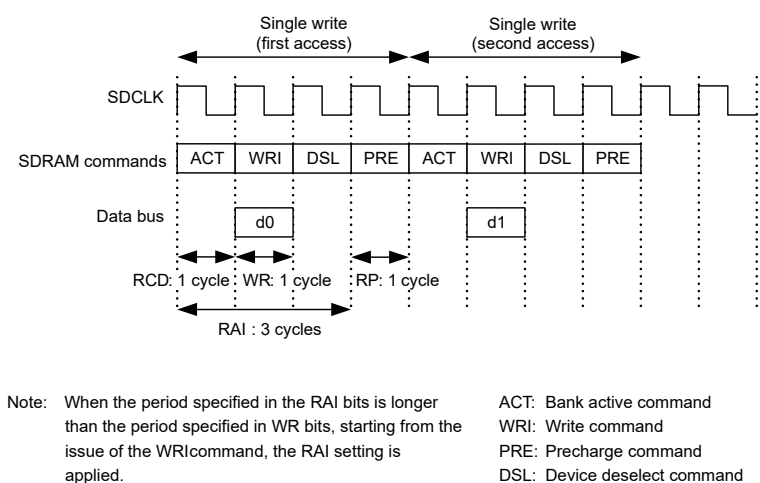


Figure 15.60 Example timing for single write (5), when two bus accesses occur for one transfer request

(3) Consecutive Read Timing Examples

Figure 15.61 to Figure 15.63 show the relationship between the consecutive read timing for four data and the SDTR register settings. Table 15.36 shows the correspondence between the figures and the SDTR register settings.

Table 15.36 Correspondence between Target Figures and STDR Register Settings (Consecutive Read Timing)

Figure No.	RAI[2:0]	Number of Cycle	RCD[1:0]	Number of Cycle	RP[2:0]	Number of Cycle	CL[2:0]	Number of Cycle
	Settings		Settings		Settings		Settings	
Figure 15.61	010	3	00	1	001	2	010	2
Figure 15.62	000	1	01	2	001	2	010	2
Figure 15.63	000	1	01	2	001	2	011	3

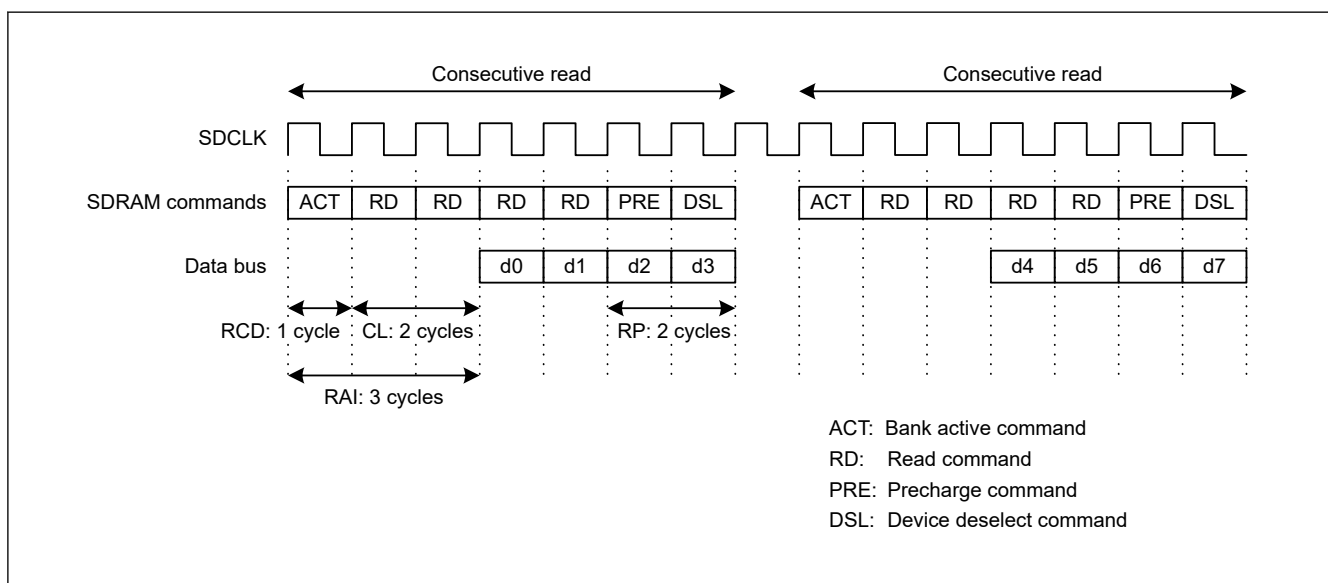


Figure 15.61 Timing example of consecutive read (1)

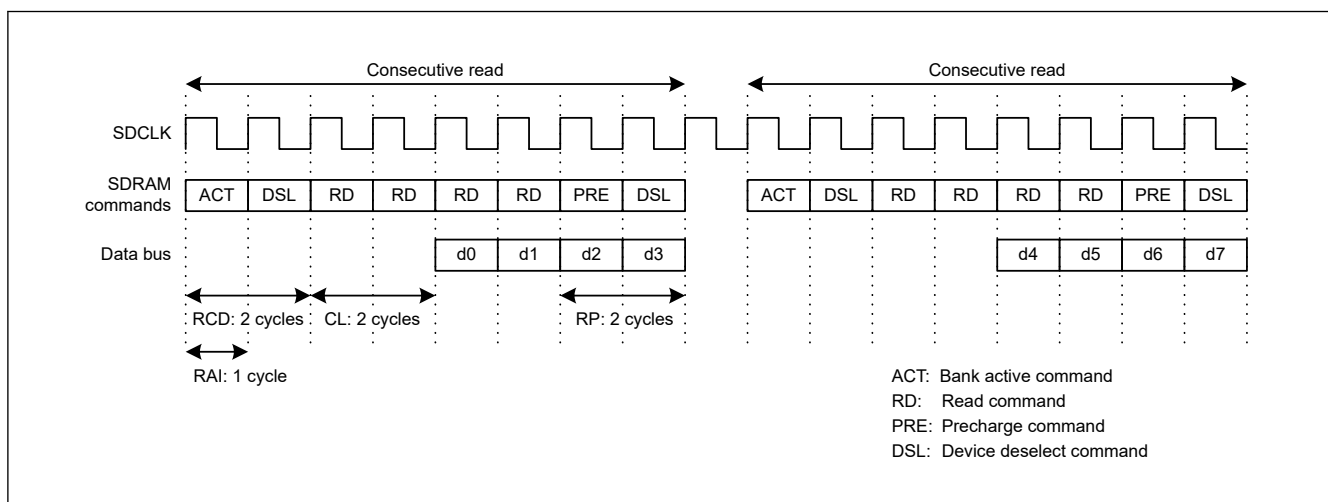


Figure 15.62 Timing example of consecutive read (2)

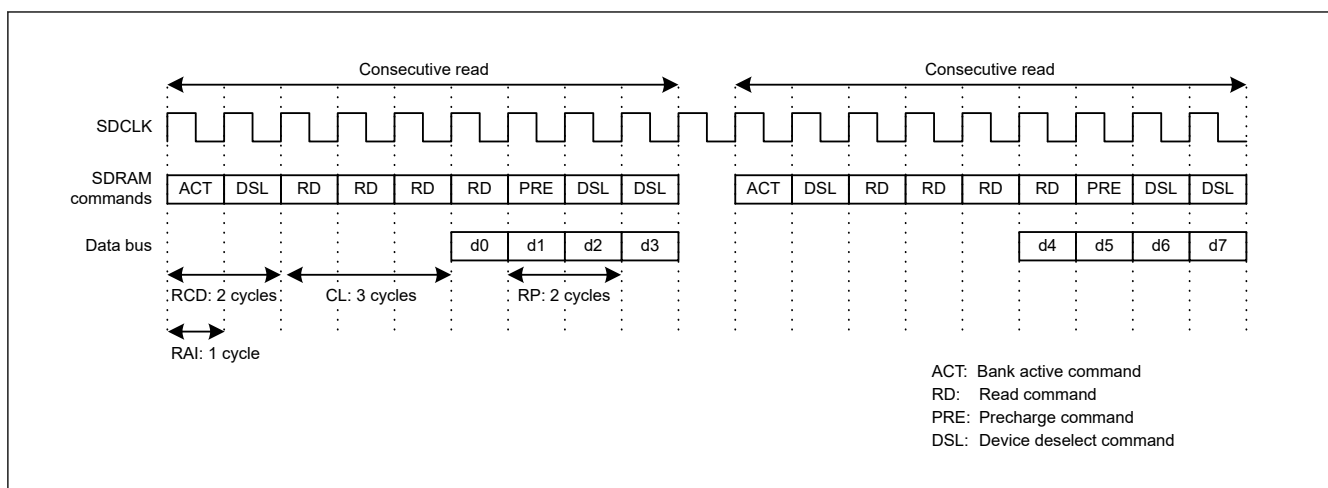


Figure 15.63 Timing example of consecutive read (3)

(4) Consecutive Write Timing Examples

Figure 15.64 to Figure 15.66 show the relationship between the consecutive write timing for four data and the SDTR register settings. Table 15.37 shows the correspondence between the figures and the SDTR register settings.

Table 15.37 Correspondence between Target Figures and STDR Register Settings (Consecutive Write Timing)

Figure No.	RAI[2:0] Settings	Number of Cycles	RCD[1:0] Settings	Number of Cycles	RP[2:0] Settings	Number of Cycles	WR Settings	Number of Cycles
Figure 15.64	010	3	00	1	001	2	0	1
Figure 15.65	000	1	01	2	001	2	0	1
Figure 15.66	000	1	01	2	001	2	1	2

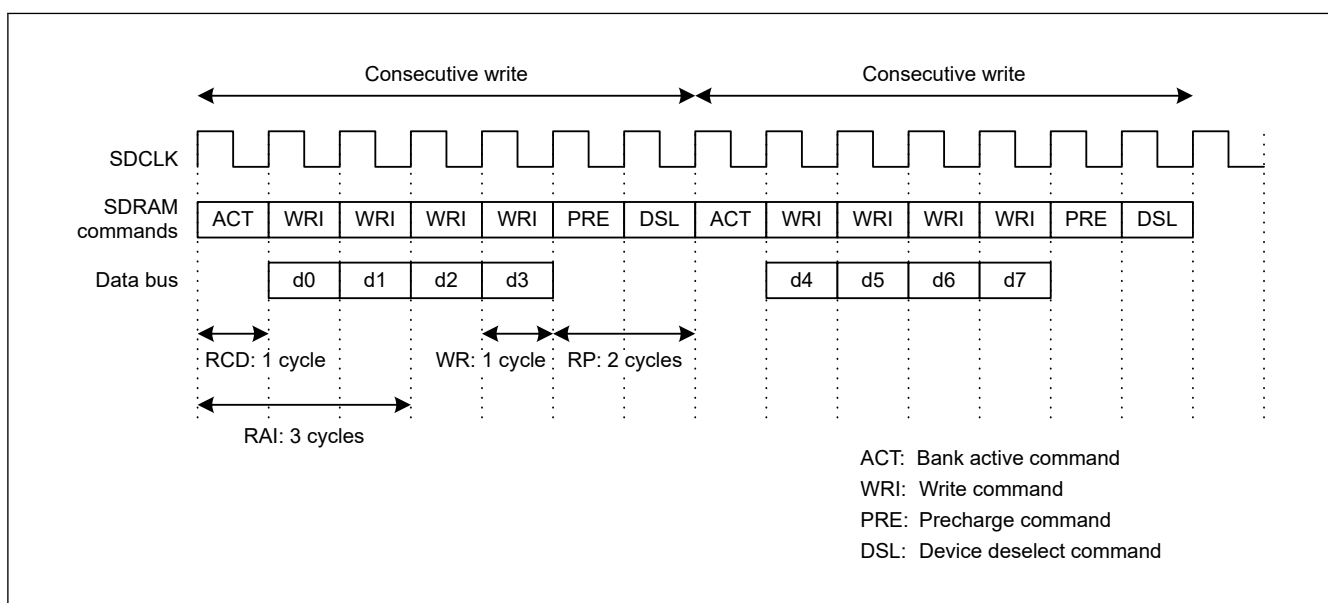


Figure 15.64 Timing example of consecutive write (1)

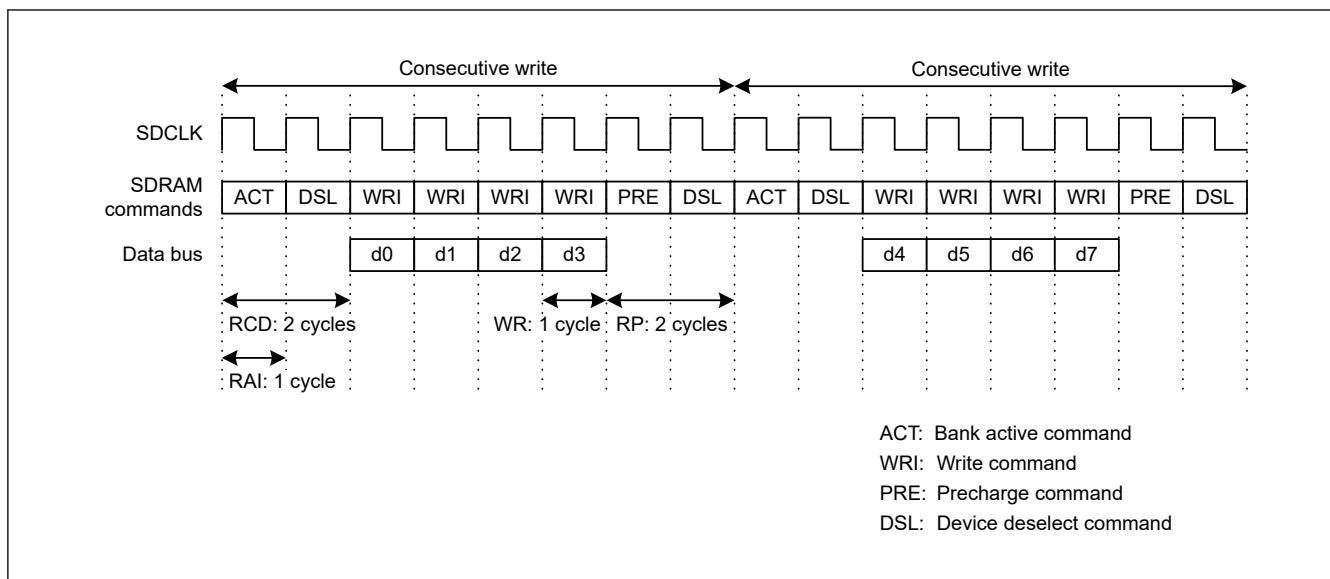


Figure 15.65 Timing example of consecutive write (2)

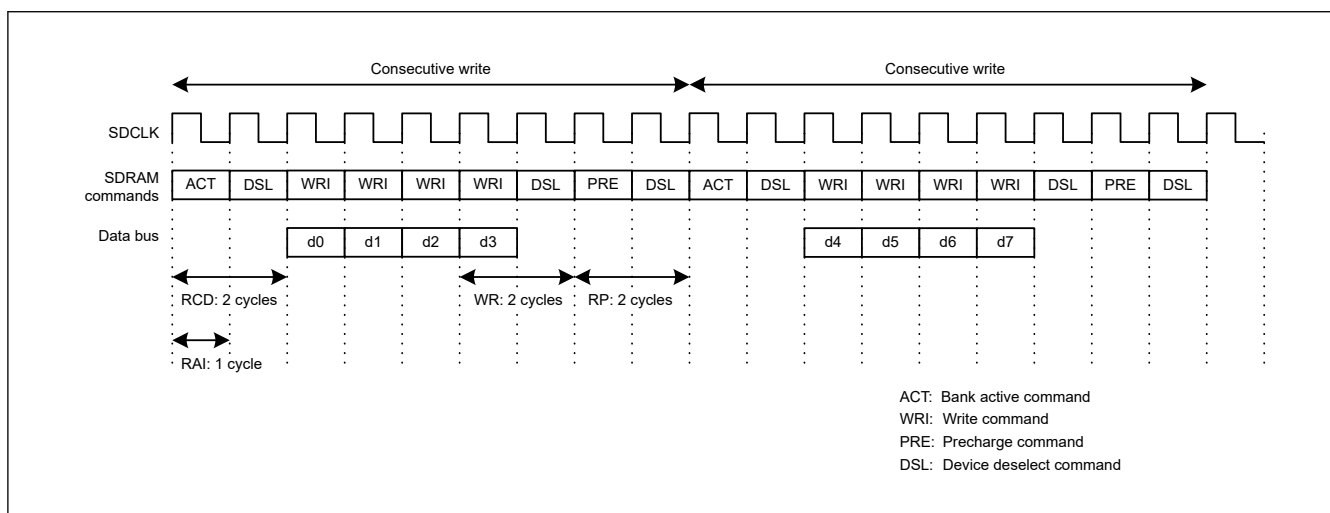


Figure 15.66 Timing example of consecutive write (3)

15.6.12 Address Multiplexing

In the SDRAM space, row and column addresses are multiplexed. The size of the shift in a row address must be specified in the Address Multiplex Select bits (SDADR.MXC[1:0]) in the SDRAM Address Register (SDADR). Additionally, in the SDRAM space, the address precharge-select command (Precharge-sel) is output to the upper bits of column addresses. [Table 15.38](#) shows the relationship between the SDADR.MXC[1:0] settings and the shift amount.

Table 15.38 Address multiplexing (1 of 2)

MXC [1:0]	Shift amount	Data bus width	Address	Address pins external to the MCU																	
				A16	A15	A14	A13	A12	A11	A10	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00	
00	8 bits	8 bits	Row	A24	A23	A22	A21	A20	A19	A18 ^{*1}	A17	A16	A15	A14	A13	A12	A11	A10	A09	A08	
			Column	A24	A23	A22	A21	A20	A19	P	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00	
		16 bits	Row	A24	A23	A22	A21	A20	A19 ^{*1}	A18	A17	A16	A15	A14	A13	A12	A11	A10	A09	A08	
			Column	A24	A23	A22	A21	A20	P	A10	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00	
		32 bits	Row	A24	A23	A22	A21	A20 ^{*1}	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10	A09	A08	
			Column	A24	A23	A22	A21	P	A11	A10	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00	

Table 15.38 Address multiplexing (2 of 2)

MXC [1:0]	Shift amount	Data bus width	Address	Address pins external to the MCU																
				A16	A15	A14	A13	A12	A11	A10	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00
01	9 bits	8 bits	Row	A25	A24	A23	A22	A21	A20	A19 ^{*1}	A18	A17	A16	A15	A14	A13	A12	A11	A10	A09
			Column	A25	A24	A23	A22	A21	A20	P	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00
		16 bits	Row	A25	A24	A23	A22	A21	A20 ^{*1}	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10	A09
			Column	A25	A24	A23	A22	A21	P	A10	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00
		32 bits	Row	A25	A24	A23	A22	A21 ^{*1}	A20	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10	A09
			Column	A25	A24	A23	A22	P	A11	A10	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00
10	10 bits	8 bits	Row	A26	A25	A24	A23	A22	A21	A20 ^{*1}	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10
			Column	A26	A25	A24	A23	A22	A21	P	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00
		16 bits	Row	A26	A25	A24	A23	A22	A21 ^{*1}	A20	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10
			Column	A26	A25	A24	A23	A22	P	A10	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00
		32 bits	Row	A26	A25	A24	A23	A22 ^{*1}	A21	A20	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10
			Column	A26	A25	A24	A23	P	A11	A10	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00
11	11 bits	8 bits	Row	—	A26	A25	A24	A23	A22	A21 ^{*1}	A20	A19	A18	A17	A16	A15	A14	A13	A12	A11
			Column	—	A26	A25	A24	A23	A10	P	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00
		16 bits	Row	—	A26	A25	A24	A23	A22 ^{*1}	A21	A20	A19	A18	A17	A16	A15	A14	A13	A12	A11
			Column	—	A26	A25	A24	A11	P	A10	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00
		32 bits	Row	—	A26	A25	A24	A23 ^{*1}	A22	A21	A20	A19	A18	A17	A16	A15	A14	A13	A12	A11
			Column	—	A26	A25	A24	P	A11	A10	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00

Note: P: Precharge-select command (Precharge-sel) is output.

Note 1. When the PALL command is issued, Precharge-sel = 1 (high) is output. When the Active command is issued, the associated address is output.

15.6.13 Example SDRAM Connections

15.6.13.1 32-Bit Bus Space

Table 15.39 shows an example for connecting to two 512-Mb SDRAMs with 13-bit row address, 10-bit column address and 16-bit bus.

Table 15.39 Example of 512-Mb SDRAM × 2 connections to a 16-bit bus (1 of 2)

This MCU (shift amount: 10 bits)				SDRAM (512 Mb, with 16-bit bus)			
No.	Pin name	Row	Column	Pin name	BA/AP	Row	Column
1	RAS			→ RAS#			
2	CAS			→ CAS#			
3	WE			→ WE#			
4	CKE			→ CKE			
5	SDCS			→ CS#			
6	SDCLK			→ CLK			
7	DQM1			→ UDQM			
8	DQM0			→ LDQM			
9	A16	A26	A26	→ BA1 (A14)	BA1	—	—
10	A15	A25	A25	→ BA0 (A13)	BA0	—	—
11	A14	A24	A24	→ A12		A12	—

Table 15.39 Example of 512-Mb SDRAM × 2 connections to a 16-bit bus (2 of 2)

This MCU (shift amount: 10 bits)					SDRAM (512 Mb, with 16-bit bus)			
No.	Pin name	Row	Column		Pin name	BA/AP	Row	Column
12	A13	A23	A23	→	A11		A11	—
13	A12	A22	P	→	A10	AP	A10	—
14	A11	A21	A11	→	A9		A9	A9
15	A10	A20	A10	→	A8		A8	A8
16	A09	A19	A9	→	A7		A7	A7
17	A08	A18	A8	→	A6		A6	A6
18	A07	A17	A7	→	A5		A5	A5
19	A06	A16	A6	→	A4		A4	A4
20	A05	A15	A5	→	A3		A3	A3
21	A04	A14	A4	→	A2		A2	A2
22	A03	A13	A3	→	A1		A1	A1
23	A02	A12	A2	→	A0		A0	A0
24	DQ15 to DQ08			↔	DQ[15:8]			
25	DQ07 to DQ00			↔	DQ[7:0]			
This MCU (shift amount 10 bits)					SDRAM (512 Mb, with 16-bit bus)			
No.*1	Pin name	Row	Column		Pin name	BA/AP	Row	Column
(1)	RAS			→	RAS#			
(2)	CAS			→	CAS#			
(3)	WE			→	WE#			
(4)	CKE			→	CKE			
(5)	SDCS			→	CS#			
(6)	SDCLK			→	CLK			
26	DQM3			→	UDQM			
27	DQM2			→	LDQM			
(9)	A16	A26	A26	→	BA1 (A14)	BA1	—	—
(10)	A15	A25	A25	→	BA0 (A13)	BA0	—	—
(11)	A14	A24	A24	→	A12		A12	—
(12)	A13	A23	A23	→	A11		A11	—
(13)	A12	A22	P	→	A10	AP	A10	—
(14)	A11	A21	A11	→	A9		A9	A9
(15)	A10	A20	A10	→	A8		A8	A8
(16)	A09	A19	A9	→	A7		A7	A7
(17)	A08	A18	A8	→	A6		A6	A6
(18)	A07	A17	A7	→	A5		A5	A5
(19)	A06	A16	A6	→	A4		A4	A4
(20)	A05	A15	A5	→	A3		A3	A3
(21)	A04	A14	A4	→	A2		A2	A2
(22)	A03	A13	A3	→	A1		A1	A1
(23)	A02	A12	A2	→	A0		A0	A0
28	DQ31 to DQ24			↔	DQ[15:8]			
29	DQ23 to DQ16			↔	DQ[7:0]			

Note 1. The number (No.) in parentheses indicates that the pin branches out to two SDRAM pins.

Table 15.40 shows an example for connecting to a 256-Mb SDRAM with 12-bit row address, 9-bit column address and 32-bit bus.

Table 15.40 Example of 256-Mb SDRAM × 1 connection to a 32-bit bus

This MCU (shift amount: 9 bits)					SDRAM (256 Mb, with 32-bit bus)			
No.	Pin name	Row	Column		Pin name	BA/AP	Row	Column
1	RAS			→	RAS#			
2	CAS			→	CAS#			
3	WE			→	WE#			
4	CKE			→	CKE			
5	SDCS			→	CS#			
6	SDCLK			→	CLK			
7	DQM3			→	DQM3			
8	DQM2			→	DQM2			
9	DQM1			→	DQM1			
10	DQM0			→	DQM0			
11	A15	A24	A24	→	BA1 (A13)	BA1	—	—
12	A14	A23	A23	→	BA0 (A12)	BA0	—	—
13	A13	A22	A22	→	A11		A11	—
14	A12	A21	P	→	A10	AP	A10	—
15	A11	A20	A11	→	A9		A9	—
16	A10	A19	A10	→	A8		A8	A8
17	A09	A18	A9	→	A7		A7	A7
18	A08	A17	A8	→	A6		A6	A6
19	A07	A16	A7	→	A5		A5	A5
20	A06	A15	A6	→	A4		A4	A4
21	A05	A14	A5	→	A3		A3	A3
22	A04	A13	A4	→	A2		A2	A2
23	A03	A12	A3	→	A1		A1	A1
24	A02	A11	A2	→	A0		A0	A0
25	DQ31 to DQ24			↔	DQ[31:24]			
26	DQ23 to DQ16			↔	DQ[23:16]			
27	DQ15 to DQ08			↔	DQ[15:8]			
28	DQ07 to DQ00			↔	DQ[7:0]			

Table 15.41 shows an example for connecting to two 128-Mb SDRAMs with 12-bit row address, 9-bit column address and 16-bit bus.

Table 15.41 Example of 128-Mb SDRAM × 2 connections to a 16-bit bus (1 of 3)

This MCU (shift amount: 9 bits)					SDRAM (128 Mb, with 16-bit bus)			
No.	Pin name	Row	Column		Pin name	BA/AP	Row	Column
1	RAS			→	RAS#			
2	CAS			→	CAS#			
3	WE			→	WE#			
4	CKE			→	CKE			

Table 15.41 Example of 128-Mb SDRAM × 2 connections to a 16-bit bus (2 of 3)

This MCU (shift amount: 9 bits)					SDRAM (128 Mb, with 16-bit bus)			
No.	Pin name	Row	Column		Pin name	BA/AP	Row	Column
5	SDCS			→	CS#			
6	SDCLK			→	CLK			
7	DQM1			→	UDQM			
8	DQM0			→	LDQM			
9	A15	A24	A24	→	BA1 (A13)	BA1	—	—
10	A14	A23	A23	→	BA0 (A12)	BA0	—	—
11	A13	A22	A22	→	A11		A11	—
12	A12	A21	P	→	A10	AP	A10	—
13	A11	A20	A11	→	A9		A9	—
14	A10	A19	A10	→	A8		A8	A8
15	A09	A18	A9	→	A7		A7	A7
16	A08	A17	A8	→	A6		A6	A6
17	A07	A16	A7	→	A5		A5	A5
18	A06	A15	A6	→	A4		A4	A4
19	A05	A14	A5	→	A3		A3	A3
20	A04	A13	A4	→	A2		A2	A2
21	A03	A12	A3	→	A1		A1	A1
22	A02	A11	A2	→	A0		A0	A0
23	DQ15 to DQ08			↔	DQ[15:8]			
24	DQ07 to DQ00			↔	DQ[7:0]			
This MCU (shift amount 9 bits)					SDRAM (128 Mb, with 16-bit bus)			
No.*1	Pin name	Row	Column		Pin name	BA/AP	Row	Column
(1)	RAS			→	RAS#			
(2)	CAS			→	CAS#			
(3)	WE			→	WE#			
(4)	CKE			→	CKE			
(5)	SDCS			→	CS#			
(6)	SDCLK			→	CLK			
25	DQM3			→	UDQM			
26	DQM2			→	LDQM			
(9)	A15	A24	A24	→	BA1 (A13)	BA1	—	—
(10)	A14	A23	A23	→	BA0 (A12)	BA0	—	—
(11)	A13	A22	A22	→	A11		A11	—
(12)	A12	A21	P	→	A10	AP	A10	—
(13)	A11	A20	A11	→	A9		A9	—
(14)	A10	A19	A10	→	A8		A8	A8
(15)	A09	A18	A9	→	A7		A7	A7
(16)	A08	A17	A8	→	A6		A6	A6
(17)	A07	A16	A7	→	A5		A5	A5
(18)	A06	A15	A6	→	A4		A4	A4
(19)	A05	A14	A5	→	A3		A3	A3

Table 15.41 Example of 128-Mb SDRAM × 2 connections to a 16-bit bus (3 of 3)

This MCU (shift amount: 9 bits)					SDRAM (128 Mb, with 16-bit bus)			
No.	Pin name	Row	Column		Pin name	BA/AP	Row	Column
(20)	A04	A13	A4	→	A2		A2	A2
(21)	A03	A12	A3	→	A1		A1	A1
(22)	A02	A11	A2	→	A0		A0	A0
27	DQ31 to DQ24			↔	DQ[15:8]			
28	DQ23 to DQ16			↔	DQ[7:0]			

Note 1. The number (No.) in parentheses indicates that the pin branches out to two SDRAM pins.

15.6.13.2 16-bit Bus Space

Table 15.42 shows an example connection to two 512-Mb SDRAMs with a 13-bit row address, 11-bit column address, and 8-bit bus.

Table 15.42 Example of 512-Mb SDRAM × 2 connections to an 8-bit bus (1 of 2)

This MCU (shift amount: 11 bits)					SDRAM (512 Mb, with 8-bit bus)			
No.	Pin name	Row	Column		Pin name	BA/AP	Row	Column
1	RAS			→	RAS#			
2	CAS			→	CAS#			
3	WE			→	WE#			
4	CKE			→	CKE			
5	SDCS			→	CS#			
6	SDCLK			→	CLK			
7	DQM0			→	LDQM			
8	A15	A26	A26	→	BA1 (A14)	BA1	—	—
9	A14	A25	A25	→	BA0 (A13)	BA0	—	—
10	A13	A24	A24	→	A12		A12	—
11	A12	A23	A11	→	A11		A11	A11
12	A11	A22	P	→	A10	AP	A10	—
13	A10	A21	A10	→	A9		A9	A9
14	A09	A20	A9	→	A8		A8	A8
15	A08	A19	A8	→	A7		A7	A7
16	A07	A18	A7	→	A6		A6	A6
17	A06	A17	A6	→	A5		A5	A5
18	A05	A16	A5	→	A4		A4	A4
19	A04	A15	A4	→	A3		A3	A3
20	A03	A14	A3	→	A2		A2	A2
21	A02	A13	A2	→	A1		A1	A1
22	A01	A12	A1	→	A0		A0	A0
23	DQ07 to DQ00			↔	DQ[7:0]			
This MCU (shift amount 11 bits)					SDRAM (512 Mb, with 8-bit bus)			
No.*1	Pin name	Row	Column		Pin name	BA/AP	Row	Column
(1)	RAS			→	RAS#			
(2)	CAS			→	CAS#			
(3)	WE			→	WE#			

Table 15.42 Example of 512-Mb SDRAM × 2 connections to an 8-bit bus (2 of 2)

This MCU (shift amount: 11 bits)					SDRAM (512 Mb, with 8-bit bus)			
No.	Pin name	Row	Column		Pin name	BA/AP	Row	Column
(4)	CKE			→	CKE			
(5)	SDCS			→	CS#			
(6)	SDCLK			→	CLK			
24	DQM1			→	LDQM			
(8)	A15	A26	A26	→	BA1 (A14)	BA1	—	—
(9)	A14	A25	A25	→	BA0 (A13)	BA0	—	—
(10)	A13	A24	A24	→	A12		A12	—
(11)	A12	A23	A11	→	A11		A11	A11
(12)	A11	A22	P	→	A10	AP	A10	—
(13)	A10	A21	A10	→	A9		A9	A9
(14)	A09	A20	A9	→	A8		A8	A8
(15)	A08	A19	A8	→	A7		A7	A7
(16)	A07	A18	A7	→	A6		A6	A6
(17)	A06	A17	A6	→	A5		A5	A5
(18)	A05	A16	A5	→	A4		A4	A4
(19)	A04	A15	A4	→	A3		A3	A3
(20)	A03	A14	A3	→	A2		A2	A2
(21)	A02	A13	A2	→	A1		A1	A1
(22)	A01	A12	A1	→	A0		A0	A0
25	DQ15 to DQ08			↔	DQ[7:0]			

Note 1. The number (No.) in parentheses indicates that the pin branches out to two SDRAM pins.

Table 15.43 shows an example connection to a 512-Mb SDRAMs with a 13-bit row address, 10-bit column address, and 16-bit bus.

Table 15.43 Example of 512-Mb SDRAM × 1 connection to a 16-bit bus (1 of 2)

This MCU (shift amount: 10 bits)					SDRAM (512 Mb, with 16-bit bus)			
No.	Pin name	Row	Column		Pin name	BA/AP	Row	Column
1	RAS			→	RAS#			
2	CAS			→	CAS#			
3	WE			→	WE#			
4	CKE			→	CKE			
5	SDCS			→	CS#			
6	SDCLK			→	CLK			
7	DQM1			→	UDQM			
8	DQM0			→	LDQM			
9	A15	A25	A25	→	BA1 (A14)	BA1	—	—
10	A14	A24	A24	→	BA0 (A13)	BA0	—	—
11	A13	A23	A23	→	A12		A12	—
12	A12	A22	A22	→	A11		A11	—
13	A11	A21	P	→	A10	AP	A10	—
14	A10	A20	A10	→	A9		A9	A9

Table 15.43 Example of 512-Mb SDRAM × 1 connection to a 16-bit bus (2 of 2)

This MCU (shift amount: 10 bits)					SDRAM (512 Mb, with 16-bit bus)			
No.	Pin name	Row	Column		Pin name	BA/AP	Row	Column
15	A09	A19	A9	→	A8		A8	A8
16	A08	A18	A8	→	A7		A7	A7
17	A07	A17	A7	→	A6		A6	A6
18	A06	A16	A6	→	A5		A5	A5
19	A05	A15	A5	→	A4		A4	A4
20	A04	A14	A4	→	A3		A3	A3
21	A03	A13	A3	→	A2		A2	A2
22	A02	A12	A2	→	A1		A1	A1
23	A01	A11	A1	→	A0		A0	A0
24	DQ15 to DQ08			↔	DQ[15:8]			
25	DQ07 to DQ00			↔	DQ[7:0]			

Table 15.44 shows an example connection to a 256-Mb SDRAMs with a 13-bit row address, 9-bit column address, and 16-bit bus.

Table 15.44 Example of 256-Mb SDRAM × 1 connection to a 16-bit bus (1 of 2)

This MCU (shift amount: 9 bits)					SDRAM (256 Mb, with 16-bit bus)			
No.	Pin name	Row	Column		Pin name	BA/AP	Row	Column
1	RAS			→	RAS#			
2	CAS			→	CAS#			
3	WE			→	WE#			
4	CKE			→	CKE			
5	SDCS			→	CS#			
6	SDCLK			→	CLK			
7	DQM1			→	UDQM			
8	DQM0			→	LDQM			
9	A15	A24	A24	→	BA1 (A14)	BA1	—	—
10	A14	A23	A23	→	BA0 (A13)	BA0	—	—
11	A13	A22	A22	→	A12		A12	—
12	A12	A21	A21	→	A11		A11	—
13	A11	A20	P	→	A10	AP	A10	—
14	A10	A19	A10	→	A9		A9	—
15	A09	A18	A9	→	A8		A8	A8
16	A08	A17	A8	→	A7		A7	A7
17	A07	A16	A7	→	A6		A6	A6
18	A06	A15	A6	→	A5		A5	A5
19	A05	A14	A5	→	A4		A4	A4
20	A04	A13	A4	→	A3		A3	A3
21	A03	A12	A3	→	A2		A2	A2
22	A02	A11	A2	→	A1		A1	A1
23	A01	A10	A1	→	A0		A0	A0
24	DQ15 to DQ08			↔	DQ[15:8]			

Table 15.44 Example of 256-Mb SDRAM × 1 connection to a 16-bit bus (2 of 2)

This MCU (shift amount: 9 bits)					SDRAM (256 Mb, with 16-bit bus)			
No.	Pin name	Row	Column		Pin name	BA/AP	Row	Column
25	DQ07 to DQ00			↔	DQ[7:0]			

15.6.14 Constraints

(1) Prohibition of access across areas of the address space

Access across multiple areas of the address space is prohibited with one access, and operation in that case is not guaranteed. Make sure that one word, longword access does not span two areas across the boundaries of each area of the address space.

(2) Low Power condition

In Software Standby and Deep Software Standby modes, auto-refresh operation is not available because the clock supply to the SDRAMC is stopped. To retain the data in the SDRAM when the SDRAM is externally connected, use the self-refresh function. For the procedure for transitioning to and recovering from self-refresh mode, see [section 15.6.6. Self-Refresh](#).

(3) Setting the SDRAM Timing Register

Set the RAI[2:0] bits in the SDRAM Timing Register (SDTR) to a value less than or equal to the sum of the row column latency (SDTR.RCD[1:0]) and column latency (SDTR.CL[2:0]) settings. Operation is not guaranteed if this condition is not satisfied.

(4) Instruction code constraint

You must fix the instruction code to little-endian order.

See [section 15.2.5.1. Endianness Constraint](#) for details.

(5) Consecutive access mode

When SDAMOD.BE is "1" (continuous access enable), setting SDTR.CL = "1" is prohibited, and the operation is not guaranteed when SDTR.CL = "1" is set.

15.7 Bus Error Monitoring Section

This monitoring system monitors each individual area, and whenever it detects an error, it returns the error to the requesting master IP.

15.7.1 Bus Error Types

The following types of errors can occur on each bus:

- Master Security Attribution Unit error
- Bus master MPU error.
- Illegal address access.
- Slave TrustZone Filter error
- Bus error transmitted from each slave IP.

[Table 15.48](#) lists the address ranges where access leads to illegal address access errors. The reserved area in the slave does not trigger an illegal address access error. For more information on the bus master MPU, see [section 16, Memory Protection Unit \(MPU\)](#).

15.7.2 Operations When a Bus Error Occurs

15.7.2.1 Operations When a Bufferable Write Error Occurs

If an access violation occurs due to the bufferable write access an error response may not be returned to the bus master.

When a bufferable write access error that does not return an error response is detected, access is blocked and 1 is set in the bit corresponding to the error occurrence location of MBWERRSTAT. Furthermore, the bus error interrupt request or reset

request is generated according to the setting of the BUSOAD.BWERROAD. Since MBWERRSTAT is held until reset other than Bus and Memory Error Reset or cleared by MBWERRCLR, they can be confirmed in the bus error interrupt handler.

A bufferable write access error can occur in the following cases.

1. TDRW accesses illegal address area.
2. TDRW accesses accessible area and a slave bus error or a slave TrustZone Filter error occurs.
3. CPU0 M-AXI accesses CPU1TCMBI and slave bus error or a slave TrustZone Filter error occurs.
4. CPU0 P-AHB accesses illegal address area.
5. CPU0 P-AHB accesses accessible area and slave bus error or a slave TrustZone Filter error occurs.
6. CPU1 C-AHB accesses illegal address area.
7. CPU1 C-AHB accesses accessible area and slave bus error or a slave TrustZone Filter error occurs.
8. CPU1 S-AHB accesses illegal address area.
9. CPU1 S-AHB accesses accessible area and slave bus error or a slave TrustZone Filter error occurs.
10. DMA accesses PABI, PBBI, PIBI, PSBI, CPU0SAHBI and CPU1TCMBI and slave bus error or a slave TrustZone Filter error occurs.

Table 15.45 Bus error notice method for Bufferable write access

OAD Setting	DAP	CPU0, CPU1	Other
BUSOAD.BWERROAD = 0	Interrupt	Interrupt	Interrupt
BUSOAD.BWERROAD = 1	Reset	Reset	Reset

The bus error interrupt request BUS_ERR can be assigned as Non-maskable interrupt request. See the ICU.NMIER register specification for details.

15.7.2.2 Operations When an Error except Bufferable Write Error Occurs

When a bus error occurs for an access other than the bufferable write, error access is blocked, and error operations determine by OAD setting and the bus master are performed. Table 15.46 shows OAD setting corresponding to each error. Table 15.47 shows how to notify when a bus error occurs.

- When OAD setting is interrupt.
Error response is returned to the bus master.
If BUSIRQEN<Master Name>.EN=1, the bus error interrupt BUS_ERR is generated.
The bus error interrupt request BUS_ERR can also be used as Non-maskable interrupt request. See the ICU.NMIER register specifications for details.
- When OAD setting is Reset.
Error response is returned to the bus master and reset request is generated.

Table 15.46 OAD setting corresponding to each error

Error type	OAD bit
MSAU Error	MSAOAD.OAD
MPU Error	MMPUOAD.OAD
Illegal Address Access Error	BUSOAD.ILERROAD
Slave Bus Error	BUSOAD.SLERROAD
Slave TrustZone Filter error	BUSOAD.SLERROAD

Table 15.47 Bus error notice method for Non-bufferable write access (1 of 3)

OAD setting		DAP*1	CPU _n (n = 0, 1)	DMAC/DTC _n (n = 0, 1)	EDMAC	NPU
Interrupt	Bus	Error response only	BUS_ERR	BUS_ERR	BUS_ERR	BUS_ERR
	Bus Master		BusFault	DMAn_TRANSER R	ETHER_GWEI	NPU_IRQ

Table 15.47 Bus error notice method for Non-bufferable write access (2 of 3)

OAD setting		DAP*1	CPU _n (n = 0, 1)	DMAC/DTC _n (n = 0, 1)	EDMAC	NPU
Reset	Bus/Bus Master	Error response only	Reset	Reset	Reset	Reset

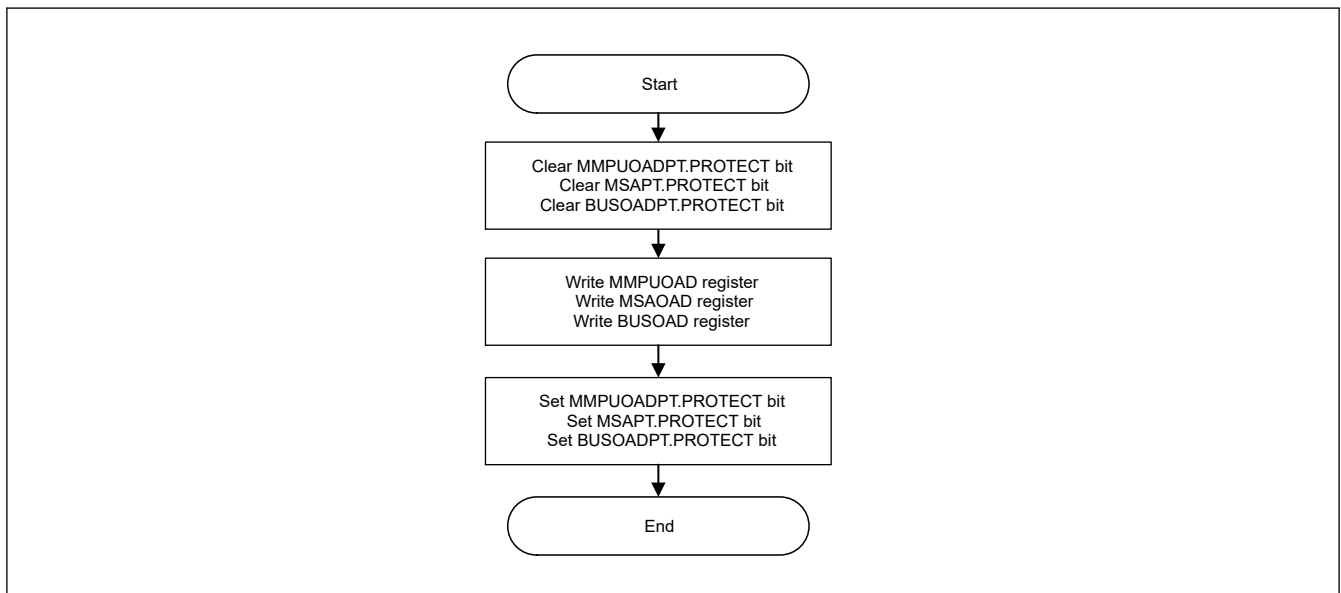
Table 15.47 Bus error notice method for Non-bufferable write access (3 of 3)

OAD setting		GLCDC0/1	DRW0/1	MIPI-DSI	MIPI-CSI via VIN	CEU
Interrupt	Bus	BUS_ERR	BUS_ERR	BUS_ERR	BUS_ERR	BUS_ERR
	Bus Master	—	DRW_IRQ	DSI_SEQ0 DSI_SEQ1 DSI_RCV	VIN0_IRQ	—
Reset	Bus/Bus Master	Reset	Reset	Reset	Reset	Reset

Note: NPU, DRW, MIPI-DSI, MIPI-CSI, DMAC/DTC0, DMAC/DTC1 and EDMAC can issue interrupt when it receives error response. This interrupt is output to ICU and transmitted from ICU to CPU.

Note 1. Only when the bus master is DAP, only an error response is returned regardless of the BUSOAD.OAD setting, and Interrupt and Reset requests are not generated.

Figure 15.67 shows the OAD setting flow after reset. During this register setting, stop the Bus Master.

**Figure 15.67 OAD setting flow after reset**

For details of MMPUOADPT register and MMPUOAD register, see [section 16, Memory Protection Unit \(MPU\)](#).

For details of MSAPT register and MSAOAD register, see [section 52, Security Features](#).

(1) Master Security Attribution Unit Error

The bus master of other than CPU and DAP has MSAU to block access to secure alias address from non-secure masters. The CPU does not have MSAU since it has IDAU and SAU. When MSAU Error is detected, an Error response is returned to the Master. At the same time, set 1 to MSERRSTAT bit of BUSERRSTAT<Master Name>. Furthermore, the bus error interrupt request or reset request is generated according to the setting of the MSAOAD.OAD and BUSIRQEN<Master Name>.EN. Since BUSERRSTAT<Master Name> is held until reset other than Bus error reset, Common memory reset, and CPU local memory reset Error Reset or cleared by BUSERRCLR<Master Name>, they can be confirmed in the bus error interrupt handler or after reset. The bus error interrupt is generated only at the first MSAU Error after reset or clearing BUSERRSTAT<Master Name>.MSERRSTAT bit by BUSERRCLR<Master Name>.

(2) Bus Master MPU Error

The bus master of other than CPU and DAP has Master-MPU for access control of the set address area. The CPU does not have Master-MPU since it has ARM MPU. When Master-MPU Error is detected, an Error response is returned to

the Master. At the same time, set 1 to MMERRSTAT bit of BUSERRSTAT<Master Name>. Furthermore, the bus error interrupt request or reset request is generated according to the setting of the MMPUOAD.OAD and BUSIRQEN<Master Name>.EN. Since BUSERRSTAT<Master Name> is held until reset other than Bus error reset, Common memory reset, and CPU local memory reset Error Reset or cleared by BUSERRCLR<Master Name>, they can be confirmed in the bus error interrupt handler or after reset. The bus error interrupt is generated only at the first Master-MPU Error after reset or clearing BUSERRSTAT<Master Name>.MMERRSTAT bit by BUSERRCLR<Master Name>.

(3) Illegal Access Error

According to [section 15.7.3. Conditions Leading to Illegal Address Access Errors](#), the illegal address access error occurs. When illegal address access error is detected, an Error response is returned to the Master. When a master other than DAP caused the illegal address access error, set 1 to ILERRSTAT bit of BUSERRSTAT<Master Name>. Furthermore, the bus error interrupt request or reset request is generated according to the setting of the BUSOAD.ILERRROAD and BUSIRQEN<Master Name>.EN. Since BUSERRSTAT<Master Name> is held until reset other than Bus error reset, Common memory reset, and CPU local memory reset Error Reset or cleared by BUSERRCLR<Master Name>, they can be confirmed in the BusFault Handler or the bus error interrupt handler. The bus error interrupt is generated only at the first Illegal Error after reset or clearing BUSERRSTAT<Master Name>.ILERRSTAT bit by BUSERRCLR<Master Name>.

(4) Slave TrustZone Filter Error

When Slave TrustZone Filter Error is detected, an Error response is returned to the Master. When a master other than DAP caused Slave TrustZone Filter error, set 1 to SLERRSTAT bit of BUSERRSTAT<Master Name>. Furthermore, the bus error interrupt request or reset request is generated according to the setting of the BUSOAD.SLERROAD and BUSIRQEN<Master Name>.EN. Since BUSERRSTAT<Master Name> is held until reset other than Bus error reset, Common memory reset, and CPU local memory reset or cleared by BUSERRCLR<Master Name>, they can be confirmed in the bus error interrupt handler or after reset. The bus error interrupt is generated only at the first Slave TrustZone Filter Error after reset or clearing BUSERRSTAT<Master Name>.SLERRSTAT bit by BUSERRCLR<Master Name>.

(5) Slave Bus Error

Slave Bus Error occurs in the slave. When Slave Bus Error is detected, an Error response is returned to the Master. When a master other than DAP caused Slave Bus error, set 1 to SLERRSTAT bit of BUSERRSTAT<Master Name>. Furthermore, the bus error interrupt request or reset request is generated according to the setting of the BUSOAD.SLERROAD and BUSIRQEN<Master Name>.EN. Since BUSERRSTAT<Master Name> is held until reset other than Bus and Memory Error Reset or cleared by BUSERRCLR<Master Name>, they can be confirmed in the BusFault Handler or bus error interrupt handler. The bus error interrupt is generated only at the first Slave-Bus Error after reset or clearing BUSERRSTAT<Master Name>.SLERRSTAT bit by BUSERRCLR<Master Name>.

15.7.3 Conditions Leading to Illegal Address Access Errors

Illegal address access error by bus access occurs in the address space shown in [Table 15.48](#).

Table 15.48 Conditions leading to illegal address access errors (1 of 4)

Address		SLAVE bus	MASTER bus											
Start	End		CPU0 MAXI	CPU0 PAHB	CPU1 CAHB	CPU1 SAHB	NPU		DMAC/DTC		EDMAC	GRAPHIC		
							0	1	0	1		GLCD C	DRW/ MIPDSI	CEU/ MIPCSI
0x0000_0000	0x01FF_FFFF	Reserved	E	N/A	E	N/A	E	E	E	E	E	E	E	E
0x0200_0000	0x0207_FFFF	MRC0BI (Read)	—	N/A	—	N/A	E	—	—	—	—	—	—	N/A
		MRC0BI (Write)	—	N/A	—	N/A	E	N/A	—	—	E	N/A	E	E
0x0208_0000	0x020B_FFFF	MRC0BI (Read)	—/E*1	N/A	—/E*1	N/A	E	—/E*1	—/E*1	—/E*1	—/E*1	—/E*1	—/E*1	N/A
		MRC0BI (Write)	—/E*1	N/A	—/E*1	N/A	E	N/A	—/E*1	—/E*1	E	N/A	E	E
0x020C_0000	0x020F_FFFF	MRC0BI (Read)	—/E*2	N/A	—/E*2	N/A	E	—/E*2	—/E*2	—/E*2	—/E*2	—/E*2	—/E*2	N/A
		MRC0BI (Write)	—/E*2	N/A	—/E*2	N/A	E	N/A	—/E*2	—/E*2	E	N/A	E	E

Table 15.48 Conditions leading to illegal address access errors (2 of 4)

Address		SLAVE bus	MASTER bus											
Start	End		CPU0 MAXI	CPU0 PAHB	CPU1 CAHB	CPU1 SAHB	NPU		DMAC/DTC		EDMAC	GRAPHIC		
							0	1	0	1		GLCD C	DRW/ MIPIDSI	CEU/ MIPICSI
0x0210_0000	0x027F_FFFF	Reserved	E	N/A	E	N/A	E	E	E	E	E	E	E	E
0x0280_0000	0x0280_1FFF	Reserved	E	N/A	E	N/A	E	E	E	E	E	E	E	E
0x0280_2000	0x029F_FFFF	Reserved	E	N/A	E	N/A	E	E	E	E	E	E	E	E
0x02A0_0000	0x02CF_FFFF	MRE0BI	—	N/A	—	N/A	E	—	—	—	—	—	—	E
0x02D0_0000	0x02DF_FFFF	Reserved	E	N/A	E	N/A	E	E	E	E	E	E	E	E
0x02E0_0000	0x02FF_FFFF	MRE0BI	—	N/A	—	N/A	E	—	—	—	—	—	—	E
0x0300_0000	0x07FF_FFFF	Reserved	E	N/A	E	N/A	E	E	E	E	E	E	E	E
0x0800_0000	0x0FFF_FFFF	OSPI1BI	—	N/A	—	N/A	—	—	—	—	—	—	—	—
0x1000_0000	0x11FF_FFFF	Reserved	E	N/A	E	N/A	E	E	E	E	E	E	E	E
0x1200_0000	0x1207_FFFF	MRC0BI (Read)	—	N/A	—	N/A	E	—	—	—	—	—	—	N/A
		MRC0BI (Write)	—	N/A	—	N/A	E	N/A	—	—	E	N/A	E	E
0x1208_0000	0x120B_FFFF	MRC0BI (Read)	—/E*1	N/A	—/E*1	N/A	E	—/E*1	—/E*1	—/E*1	—/E*1	—/E*1	—/E*1	N/A
		MRC0BI (Write)	—/E*1	N/A	—/E*1	N/A	E	N/A	—/E*1	—/E*1	E	N/A	E	E
0x120C_0000	0x120F_FFFF	MRC0BI (Read)	—/E*2	N/A	—/E*2	N/A	E	—/E*2	—/E*2	—/E*2	—/E*2	—/E*2	—/E*2	N/A
		MRC0BI (Write)	—/E*2	N/A	—/E*2	N/A	E	N/A	—/E*2	—/E*2	E	N/A	E	E
0x1210_0000	0x127F_FFFF	Reserved	E	N/A	E	N/A	E	E	E	E	E	E	E	E
0x1280_0000	0x1280_1FFF	Reserved	E	N/A	E	N/A	E	E	E	E	E	E	E	E
0x1280_2000	0x129F_FFFF	Reserved	E	N/A	E	N/A	E	E	E	E	E	E	E	E
0x12A0_0000	0x12CF_FFFF	MRE0BI	—	N/A	—	N/A	E	—	—	—	—	—	—	E
0x12D0_0000	0x12DF_FFFF	Reserved	E	N/A	E	N/A	E	E	E	E	E	E	E	E
0x12E0_0000	0x12FF_FFFF	MRE0BI	—	N/A	—	N/A	E	—	—	—	—	—	—	E
0x1300_0000	0x17FF_FFFF	Reserved	E	N/A	E	N/A	E	E	E	E	E	E	E	E
0x1800_0000	0x1FFF_FFFF	OSPI1BI	—	N/A	—	N/A	—	—	—	—	—	—	—	—
0x2000_0000	0x21FF_FFFF	Reserved	E	N/A	N/A	E	E	E	E	E	E	E	E	E
0x2200_0000	0x2207_FFFF	S0BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x2208_0000	0x220F_FFFF	S1BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x2210_0000	0x2217_FFFF	S2BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x2218_0000	0x2219_FFFF	S3BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x221A_0000	0x221A_FFFF	S0BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x221B_0000	0x221B_FFFF	S1BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x221C_0000	0x221C_FFFF	S2BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x221D_0000	0x221D_3FFF	S3BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x221D_4000	0x27FF_FFFF	Reserved	E	N/A	N/A	E	E	E	E	E	E	E	E	E
0x2800_0000	0x2803_FFFF	CPU0SAHBI	E	N/A	N/A	—	E	E	—	—	E	E	E	E
0x2804_0000	0x29FF_FFFF	Reserved	E	N/A	N/A	E	E	E	E	E	E	E	E	E
0x2A00_0000	0x2A01_FFFF	CPU1TCMBI	—	N/A	N/A	E	E	E	—	—	E	E	E	E
0x2A02_0000	0x2FFF_FFFF	Reserved	E	N/A	N/A	E	E	E	E	E	E	E	E	E
0x3000_0000	0x31FF_FFFF	Reserved	E	N/A	N/A	E	E	E	E	E	E	E	E	E
0x3200_0000	0x3207_FFFF	S0BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x3208_0000	0x320F_FFFF	S1BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—

Table 15.48 Conditions leading to illegal address access errors (3 of 4)

Address		SLAVE bus	MASTER bus											
Start	End		CPU0 MAXI	CPU0 PAHB	CPU1 CAHB	CPU1 SAHB	NPU		DMAC/DTC		EDMAC	GRAPHIC		
							0	1	0	1		GLCD C	DRW/ MIPDSI	CEU/ MIPICSI
0x3210_0000	0x3217_FFFF	S2BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x3218_0000	0x3219_FFFF	S3BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x321A_0000	0x321A_FFFF	S0BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x321B_0000	0x321B_FFFF	S1BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x321C_0000	0x321C_FFFF	S2BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x321D_0000	0x321D_3FFF	S3BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x321D_4000	0x37FF_FFFF	Reserved	E	N/A	N/A	E	E	E	E	E	E	E	E	E
0x3800_0000	0x3803_FFFF	CPU0SAHBI	E	N/A	N/A	—	E	E	—	—	E	E	E	E
0x3804_0000	0x39FF_FFFF	Reserved	E	N/A	N/A	E	E	E	E	E	E	E	E	E
0x3A00_0000	0x3A01_FFFF	CPU1TCMBI	—	N/A	N/A	E	E	E	—	—	E	E	E	E
0x3A02_0000	0x3FFF_FFFF	Reserved	E	N/A	N/A	E	E	E	E	E	E	E	E	E
0x4000_0000	0x4000_1FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4000_2000	0x4000_2FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4000_3000	0x4000_3FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4000_4000	0x4000_4FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4000_5000	0x4000_5FFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x4000_6000	0x4000_6FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4000_7000	0x4000_7FFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x4000_8000	0x4000_8FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4000_9000	0x4000_9FFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x4000_A000	0x4000_AFFF	PS0BI	N/A	—	N/A	N/A	E	E	E	E	E	E	E	E
		PS1BI	N/A	N/A	N/A	—	E	E	E	E	E	E	E	E
0x4000_B000	0x4000_BFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x4000_C000	0x4000_CFFF	PS0BI	N/A	—	N/A	N/A	E	E	E	E	E	E	E	E
		PS1BI	N/A	N/A	N/A	—	E	E	E	E	E	E	E	E
0x4000_D000	0x4000_EFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x4000_F000	0x4000_FFFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4001_0000	0x4001_AFFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4001_B000	0x4001_BFFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4001_C000	0x4001_CFFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4001_D000	0x4001_DFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x4001_E000	0x4001_EFFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4001_F000	0x4001_FFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x4002_0000	0x4002_0FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4002_1000	0x4011_FFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x4012_0000	0x4013_FFFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4014_0000	0x4014_0FFF	NPUCBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4014_1000	0x401F_FFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x4020_0000	0x402F_FFFF	PBBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4030_0000	0x403F_FFFF	PABI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4040_0000	0x404F_FFFF	PIBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x4050_0000	0x4FFF_FFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x5000_0000	0x5000_1FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5000_2000	0x5000_2FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E

Table 15.48 Conditions leading to illegal address access errors (4 of 4)

Address		SLAVE bus	MASTER bus											
Start	End		CPU0 MAXI	CPU0 PAHB	CPU1 CAHB	CPU1 SAHB	NPU		DMAC/DTC		EDMAC	GRAPHIC		
							0	1	0	1		GLCD C	DRW/ MIPDSI	CEU/ MIPICSI
0x5000_3000	0x5000_3FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5000_4000	0x5000_4FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5000_5000	0x5000_5FFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x5000_6000	0x5000_6FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5000_7000	0x5000_7FFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x5000_8000	0x5000_8FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5000_9000	0x5000_9FFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x5000_A000	0x5000_AFFF	PS0BI	N/A	—	N/A	E	E	E	E	E	E	E	E	E
		PS1BI	N/A	E	N/A	—	E	E	E	E	E	E	E	E
0x5000_B000	0x5000_BFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x5000_C000	0x5000_CFFF	PS0BI	N/A	—	N/A	E	E	E	E	E	E	E	E	E
		PS1BI	N/A	E	N/A	—	E	E	E	E	E	E	E	E
0x5000_D000	0x5000_EFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x5000_F000	0x5000_FFFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5001_0000	0x5001_AFFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5001_B000	0x5001_BFFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5001_C000	0x5001_CFFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5001_D000	0x5001_DFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x5001_E000	0x5001_EFFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5001_F000	0x5001_FFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x5002_0000	0x5002_0FFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5002_1000	0x5011_FFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x5012_0000	0x5013_FFFF	PSBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5014_0000	0x5014_0FFF	NPUCBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5014_1000	0x501F_FFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x5020_0000	0x502F_FFFF	PBBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5030_0000	0x503F_FFFF	PABI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5040_0000	0x504F_FFFF	PIBI	N/A	—	N/A	—	E	E	—	—	E	E	E	E
0x5050_0000	0x5FFF_FFFF	Reserved	N/A	E	N/A	E	E	E	E	E	E	E	E	E
0x6000_0000	0x6FFF_FFFF	ECBI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0x7000_0000	0x7FF_FFFFF	OSPI1BI	—/E*3	N/A	N/A	—/E*3	—/E*3	—/E*3	—/E*3	—/E*3	—/E*3	—/E*3	—/E*3	—/E*3
0x8000_0000	0x9FF_FFFFF	OSPI0BI	—	N/A	N/A	—	—	—	—	—	—	—	—	—
0xA000_0000	0xFFFF_FFFF	Reserved	E	N/A	N/A	E	E	E	E	E	E	E	E	E

Note: "E" : A bus error is occurred.

"N/A" : Transfer does not occur.

"—" : A bus error has not occurred. Even if there is reserved area, a bus error has not occurred.

Do not access the reserved area in Code MRAM, Extra MRAM, SRAM0, SRAM1, SRAM2, and SRAM3. If accessed, a slave TrustZone Filter error may occur.

Note 1. The product with Code MRAM size of 512 KB is "E", others is "—".

Note 2. The product with Code MRAM size of 512 KB or 768 KB is "E", others is "—".

Note 3. The product without SIP-FLASH is "—", and the product with SIP-FLASH is "E".

15.7.4 Timeout

As for some peripheral module, a timeout error can occur in the module-stop function.

When there is no response from the slave for a certain time period, a timeout error is detected.

Timeout error is returned to the request master IP by Error response of the AHB-Lite protocol.

15.8 References

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3. ARM Limited, *Arm[®] Cortex[®]-M85 Processor Technical Reference Manual Revision:r0p2* (ARM 101924_0002_05_en)
4. ARM Limited, *Arm[®] Cortex[®]-M85 Processor User Guide Reference Material Revision:r0p2* (ARM 101927_0002_05_en)
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8. ARM Limited, *ARM AMBA 5 AHB Protocol Specification AHB5, AHB-Lite* (ARM IHI 0033B.b)
9. ARM Limited, *ARM AMBA AXI and ACE Protocol Specification AXI3, AXI4, and AXI4-Lite, ACE and ACE-Lite* (ARM IHI 0022H.c)
10. ARM Limited, *ARM AMBA APB Protocol Specification Version: 2.0* (ARM IHI 0024D)